Occupation counts of rarely switching Markov chains: stratified local laws, modal supports and reconstruction

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Abstract

A Markov chain on finitely many states makes about $\tau \log N$ switches in N steps. We ask which states its most likely vector of occupation counts uses. On each face of the simplex the law weighs a lattice entropy of one power of N per dimension against the Donsker and Varadhan cost of the chain killed on leaving the face. A local limit theorem in each face gives an explicit constant, locates every crossing between irreducible faces up to $o(1)$, and describes the window of tied faces. Near a facet the law follows a Bessel kernel, which we prove for faces with at most three states and for larger faces under one stated hypothesis. The complete graph, where all faces tie at first order, is solved exactly. We then ask which coarse functionals of the law determine the generator. The Donsker and Varadhan functional determines exactly the principal minors of the negative generator, and so does the support of the most likely composition under killing at known rates. These minors fix a reversible generator, and in general a finite set of generators, which is the generator and its time reversal when there is no cut. At low temperature the winners are Freidlin and Wentzell cycles of least size away from finitely many heights. At a critical height explicit prefactors decide, and noncycles win on open windows of the rescaled time scale. Rewards of known size read off the cycle tree.

Keywords: occupation times of Markov chains; mode; principal Dirichlet eigenvalue; Donsker and Varadhan functional; principal minors; Freidlin and Wentzell cycles.
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1 Introduction

Which states does the most likely occupation-count vector of a slowly switching Markov chain use? Fix a generator Q on d states and a start law, and take N steps with transition matrix $I+(T/N)Q$, so that the chain makes about T switches. The occupation counts $n_i = \#\{t \le N : X_t = i\}$ have support $F = \{i : n_i > 0\}$, which we call a face. Write M_F for the largest probability on that exact support and λ_F for its principal exit rate (Definition 2.1). A face is admissible if a path of positive probability has exactly that support. When $T = \tau \log N$ the cost $e^{-T\lambda_F}$ of staying in F and the cost $N^{-(|F|-1)}$ of sharing the mass of F among its count vectors are both powers of N, and they compete. The paper describes the law of the count vector on every face at this scale, reads off the face that carries the mode and the times at which it changes, and then turns the description around and asks which generators are consistent with it.

1.1 The law on a face

Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$.

Theorem A (first-order face selection)**.** *Fix the rates q_{ij} and the law of X_1, and let $T = \tau \log N$ with $\tau > 0$ fixed. Every admissible face F has $M_F = N^{-\Phi_\tau(F)+o(1)}$. For large N the support of every most likely composition is an admissible face that minimizes Φ_τ among the admissible faces. If the chain can pass between any two states of each such minimizing face without leaving it, then the visit fractions of every most likely composition are within $O(1/\log N)$ of the quasi-ergodic law of its support (Definition 2.1).*

Theorem A is Theorem 4.2(ii)–(iv). Theorem 4.4 reads the winners off the lower-left boundary of the convex hull of the points $(\lambda_F, |F| - 1)$, so as τ grows the winning size can only grow. The exponent is one functional. The rate J_F inside a face is the Donsker and Varadhan functional of the chain killed on leaving F, and Proposition 3.1 shows that J_F is the restriction to the face of the functional J_V of the whole chain, where $V = [d]$ is the face of all states. So Theorem 5.6 states the law of the count vector on every face in one display, as the lattice entropy $(|F| - 1)\log N$ of the face plus T times the Donsker and Varadhan cost of the point.

First order fixes a crossing only up to a factor $1 + o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F, and the start gives F positive mass. For such faces we prove a local limit theorem inside the face (Theorem 5.4), and it gives the following result.

Theorem B (face maxima and crossings)**.** *Fix the rates q_{ij} and the law of X_1.*

(i) Let F satisfy (H)*, with $k = |F|$ states. Uniformly for $1 \le T \le N^{1/3}$,*

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F\, e^{-T\lambda_F}\big(1 + O(1/T + T^2/N)\big),$$

where $K_F > 0$ is the explicit constant of Definition 5.1 (for $k = 1$ it is the start mass of the state).

(ii) Let F and G satisfy (H)*, and suppose that F has $\Delta \ge 1$ more states than G and an exit rate smaller by $\delta > 0$. For large N the equation $M_F = M_G$ has a root T in $[1, N^{1/3}]$, and every such root satisfies*

$$\delta T + \frac{\Delta}{2}\log T = \Delta \log N - \log\frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

Part (i) is Theorem 5.7 and part (ii) is Theorem 5.8. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2}\log\log N + O(1)]$, so the coefficient $\frac{1}{2}$ of $\log\log N$ holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 5.2), and on a general face two counts of trees (Proposition 5.11). When several faces tie at first order, Theorem 5.9 describes the window of T in which they compete, and a weak field shifts the line of each face in that window by the field averaged against its quasi-ergodic law (Proposition 5.15).

The local law stays away from the boundary of the face. At the boundary a new scale appears. Let $x = Tn_i/N$ be the time spent in a state i whose count is small, and $S = T - x$. Section 6 pulls the visits to i out of the exact run expansion and writes the density on F as a contour

integral against the density on the facet $F \setminus \{i\}$ with a complex weight. For a pair face the integral is a Bessel kernel, $e^{-\gamma x}\sqrt{abS/x}\, I_1(2\sqrt{abSx})$ times the density on the facet, which is the facet form of Pedler's formula for two-state chains, and we prove it with an explicit error for all x (Proposition 6.3). The constants a, b and γ are explicit for every irreducible chain through the left and right tilt vectors at the facet profile, and $\gamma - q_i$ is $ab/2$ times an effective resistance (Proposition 6.4). For every face with at most three states the layer theorem holds without further hypothesis. Its Bessel form is exact up to a factor e^E with E of order $1/T + \sqrt{x/T}(1+x)$ up to $x \le \delta T$, so the relative form holds for $x \ll T^{1/3}$ when the chain is not reversible, and a generalized kernel holds with relative error $O(1/T)$ on the whole range. For faces with four or more states the same statements are conditional on one hypothesis, a local law with complex weights, which we state, prove for two-state facets, and leave open in general (Hypothesis L and Theorem 6.7). The near-vertex law of Section 8, where one count is large and a few counts are of order one and the crossing with the vertex moves to T of order $\sqrt{N}$, is the discrete pair-face case of this layer, in the regime where the Laguerre kernel cannot yet be replaced by the Bessel kernel (Theorem 8.2).

1.2 The complete graph

On the complete graph with the uniform start, $Q = J - dI$ with J the all-ones matrix, every face ties at first order and the constants decide. A face of k states has $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$, and the k^{k-2} spanning trees of K_k enter through it. Put $a_k = \frac{1}{2}\log(4\pi) - \frac{k+1/2}{k-1}\log k$, write V_N for the probability of a vertex, and call a composition balanced if its positive coordinates differ by at most 1.

Theorem C (the complete graph)**.** *Let $Q = J - dI$ and let the start be uniform.*

(i) Let F be a face of $2 \le k \le d$ states. For every $N \ge k$ the equation $M_F = V_N$ has exactly one root $T_{N,k}$ in $(0, N/(d-1))$, and the switch probability $T_{N,k}/N$ at the root decreases strictly as N grows. As $N \to \infty$,

$$T_{N,k} + \tfrac{1}{2}\log T_{N,k} = \log N + a_k + O\Big(\frac{1}{\log N}\Big).$$

(ii) For every N and every $0 < T \le N/d$, the most likely compositions with a given support are exactly the balanced ones.

(iii) Since $a_2 > a_3 > \cdots > a_d$, for large N and T up to any fixed multiple of $\log N$ the mode changes exactly once, from the d vertices straight to the balanced compositions with d positive coordinates.

Part (i) is Theorem 7.5, part (ii) is Theorem 7.6 and part (iii) is Corollary 7.7. Figure 1 shows the window. A start law changes the constant of a face through its start mass, and a weak field changes it through the field averaged over the face. Either can let a face of intermediate size win (Example 7.10 and Corollary 7.11), and one field makes every size win in turn.

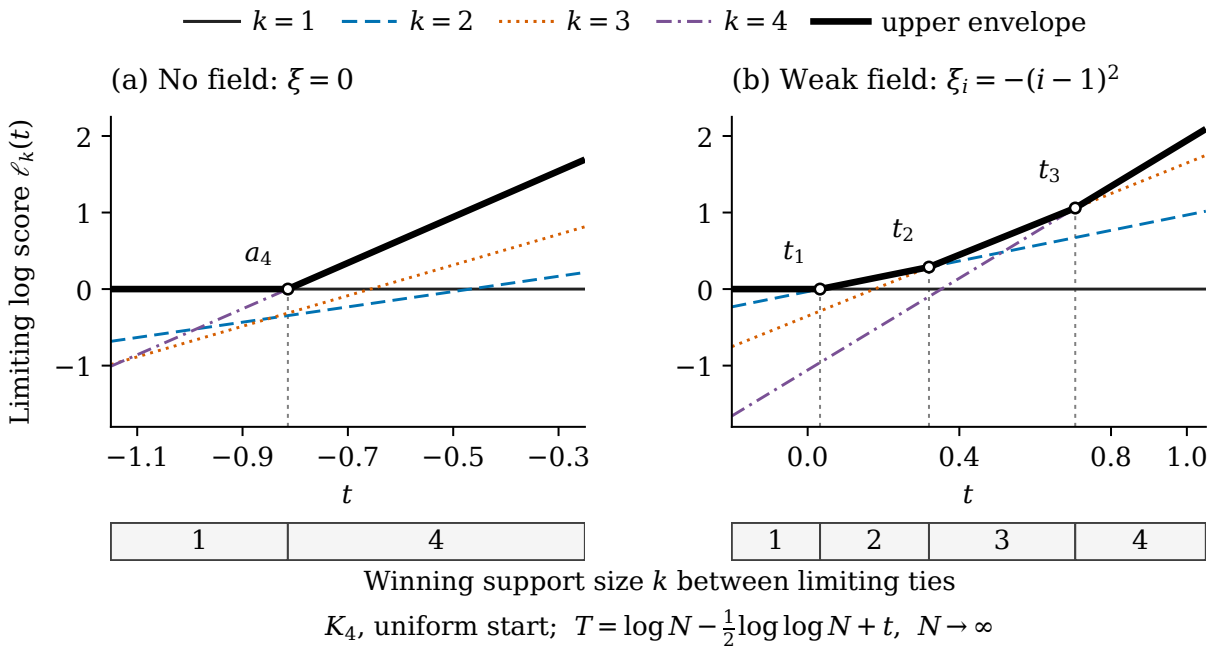


Figure 1: Support selection on the complete graph with four states, unit rates and uniform start, in the window $T = \log N - \frac{1}{2}\log\log N + t$. Thin lines are limiting log face maxima relative to an untilted vertex, and the bold curve is their upper envelope. (a) Without a field the winner jumps from one state to all four. (b) Weighting a composition by $e^{\langle \xi, n\rangle/N}$, with $\xi_i = -(i-1)^2$, lets every size win in turn. The scores and marked crossings are derived in Section 7. Field weights are unnormalized, and marked ties concern the limit, not finite-N equalities

1.3 What the occupation law determines

Section 9 turns the description around and asks which coarse functionals of the law of the counts determine the generator. The finest is the law itself. Its logarithmic scale on single compositions is, by Theorem 5.6, the Donsker and Varadhan functional J_V. Write $Q^* = D_\pi^{-1} Q^\top D_\pi$ for the time reversal of an irreducible generator with stationary law π, and call a partition $[d] = X \cup Y$ with $|X|, |Y| \geq 2$ a cut of Q if both blocks $Q[X, Y]$ and $Q[Y, X]$ have rank at most 1.

Theorem D (what the Donsker and Varadhan functional determines)**.** *Let Q and Q' be irreducible generators on $[d]$.*

(i) $J_V = J'_V$ on the simplex if and only if $-Q$ and $-Q'$ have the same principal minors.

(ii) The generators with the principal minors of $-Q$ form a finite set. It contains Q and Q^*. It is $\{Q, Q^*\}$ when Q has no cut, in particular when $d \leq 3$, and it is $\{Q\}$ when Q is reversible.

Part (i) is the equivalence of (i) and (iii) in Corollary 9.3, and part (ii) is Theorem 9.13(b)–(e). The proof of (i) needs three inputs. They are the variational formula $\lambda_V(v) = \min_\alpha(J_V(\alpha) - v \cdot \alpha)$ for $\lambda_V(v) = -\varrho(Q + \operatorname{diag} v)$ (Proposition 3.4), the irreducibility of the multiaffine polynomial $\det(-Q_F - \operatorname{diag} w)$ for every irreducible face (Lemma 9.9), and the Perron-branch lemma, by which the Perron root on an open set of fields determines that polynomial (Lemma 9.10). Part (ii) rests on the matrix theory of Loewy, Hartfiel and Loewy, Al Ahmadieh and Chatterjee, Ghosh,

Gurjar and Raj, who describe the matrices with given principal minors up to diagonal similarity, through transposition and a cut-transpose move across each cut. Theorem D answers the question left open by Example 5.4 of Kim and Landim [24], a three-cycle and its reversal with the same J_V. By their Corollary 5.8 the joint functional of the empirical measure and the empirical flow determines the generator, so adding the flow removes the whole fiber.

The modal supports give a strengthening. Weight each composition by $e^{Tv\cdot n/N}$ for a field $v \le 0$, which is the chance that the chain survives when it is killed at known rates, and observe only which face carries the mode of the surviving composition, as a function of τ and v. Throughout Section 9 the field and the time scale are calibrated, that is, known in the units of Q. The data are the boundaries between the regions of (τ, v) where one face wins strictly. Corollary 9.3 shows that these boundaries determine the principal minors too, so a single label at each point loses nothing that J_V keeps. The survival probability itself is a finer observation. It equals $e^{-T\lambda_V(v)}$ up to a bounded factor, so it gives λ_V on all fields and, through the Perron-branch lemma, every principal minor at once, with no descent through the faces. The coarser label does not require measuring a probability, only the support of the most likely composition, and it is the datum that the forward theory describes. Reading the minors from it needs a descent through the faces, which exposes the Perron root of every face on an open set of fields (Lemma 9.12). The fields used there depend on the generators being compared, so this part is a statement of identifiability, with no design fixed in advance.

For a reversible chain on a known connected graph with m edges a design is fixed in advance. Fields with constants built from a norm bound and an edge bound alone expose $d-1$ singleton switches and m pair-entry times, and these $m+d-1$ numbers determine Q by a Perron reconstruction (Theorem 9.5). No family of fewer than $m+d-1$ smooth observations can do this (Theorem 9.6), and the reconstruction is Lipschitz in the observed coordinates (Theorem 9.8). At a finite horizon the pair-entry times carry a correction of relative size $\log\log N/(2\log N)$ with an explicit next term, uniformly over a compact class of chains and starts (Theorem 9.16), and the start bound cannot be dropped (Remark 9.17).

What is borrowed and what is new. The fiber of the principal-minor map of an irreducible matrix is known up to diagonal similarity and transposition. The paper adds three things. The first, and the shortest, is the translation to generators. A Perron and Frobenius normalization picks exactly one generator in each diagonal class, so the fiber is finite, every member is irreducible with the spectrum of Q, and a cut-transpose acts as Q on one side of the cut and as Q^* on the other (Theorem 9.13). The second is the data. The minors are not observed, and the exposure lemma with the Perron-branch lemma reads all of them off the modal-support map. The third is the amplitude obstruction. From the stationary start the law of the count vector is the same under Q and Q^* for every N and T with $Tq_{\max} \le N$, so from that start no count statistic separates them. For every start outside a finite union of proper subspaces the second-order amplitudes of the local law separate every member of the fiber from Q (Theorem 9.13(f)).

1.4 Low temperature

When the rates are $q_{ij} = e^{-V_{ij}/\eta}$ and the time scale is $\tau = e^{h/\eta}$, the competition between dimension and exit rate takes place on the Freidlin and Wentzell cycles of the chain. The depth of a face is the largest depth of a cycle it contains (Theorem 10.1). Away from the finite set of cycle depths the winner is a cycle of least size among those deeper than h (Theorem 10.2). At a critical depth a finite problem with explicit prefactors decides, and its winners need not be cycles. A four-state reversible example has a three-state noncycle winner on an open window of the rescaled time

$x = Te^{-\gamma/\eta}/\log N$ at a critical height γ (Theorem 10.4 and Example 10.5). The controls read the cycle tree. A field of Arrhenius size on a face rewards it. The largest field exponent at which a cycle still wins is its depth, and the depth of every face is the largest such threshold over its subfaces (Theorem 10.11). The depths determine the cycles, since a set is a cycle exactly when every proper subset is strictly shallower, for every Arrhenius family with all rates positive and with no nondegeneracy (Theorem 10.12), and the window endpoints determine the prefactors of the faces on the envelope (Theorem 10.13).

1.5 Related work

The exact law of the count vector goes back to Whittle [33], Dawson and Good [14], Goodman [19] and Good [48]. Good's formula (3) is the run expansion of Lemma 2.3, and his formula (10) is the continuum density of Lemma 2.4, up to the killing and the exponent $n_i - 1$ that our setting needs. He suggested the saddle-point analysis of the density and did not carry it out, and Theorem 5.4 is that analysis, uniform in T and with its constant. Polettini [30] and Carugno, Vivo and Coghi [8] derive the exact law from the BEST theorem and the matrix-tree theorem. Stadje [31] gives the generating functions of the joint numbers of visits and limit laws when one state becomes nearly absorbing. In continuous time, Brydges, van der Hofstad and König [7] give the joint density of the local times on the event that the range is a given set, and Huang, Kious, Sidoravicius and Tarrès [22] write it with Bessel functions and spanning trees for symmetric rates. The Bessel kernel of the boundary layer is Pedler's occupation-time density for two-state chains [47], continued by Kovchegov, Meredith and Nir [46], and Section 6 is its facet version. Hervé and Ledoux [21] prove a quantitative local limit theorem for densities of finite-state Markov additive processes, and Kontoyiannis and Meyn [25, Theorem 6.3] give exact large deviation asymptotics for a bounded functional of a fixed chain. Theorem 5.4 is a local form for the whole vector of counts, for a chain killed on leaving F whose switch probabilities shrink with N.

The rate inside a face is the Donsker and Varadhan functional of the killed chain [17]. Dupuis and Liu [18, Section 3.1] give it for reversible jump processes, Chen and Deng [10] use it for the principal Dirichlet eigenvalue, and Guillin, Nectoux and Wu [20] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. The quasi-ergodic law goes back to Darroch and Seneta [12, 13], with the book of Collet, Martínez and San Martín [11] as the standard reference. Dietz and Sethuraman [16, 15] and Bouguet and Cloez [6] study chains whose switching decays along the trajectory. Here the rate is fixed during each path and changes with N. Bertini, Gabrielli and Landim [4] and Landim [26] expand the level-two rate functions of finite chains in a parameter, and Kim and Landim [24] extend this to the joint rate of the empirical measure and the flow [2, 3]. Their Section 5 shows that the Donsker and Varadhan functional determines the exit rates and the products $q_{ij}q_{ji}$, which with the stationary law determine a reversible generator, and gives a three-cycle and its reversal as an example of non-identifiability. The matrix theory behind Theorem D is Hartfiel and Loewy [41] and Loewy [42] for the case without a cut, Engel and Schneider [39] for diagonal similarity to a symmetric matrix, and Al Ahmadieh [34] and Chatterjee, Ghosh, Gurjar and Raj [36] for the general fiber. Delmas, Dronnier and Zitt [38] prove a multiplicative analogue of the Perron-branch lemma. The reversible coordinates and the dimension count of Theorem 9.6 are those of Wolfer and Watanabe [44].

The cycle framework of Section 10 is that of Freidlin and Wentzell [40], Catoni [35], Olivieri and Vares [43], Trouvé [51] and Cirillo, Nardi and Sohier [37]. The depth identity rests on Catoni's mean exit theorem. The logarithmic asymptotics of the principal eigenvalue of a killed generator with Arrhenius rates, and its graph and determinant expressions, go back to Wentzell [52] and to

Freidlin and Wentzell [40, Chapter 6], so the prefactor proposition is classical in substance. What is new there is the score of dimension against exit rate, the finite competition of envelopes at a critical depth with its noncycle window, and the attraction design.

The question grew out of Problem 131 of Kagey's Open Problem Collection [23], a two-direction crossing problem, whose two-state case is treated in [28]. The present paper incorporates the symmetric-model preprint [29].

1.6 Plan of the paper

Section 2 fixes the notation and gives the exact law of the composition as a sum over runs. Section 3 introduces the killed functional and the restriction identity. Section 4 proves first-order face selection, Section 5 the local law with a field, the face maxima, the crossing equation and the tie windows, and Section 6 the boundary layer. Section 7 treats the two-state walk and the complete graph, and Section 8 the compositions near a vertex. Section 9 proves the reconstruction theorems and Section 10 the low-temperature results. Section 11 describes the computer checks and the formal proofs. The appendices hold the torus and Laplace proofs of the local law, the proofs for the complete graph and the proofs near a vertex. We quote a few results of [28], namely the run count, the growth of the polynomials $S_{a,b}$ toward the middle, the uniform Bessel bound recalled in (1) and the expansion of z_N.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). (The scalar $q = 1 - p$ of the two-state walk appears only where we recall [28], and the scalar q of Proposition 7.1 is a common exit rate.) The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. We write $V = [d]$ for the face of all states. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N : X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i,j \in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i \in F}$. Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [1].

Definition 2.1 (faces)**.** Let F be a face. We call F *irreducible* if G_F is strongly connected. In particular every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [12, 13]. The *face maximum* is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ and r_F is positive. So the chance that the continuous-time chain started at a state of F stays in F up to time T, which is a row sum of e^{TQ_F}, lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For a face F satisfying (H), the quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F, and the best composition on F sits near $N\alpha_F^*$ (Theorem 5.7).

Section 3 adds the rate J_F, and Definition 5.1 the tilted chain and the constant K_F.

The two-state walk. The persistent random walk on $\mathbb{Z}$ makes its first step left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p and switches with probability $q = 1 - p$. Its directions form the chain of Example 2.5(i). As on a Galton board, [28] calls a composition a *bin*. For two states we identify a composition with its number of right steps and b left steps, where $s(w)$ is the number of switches of w. Each such word has probability $\frac{1}{2} p^{N-1} (q/p)^{s(w)}$ for the walk with $N = a + b$ steps. So $S_{a,b}(q/p)$ is the probability of composition a divided by the probability $\frac{1}{2} p^{N-1}$ of an endpoint. For $N \ge 2$ let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$, which belongs to a middle composition. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N, and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. By [28], z_N satisfies the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + O(1/L^2)$. Its uniform Bessel bound [28, Theorem 3.3(i)] says, with $\psi(z) = z(I_0(z) + I_1(z))$ and the modified Bessel functions I_0 and I_1, that for $N \ge 2$ and $z > 0$

$$\Big(1 - \frac{z(z+2)}{2N}\Big)\psi(z) \le \frac{N}{2} S_N\Big(\frac{z}{N}\Big) \le \psi(z). \tag{1}$$

Conventions. Unless we say otherwise, Q, μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds "uniformly for $1 \le T \le N^{1/3}$" holds for all $N \ge N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces)**.** *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. We translate paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp}\mu$ and visits every state of F. Conversely, every such walk $v_1, \dots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \geq J$. (The path follows the walk and then stays in its last state, and a stay has positive probability.) So F is admissible exactly when G_F has a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. (If it did, y could be reached from C and C from y, so y would belong to C.) Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. So no edge joins two components, and the condition in (a) forces $r = 1$. □

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion)**.** *Let $hq_{\max} \leq 1$ and let n have support F. Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

This is Good's coefficient identity [48, (3)], with the exponents $n_i - 1$, for a chain of N letters with any initial law, here for the chain killed on leaving F. Grouping by runs is our way of writing it, and the proof is short.

Proof. The states $c_1 \cdots c_M$ of the maximal runs of a path with composition n form a word as in $W(m)$. The path starts at c_1, of probability μ_{c_1}, it has $M - 1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. The number of paths with the word c and composition n is $\prod_i \binom{n_i-1}{m_i-1}$, the number of ways to write each n_i as an ordered sum of m_i positive run lengths. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

Lemma 2.3 is checked in Lean (Section 11). For $F = \{i\}$ the only path is i^N, so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{2}$$

For $t \in (0,\infty)^F$ let $H(t) = \sum_m W(m) \prod_{i\in F} t_i^{m_i-1}/(m_i-1)!$, which is finite since $W(m) \le k^M q_{\max}^{M-1}$. In the continuum limit the binomial coefficients of Lemma 2.3 become the powers $t_i^{m_i-1}/(m_i-1)!$, and Lemma 5.5 below shows that the discrete law is close to the following density.

Lemma 2.4 (the continuum density)**.** *Let $(Y_s)_{s\ge 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\, ds$, and let E_F be the event that Y stays in F during $[0,T]$ and visits every state of F. Then $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ has the density $e^{-\langle q,t\rangle} H(t)$ in $(t_2,\dots,t_k)$ on $\{t \in (0,\infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q,\alpha\rangle} H(T\alpha)$$

in $(\alpha_2,\dots,\alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

The density is Good's formula [48, (10)]. What we add is the killing, that is, the subprobability with the full exit rates q_i on the diagonal, and the exponents $m_i - 1$, which the proof checks.

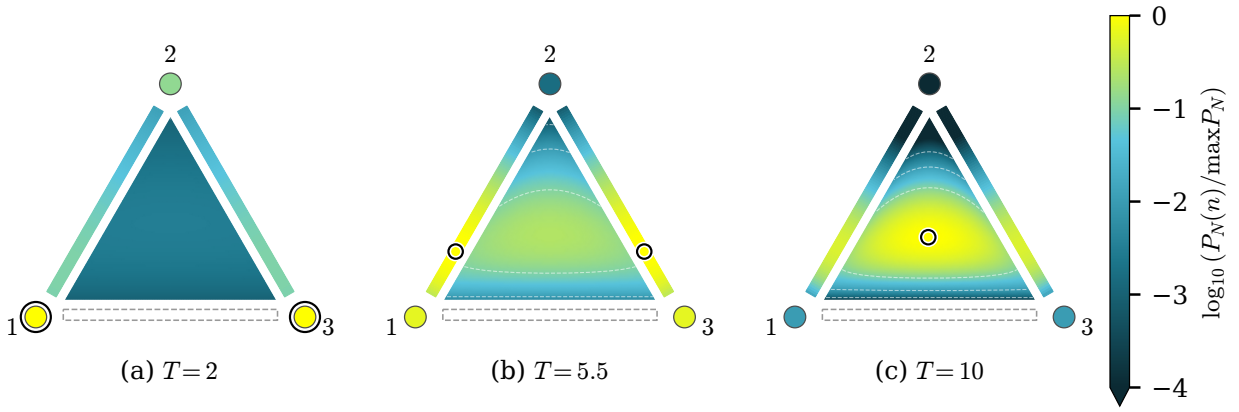
Proof. Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1,\dots,c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$. Fix c, and let s_j be the time that Y spends in its j-th run, so that $s_1 + \cdots + s_M = T$. For $j < M$ the run has the density $q_{c_j} e^{-q_{c_j} s_j}$ and ends with a jump to c_{j+1} of probability $q_{c_j c_{j+1}}/q_{c_j}$, and the last run survives with probability $e^{-q_{c_M} s_M}$. So on $\{s_j > 0\ \forall j\}$ the event E_c has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,\mathcal{O}\rangle}$ in $(s_1,\dots,s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$. For each $i \ne c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. With $\mathcal{O} = t$ fixed, the $m_i - 1$ other holding times in state i are positive with sum less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. Integrating them out gives the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_i t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \ne c_M$, and any $k-1$ of the coordinates give the same Lebesgue measure on $\{\sum_i t_i = T\}$. Summing over c and m gives $e^{-\langle q,t\rangle} H(t)$, and $t = T\alpha$ has Jacobian T^{k-1}. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (two models)**.** (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is the persistent walk, with $p = 1 - h$. So $T = N(1-p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints, and Lemma 2.3 is the run count for two states. (ii) For $Q = J - dI$, with J the all-ones matrix, the chain switches to each other state with probability h and keeps its state with probability $p = 1 - (d-1)h$. With uniform μ it is the model of [29], whose switch probability r is our h. A word $x_1 \cdots x_N$ then has probability $\frac{1}{d} h^{N-1} (p/h)^{\#\{s<N:\, x_s = x_{s+1}\}}$, so the path law is the one-dimensional d-state Potts chain with free boundaries. Here $q_i = d - 1$ for every i, so Lemma 2.3 reads

$$P_N(n) = p^{N-1} \sum_{a\in F} \mu_a S_n^{(a)}(u), \qquad u = \frac{h}{p}, \tag{3}$$

for $p > 0$ and $\operatorname{supp} n = F$, where $S_n^{(a)}(u)$ is the sum of u^j over the words with the counts n that start with a and have j switches. Indeed such a word has probability $\mu_a p^{N-1-j} h^j = \mu_a p^{N-1} u^j$. In both models Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

Figure 2 shows exact support competition for the three-state path. There λ_F is 1 for an end, $(3-\sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1+\sqrt{5})/2$ and at $\tau = (3+\sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$ for $N = 90$. At $N = 90$ it moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 5.



(a) $T=2$ (b) $T=5.5$ (c) $T=10$

Figure 2: The exact law of the composition for the path $1-2-3$ (rate 1 on each edge, uniform start, $N=90$) on the simplex of visit fractions, with the faces drawn apart. The triangle holds the compositions with three states, each band those with two, and each disk a vertex. The color is the probability relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. The mode is (a) at an end of the path, (b) on a pair of neighbors and (c) inside. The dashed band $\{1,3\}$ has probability zero

3 The killed functional

Every rate in this paper is the restriction of one functional, the Donsker and Varadhan occupation functional of the chain. This section defines it, proves the restriction identity and the variational formula with a field, and records the monotonicity of exit rates in the face.

3.1 The restriction identity

For a probability vector α on $[d]$ let

$$J_V(\alpha)=\sup_{u\in(0,\infty)^d}\sum_{i\in[d]}\alpha_i\Big(-\frac{(Qu)_i}{u_i}\Big)=\sup_{u\in(0,\infty)^d}\sum_{i\in[d]}\alpha_i\Big(q_i-\sum_{j\neq i}q_{ij}\frac{u_j}{u_i}\Big),\tag{4}$$

the Donsker and Varadhan rate of the empirical measure of the continuous-time chain with generator Q [17]. For a face F and $\alpha\in\Delta_F$ let

$$J_F(\alpha)=\sup_{u\in(0,\infty)^F}\sum_{i\in F}\alpha_i\Big(-\frac{(Q_Fu)_i}{u_i}\Big)=\sup_{u\in(0,\infty)^F}\sum_{i\in F}\alpha_i\Big(q_i-\sum_{j\in F\setminus\{i\}}q_{ij}\frac{u_j}{u_i}\Big),$$

the same functional for the chain killed on leaving F. Let $q_i^{\mathrm{out}}(F)=\sum_{j\notin F}q_{ij}$. Taking $u=\mathbf{1}$ gives $J_F(\alpha)\geq\sum_i\alpha_iq_i^{\mathrm{out}}(F)$, and dropping the terms with q_{ij} gives $J_F(\alpha)\leq\sum_i\alpha_iq_i$.

Proposition 3.1 (the restriction identity)**.** *For every face F and every $\alpha\in\Delta_F$, $J_F(\alpha)=J_V(\alpha)$. Hence $J_F=J_{F'}$ on $\Delta_{F'}$ for every nonempty $F'\subseteq F$.*

Proof. Fix $\alpha\in\Delta_F$ and $u\in(0,\infty)^d$. The terms of (4) with $i\notin F$ vanish, since $\alpha_i=0$ there. For $i\in F$ the terms with $j\notin F$ are nonnegative and are subtracted, so

$$\sum_{i\in F}\alpha_i\Big(q_i-\sum_{j\neq i}q_{ij}\frac{u_j}{u_i}\Big)\leq\sum_{i\in F}\alpha_i\Big(q_i-\sum_{j\in F\setminus\{i\}}q_{ij}\frac{u_j}{u_i}\Big)\leq J_F(\alpha).$$

So $J_V(\alpha) \le J_F(\alpha)$. Conversely, let $w \in (0,\infty)^F$ and let u equal w on F and ε off F. The terms with $j \notin F$ are then $-q_{ij}\varepsilon/w_i$, which tend to zero as $\varepsilon \downarrow 0$, and the supremum over w gives $J_V(\alpha) \ge J_F(\alpha)$. The second claim is the first for F and for F'. □

So the cost of a point of Δ_F does not depend on the face in which we read it. The competition between faces is between the lattice entropy $(|F|-1)L$ of a face and the cost TJ_V of its best point, as Theorem 5.6 states in one display. The identity itself is elementary. What it gives is the reading of every face rate as one functional, which Section 9 uses to compare J_V with the modal supports.

3.2 The variational formula

For a face F and a field $v \in \mathbb{R}^F$ let

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v\big), \tag{5}$$

so that $\lambda_F(0) = \lambda_F$. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, so that $\alpha_F^*(0) = \alpha_F^*$. Since $\sum_i \alpha_i v_i = v \cdot \alpha$,

$$J_F(\alpha) - v \cdot \alpha = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(-\frac{((Q_F + \operatorname{diag} v)u)_i}{u_i} \Big). \tag{6}$$

So a field replaces the diagonal q_i by $q_i - v_i$ and changes nothing else.

Lemma 3.2 (monotonicity in the face)**.** *If $F \subsetneq G$ are faces and Q_G is irreducible, then $\lambda_G(v_G) < \lambda_F(v_F)$ for every field $v \in \mathbb{R}^G$. In particular $\lambda_V(v) < \lambda_F(v_F)$ for every proper face F when Q is irreducible, and for every fixed $v \in \mathbb{R}^d$ there is $\gamma(v) > 0$ with $\lambda_F(v_F) \ge \lambda_V(v) + \gamma(v)$ for all proper F. So the score $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$ is least at V for $\tau > (d-1)/\gamma(v)$.*

Proof. Choose c with $B = Q_G + \operatorname{diag} v_G + cI \ge 0$. Its principal submatrix on F is $Q_F + \operatorname{diag} v_F + cI$, since the diagonal of a killed generator keeps the full exit rates. Let $\widetilde{B}$ be this submatrix padded with zeros to $G \times G$. Then $0 \le \widetilde{B} \le B$, and $\widetilde{B} \ne B$ because the irreducible matrix B has a positive entry in a row of F and a column off F. So $\varrho(\widetilde{B}) < \varrho(B)$ [1, Ch. 2], and $\varrho(\widetilde{B})$ is the Perron root of the submatrix on F, since the padding adds only the eigenvalue 0. This holds whether or not F is irreducible, and subtracting c gives the first claim. For fixed v there are finitely many proper faces, so $\gamma(v)$, the least of the numbers $\lambda_F(v_F) - \lambda_V(v)$, is positive. Finally $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(V) = |F| - d + \tau(\lambda_F(v_F) - \lambda_V(v)) \ge -(d-1) + \tau\gamma(v)$. □

Proposition 3.3 (the rate on a face)**.** *Let F be irreducible with $k = |F| \ge 2$, and let $v \in \mathbb{R}^F$.*

(a) On Δ_F, $J_F(\alpha) - v \cdot \alpha \ge \lambda_F(v)$, with equality exactly at $\alpha = \alpha_F^*(v)$. In particular $J_F \ge \lambda_F$, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate J_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$J_F(\alpha) = \sum_{\{i,j\} \subseteq F,\ i \ne j} \big(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}}\big)^2 + \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F).$$

(d) *For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F(\alpha) - v \cdot \alpha = J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'}) > \lambda_F(v)$.*

So the most likely way to stay on F is to spend the fractions α_F^* of the time in its states. Fractions that miss some states of F have the rate of the smaller face that they use, which is larger. For symmetric Q_F, $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2. On a face that is not reversible the formula in (c) is only an upper bound, by its proof. The lower bound in (a) and the value at α_F^* for $v = 0$, and the interior form of (c) for symmetric Q_F, are checked in Lean.

Proof. Put $B = Q_F + \operatorname{diag} v$, an irreducible Metzler matrix with the switching graph G_F and $\lambda_F(v) = -\varrho(B)$.

Setup. With $u = e^\psi$, (6) reads $J_F(\alpha) - v \cdot \alpha = \sup_\psi \Psi_\alpha(\psi)$ with

$$\Psi_\alpha(\psi) = \sum_i \alpha_i (q_i - v_i) - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i},$$

where i, j run over F. It is concave in ψ, affine in α, and unchanged when a constant is added to ψ. Let G^ψ be the conservative generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. It has the graph G_F, so it has exactly one stationary law, and it does not involve v. Since $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$ and a differentiable concave function attains its maximum exactly at its critical points, ψ attains the supremum if and only if α is the stationary law of G^ψ. We call this the criterion.

A lower bound on every face. Let F' be any face, irreducible or not, and $B' = Q_{F'} + \operatorname{diag} v_{F'}$. We show $J_{F'}(\alpha) - v \cdot \alpha \geq -\varrho(B')$ on $\Delta_{F'}$. Let $s > \varrho(B')$ and write $B' = B_0 - cI$ with $B_0 \geq 0$. Then $s + c > \varrho(B_0)$, so $u = (sI - B')^{-1}\mathbf{1} = \sum_{\ell \geq 0} (s+c)^{-\ell-1} B_0^\ell \mathbf{1}$ converges and is positive. Since $(sI - B')u = \mathbf{1}$, we get $-(B'u)_i/u_i = -s + 1/u_i > -s$ at every state, and letting $s \downarrow \varrho(B')$ gives the claim.

The value at $\alpha_F^*(v)$. Take $\psi = \log r$ and $R = \operatorname{diag} r$. Then $G^\psi = R^{-1}(B - \varrho(B)I)R$, since the off-diagonal entries agree and $Br = \varrho(B)r$. So $(l \circ r)^\top G^\psi = l^\top (B - \varrho(B)I)R = 0$, and by the criterion ψ attains the supremum at $\alpha_F^*(v)$. Every ratio $-(Br)_i/r_i$ equals $\lambda_F(v)$, so the value is $\lambda_F(v)$.

(b) Convexity holds because J_F is a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$ and take $v = 0$.

The supremum is attained. Normalize $\min_i \psi_i = 0$ and put $\bar\psi = \max_i \psi_i$. Let a be the least positive rate q_{ij} in F. A shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar\psi$ has at most $k - 1$ edges, so one of its edges $i \to j$ has $\psi_j - \psi_i \geq \bar\psi/(k-1)$. Keeping only this edge gives $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{\bar\psi/(k-1)}$, which tends to $-\infty$ as $\bar\psi \to \infty$. So the continuous function Ψ_α attains its supremum on a bounded set. Write $r_\alpha = e^\psi$ for a maximizer. The last claim of (b) is the criterion.

The maximizer is unique up to a factor. If ψ and $\psi + \delta$ both attain the supremum, Ψ_α is constant on the segment between them. There it is a constant minus a sum of the convex functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, so each of them is affine in s, and $\delta_j = \delta_i$ on every edge of G_F. As G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, put $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $J_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α, $J_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \leq (1-s)J_F(\alpha^0) + sJ_F(\alpha^1)$. Equality would make α^0 and α^1 both stationary for G^ψ by the criterion, which is impossible.

(d) The equality is Proposition 3.1. The lower bound on F' gives $J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'})$, and Lemma 3.2 with $G = F$ gives $\lambda_{F'}(v_{F'}) > \lambda_F(v)$.

(a) The lower bound and the value at $\alpha_F^*(v)$ are proved above. Let $\alpha \neq \alpha_F^*(v)$. If α lies on the boundary of Δ_F, then (d) with $F' = \operatorname{supp}\alpha$ gives $J_F(\alpha) - v \cdot \alpha > \lambda_F(v)$. If $\alpha \in \Delta_F^\circ$ and equality held, strict convexity would give a value below $\lambda_F(v)$ at $\frac{1}{2}(\alpha + \alpha_F^*(v))$.

(c) Since $q_i = \sum_{j \in F\setminus\{i\}} q_{ij} + q_i^{\text{out}}(F)$, the function inside the supremum is $\sum_i \alpha_i q_i^{\text{out}}(F)$ plus, for each pair $\{i,j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$, which gives the upper bound. For $\alpha \in \Delta_F^\circ$ the vector $u_i = \sqrt{\alpha_i/\nu_i}$ gives $\alpha_i q_{ij} u_j/u_i = \alpha_j q_{ji} u_i/u_j$ by $\nu_i q_{ij} = \nu_j q_{ji}$, so there is equality for every pair at once. If some α_i vanish, take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1 - \varepsilon/u_j) \to \alpha_j q_{ji}$, which is its bound, and a pair with $\alpha_i = \alpha_j = 0$ has the term 0. □

Proposition 3.4 (the variational formula)**.** *Let F be irreducible. For every $v \in \mathbb{R}^F$,*

$$\lambda_F(v) = \min_{\alpha \in \Delta_F} \big(J_F(\alpha) - v \cdot \alpha\big),$$

and the minimum is attained only at $\alpha_F^*(v)$. Equivalently, $-\lambda_F(v)$ is the Legendre transform of J_F at v, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$ is concave and real-analytic, and $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$. If Q_F is reversible with respect to ν, let $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$. Then with $\psi_i = \sqrt{\alpha_i}$,

$$J_F(\alpha) - v \cdot \alpha = \sum_{i \in F}(q_i - v_i)\alpha_i - \sum_{i \neq j \in F} \sqrt{q_{ij}q_{ji}}\,\psi_i\psi_j$$

is the Rayleigh quotient of the symmetric matrix $S_F - \operatorname{diag} v$ at the unit vector ψ, and $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v)$.

The formula is the finite form of the Donsker and Varadhan formula for the principal eigenvalue [17], close to the characterization of Collatz and Wielandt [1, Ch. 2].

Proof. The formula and the unique minimizer are Proposition 3.3(a). As a minimum of affine functions of v, $\lambda_F(v)$ is concave. It is real-analytic because the Perron root of an irreducible Metzler matrix is a simple eigenvalue, and the shift is clear.

Legendre duality. The formula says $-\lambda_F(v) = \max_\alpha(v \cdot \alpha - J_F(\alpha))$. Conversely $J_F(\alpha) \geq v \cdot \alpha + \lambda_F(v)$ for all α and v. Let $\alpha \in \Delta_F^\circ$, take r_α from Proposition 3.3(b), and put $v = q - A_F r_\alpha / r_\alpha$ (entrywise). Then $Q_F + \operatorname{diag} v$ has the off-diagonal entries of Q_F and the diagonal $-A_F r_\alpha / r_\alpha$, so $r_\alpha > 0$ is a null vector of it. Only the Perron root has a positive eigenvector, so $\lambda_F(v) = 0$. Also $J_F(\alpha) = \langle q - A_F r_\alpha / r_\alpha, \alpha\rangle = v \cdot \alpha$, so equality holds at this v.

The reversible form. Reversibility gives $\sqrt{\nu_i}\, q_{ij}/\sqrt{\nu_j} = \sqrt{q_{ij}q_{ji}}$, so S_F is symmetric with diagonal q_i and off-diagonal entries $-\sqrt{q_{ij}q_{ji}}$. Expanding the squares in Proposition 3.3(c) gives $J_F(\alpha) = \langle \psi, S_F\psi\rangle$, and subtracting $v \cdot \alpha = \langle \psi, \operatorname{diag}(v)\psi\rangle$ gives the display. Every unit vector $\psi \geq 0$ arises from $\alpha = \psi \circ \psi$, and the off-diagonal entries of $S_F - \operatorname{diag} v$ are nonpositive, so passing to the entrywise modulus does not increase the quadratic form. Hence the minimum over Δ_F is $\lambda_{\min}(S_F - \operatorname{diag} v)$. □

4 First-order face selection

The mode sits on a face that minimizes Φ_τ (Theorem 4.2), and the winners can be read off a convex hull (Theorem 4.4).

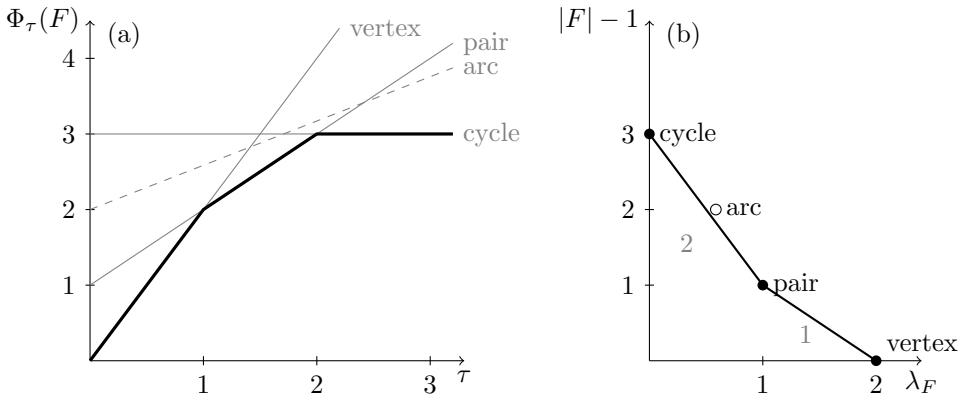


Figure 3: The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 4.4). The arc (open) lies above the boundary

4.1 An example with four states on a cycle

Take $d = 4$ states on a cycle, with rate 1 between neighbors and the uniform start. Up to symmetry the admissible faces are a vertex, an adjacent pair, an arc of three states and the whole cycle, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. (For the arc, $-Q_F = 2I - A$ with A the adjacency matrix of the path $1 - 2 - 3$, whose largest eigenvalue is $\sqrt{2}$.) The chain stays in a face F up to time T with probability about $e^{-T\lambda_F}$. Given that it stays, its visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $|F| - 1$ free directions, so this mass is shared by about $(N/\sqrt{T})^{|F|-1}$ compositions. Hence $M_F \approx N^{-(|F|-1)} T^{(|F|-1)/2} e^{-T\lambda_F}$, and with $T = \tau \log N$ this is $N^{-\Phi_\tau(F)}$ up to powers of $\log N$. Figure 3(a) plots the four exponents. The vertices win for $\tau < 1$, the pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs never win, since they beat the pairs only for $\tau > 1 + \sqrt{2}$, where the cycle beats them. The whole cycle carries almost all the mass for every $\tau > 0$, and it carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions. Theorems 4.2 and 5.7 make this rule exact, and for a pair Corollary 5.3 gives the constant $1/\sqrt{2\pi}$ in front.

4.2 Face selection

We also need an upper bound on a whole face, including near its boundary, where the local law does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha\rangle$, and J_F is the supremum of these linear rates. Indeed, for $u \in (0,\infty)^F$ the vector $\theta = A_F u/u$ (entrywise) lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a positive eigenvector. Then $\langle q - \theta, \alpha\rangle$ is the function of u inside the supremum that defines $J_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector of $A_F - \operatorname{diag}\theta$. For a field $v \in \mathbb{R}^F$ the vector $\theta = q - v - \lambda_F(v)\mathbf{1}$ lies in V_F, since $A_F - \operatorname{diag}\theta = Q_F + \operatorname{diag} v + \lambda_F(v) I$.

Lemma 4.1 (a uniform upper bound)**.** *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ and bounded for θ in compact subsets of V_F, such that for*

every $n \in \mathbb{Z}^F_{\geq 0}$ with $\operatorname{supp} n = F$ and every h with $hq_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp\bigl(-h\langle n, q-\theta\rangle + C_\theta h^2 |n|\bigr)$.

The proof is in Appendix A. The lemma holds for a path of any length $|n|$, so it also applies to the blocks of a path. With $\theta = q - v - \lambda_F(v)\mathbf{1}$ and $|n| = N$ it gives

$$P_N(n) e^{Tv\cdot n/N} \leq C_\theta h^{k-1} e^{-T\lambda_F(v) + C_\theta T^2/N} \tag{7}$$

for every n with support F. For $v = 0$ this is the upper half of the rule $M_F \approx N^{-(k-1)} e^{-T\lambda_F}$ of Section 4.1, up to a power of T. The next theorem uses the local law and the face maxima of Section 5, which are proved there without it.

Theorem 4.2 (first-order face selection)**.** *Fix Q and μ.*

(i) Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,

$$\log P_N(n) = -(k-1)L - T\, J_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$

$$\log M_F \geq -(k_F - 1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \leq -(k_F - 1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp} \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp} \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 4.4(c)).

In words, a composition whose fractions α stay in a compact part of the open face has probability $N^{-(k-1)} e^{-TJ_F(\alpha)}$ up to a bounded factor and a power of T. The best composition on an admissible face has probability $N^{-(k_F-1)} e^{-T\lambda_F}$ times a power of T with exponent between 0 and $k_F - 1$. With $T = \tau L$ the exponent of N is $-\Phi_\tau(F)$, and the mode lies on a face that minimizes it, whatever the start.

Proof. (i) For $k = 1$ this is (2), since $J_{\{i\}} = q_i$. For $k \geq 2$, Lemma 5.5 gives $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$. For $T \geq T_1$ Theorem 5.4 gives $\log f_T(\alpha) = \frac{k-1}{2}\log T + \log C_F(\alpha) - TJ_F(\alpha) + O_{\mathcal{K}}(1)$ on $\mathcal{K}$, and for $1 \leq T \leq T_1$ this holds because f_T and C_F are positive and continuous on $[1, T_1] \times \mathcal{K}$. As $\log C_F$ is bounded on $\mathcal{K}$, adding $\log N^{-(k-1)} = -(k-1)L$ gives (i).

(ii) Let F be admissible. A path visits each component in one block of consecutive times (proof of Lemma 2.2), and an irreducible face is one component, with $k^* = k_F$. For the upper bound, Lemma 4.1 (or a direct count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1} e^{-h\lambda_C N_C})$, and each of the $r - 1$ switches between blocks gives a factor h. As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F - 1} e^{-T\lambda_F})$. For the lower bound the path spends about N/T visits in each

component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) For large N we have $T_0 \le T = \tau L \le N^{1/3}$, and both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ over finitely many faces, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \ge m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \le N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which satisfies (H) by Lemma 2.2(b). So Theorem 5.7 with $T = \tau L \le N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

For a reducible face we only have the bounds in (ii), and in Example 5.14 the power of T differs from $(k_F - 1)/2$.

Remark 4.3 (a field). Let $v \in \mathbb{R}^d$, put $M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\}$ and $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$. On an admissible face, $Q_F + \operatorname{diag} v_F$ is block triangular in the order of the components, so $\lambda_F(v_F) = \min_j \lambda_{C_j}(v_{C_j})$. The upper bound in (ii) holds for $M_F(T, v)$ with $\lambda_F(v_F)$ in place of λ_F. The proof in Appendix A.6 applies with (7) on each block and the exact weight $e^{hv_iN_j}$ on a block $\{i\}$. For F satisfying (H), Theorem 5.7 gives the matching lower bound. If μ has full support, a reducible admissible face F has a component C with $\lambda_C(v_C) = \lambda_F(v_F)$, which is admissible with fewer states, so $\Phi_{\tau,v}(C) \le \Phi_{\tau,v}(F) - 1$. So if μ has full support, then for fixed τ and v, (iii) and (iv) hold for the weighted law with $\Phi_{\tau,v}$ and $\alpha_F^*(v)$, every face that minimizes $\Phi_{\tau,v}$ is irreducible, and a strict minimizer carries every maximizer of the weighted law for large N.

4.3 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of S that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv} S$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv} S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv} S$. Figure 3(b) shows it for the 4-cycle.

Theorem 4.4 (geometry of the winners)**.** *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv} S$ *is a path whose vertices lie in S. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \le i \le m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \le i \le m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \ge \lambda_G$ and $|F| \le |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

In words, as τ grows the winners walk up the lower-left boundary of the hull, from small faces with large exit rates to large faces with small exit rates, and they change at the values τ_i, minus the slopes of its edges.

Proof. (a) We study the least value $g(\tau) = \min_{(x,y)\in S}(\tau x + y)$ for $\tau > 0$. Each point (x, y) of S gives the affine function $\tau \mapsto \tau x + y$, whose slope is x, and different points give different functions. So g is a minimum of finitely many affine functions. It is concave and piecewise affine. Let $\tau_1 < \cdots < \tau_m$ be the points of $(0, \infty)$ where its slope changes, and put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Between two kinks. On (τ_i, τ_{i+1}) the function g is affine. The affine function of a point of S is nowhere below g. So if it equals g at one τ inside this interval, it equals g on the whole interval. Hence exactly one point of S attains g there. We call it $v_i = (\lambda^{(i)}, y^{(i)})$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $P_F = v_i$. The slope of g on this interval is $\lambda^{(i)}$. The slope of a concave function decreases, and it changes at each τ_i, so $\lambda^{(0)} > \cdots > \lambda^{(m)}$. By continuity both v_{i-1} and v_i attain $g(\tau_i)$, namely

$$\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}.$$

This is the formula for τ_i in the statement, and since $\tau_i > 0$ it gives $y^{(i)} > y^{(i-1)}$.

At a kink. The points (x, y) of S that attain $g(\tau_i)$ are those on the line $\tau_i x + y = g(\tau_i)$, which passes through v_{i-1} and v_i. Let (x, y) be one of them. For τ slightly below τ_i we have $\tau x + y \geq g(\tau) = \tau \lambda^{(i-1)} + y^{(i-1)}$, with equality at $\tau = \tau_i$. Subtracting gives $(\tau - \tau_i)(x - \lambda^{(i-1)}) \geq 0$, so $x \leq \lambda^{(i-1)}$. In the same way, τ slightly above τ_i gives $x \geq \lambda^{(i)}$. So (x, y) lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of S on this edge lies on the line and so attains $g(\tau_i)$.

The boundary. On $\operatorname{conv} S$ the function $\tau x + y$ is least exactly on the convex hull of the points of S where it is least. By the two steps above this hull is the vertex v_i for $\tau_i < \tau < \tau_{i+1}$ and the edge $[v_{i-1}, v_i]$ for $\tau = \tau_i$. So the lower-left boundary is the path $v_0 \cdots v_m$, and its vertices lie in S.

(b) We add two inequalities. Since F wins at τ and G wins at τ', we have $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, so $\lambda_F \geq \lambda_G$. Then the first inequality gives $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If $\lambda_F = \lambda_G$, the two inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$. So if $\mathcal{F}_\tau$ and $\mathcal{F}_{\tau'}$ are disjoint, then $\lambda_F > \lambda_G$, and the first inequality gives $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

Lean checks part (b), the fact that a face with a proper subface of the same exit rate never wins, and part (a) when the least exit rate of each size decreases strictly with the size. In that case the vertices of the hull are sizes, and Lean checks which size wins between consecutive kinks and that consecutive vertices tie at τ_i. The general path structure in (a) and the ties along a closed edge are proved only here. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community. For symmetric Q, λ_F is the least value of $\sum_{i<j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Lemma 3.2. Since faces of one size share the term $|F| - 1$, only the faces with the least exit rate of each size can win. Figure 4(a) shows S for the 6-cycle.

5 The second-order theory

The first-order rule locates a crossing only up to a factor $1 + o(1)$ in T. To locate it up to $o(1)$ we need the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H), together with the

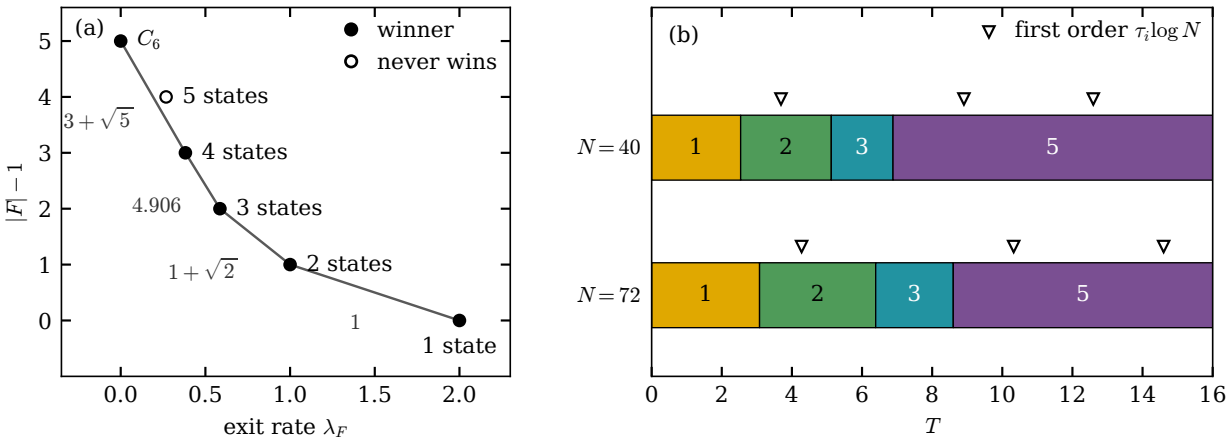


Figure 4: (a) The points $(\lambda_F, |F|-1)$ of the arcs of the 6-cycle C_6 and of C_6 itself, with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 4.4. The winners (filled) are the arcs of one to four states and C_6, and the arc of five states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T, at $N = 40$ and $N = 72$. The mode skips the arc of four states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1$, $1+\sqrt{2}$ and $2+\sqrt{2}$

power of T. Section 5.1 defines the constant K_F, Section 5.2 proves the local law and the face maxima with a field, and Section 5.3 compares two faces. Section 5.4 gives a tree formula for K_F, Section 5.5 shows what can change without (H), and Section 5.6 treats a weak field. In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i \in F}$.

5.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $J_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TJ_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 5.1 (tilt and constants)**.** Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $J_F(\alpha)$ (Proposition 3.3(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \operatorname{diag}\theta(\alpha))R$ with $R = \operatorname{diag} r_\alpha$. With $D = \operatorname{diag}\alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \qquad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha}}, \qquad K_F = C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}} = \mu_i$. Nothing in Definition 5.1 changes when r_α is scaled, and nothing in it involves the diagonal of Q_F. The objects depend on A_F and μ only, so a field does not change them. For $v \in \mathbb{R}^F$ we put $K_F(v) = C_F(\alpha_F^*(v))$, so that $K_F(0) = K_F$, and $K_{\{i\}}(v) = \mu_i$.

Note that $l_\alpha \cdot r_\alpha = 1$ and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$. Since r_α minimizes $u \mapsto \sum_{i\neq j} \alpha_i q_{ij} u_j/u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top \big(A_F - \operatorname{diag}\theta(\alpha)\big) = 0. \tag{8}$$

Also $(A_F - \operatorname{diag}\theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [1]. So $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (8) its stationary law is α. If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$. Hence v is a multiple of $\mathbf{1}$, and $\alpha^\top v = 0$ gives $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \qquad \alpha^\top Z_\alpha = 0, \qquad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \tag{9}$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = J_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F. In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\operatorname{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\}\in\mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 5.2 (the constant)**.** *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $J_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 J_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})\sqrt{\det \nabla^2 J_F(\alpha)/(2\pi)^{k-1}}$.

(b) Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0,\infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i\alpha_j}\, q_{ij}\sqrt{\nu_i/\nu_j}$. Also $J_F(\alpha) = \sum_i q_i\alpha_i - \sum_{i\neq j} c_{ij}(\alpha)$, and

$$\det \Sigma_\alpha = \frac{2^{k-1}(\prod_i \alpha_i)^2}{\operatorname{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\operatorname{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

In words, the Gaussian factor in C_F is, up to a power of 2π, the square root of the determinant of the Hessian of the rate. For a reversible face this determinant is a weighted count of spanning trees divided by $2^{k-1}(\prod_i \alpha_i)^2$.

Proof. The proof of (a) is in Appendix A and uses Lemma A.1.

(b) We guess the vector r_α and check that the guess is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ (entrywise). As we saw before (8), the derivative of $\sum_{i\neq j} \alpha_i q_{ij} u_j/u_i$ in u_j is $\sum_i l_i q_{ij} - l_j(A_F u)_j/u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j(A_F u)_j}{u_j} = \nu_j \sum_i q_{ji}\sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. So the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.3(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij}\sqrt{\nu_i/\nu_j} = q_{ji}\sqrt{\nu_j/\nu_i}$. So $\widehat{Q}_\alpha$ is reversible with these

conductances. Also $\alpha_i\theta_i(\alpha) = l_i(A_F u)_i = \sum_{j\neq i} c_{ij}(\alpha)$, and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$ gives the formula for J_F.

For the determinant we express the quadratic form of Σ_α through the Laplacian of the conductances, namely $\mathcal{L} = -D\widehat{Q}_\alpha$. It is symmetric, its off-diagonal entries are $-c_{ij}(\alpha)$, and $\mathcal{L}\mathbf{1} = 0$. Fix $\eta \in \mathbb{R}^F$ with $\eta_1 = 0$, and put $w = Z_\alpha\eta$ and $\zeta = D\eta - (\alpha\cdot\eta)\alpha$. By (9), $\mathcal{L}w = D(I - \mathbf{1}\alpha^\top)\eta = \zeta$ and $\alpha\cdot w = 0$. So $\eta^\top\Gamma_\alpha\eta = 2(D\eta)^\top w = 2\zeta^\top w$, where the second step uses $\alpha\cdot w = 0$. Next we remove state 1. Let $v = w - w_1\mathbf{1}$, so that $v_1 = 0$. Since $\zeta\cdot\mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we have $\zeta^\top w = \zeta^\top v$ and $\mathcal{L}v = \zeta$. We write primes for the coordinates $2,\dots,k$, $D' = \operatorname{diag}\alpha'$, and $\mathcal{L}_{(1)}$ for $\mathcal{L}$ without row and column 1. As $v_1 = 0$, the rows $2,\dots,k$ of $\mathcal{L}v = \zeta$ say that $\mathcal{L}_{(1)}v' = \zeta'$, and as $\eta_1 = 0$ we have $\zeta' = (D' - \alpha'\alpha'^\top)\eta'$. The matrix-tree theorem [5] gives $\det\mathcal{L}_{(1)} = \operatorname{tree}(c(\alpha))$, which is positive because F is irreducible. So $\mathcal{L}_{(1)}$ is invertible, and

$$\eta'^\top\Sigma_\alpha\eta' = 2\zeta'^\top v' = 2\eta'^\top(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)\eta'$$

for every $\eta' \in \mathbb{R}^{k-1}$. Both matrices are symmetric, hence $\Sigma_\alpha = 2(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)$. Now we take determinants. The matrix determinant lemma gives $\det(D' - \alpha'\alpha'^\top) = \prod_{i\geq 2}\alpha_i\,(1 - \sum_{i\geq 2}\alpha_i) = \prod_i \alpha_i$. So $\det\Sigma_\alpha = 2^{k-1}(\prod_i\alpha_i)^2/\operatorname{tree}(c(\alpha))$, and putting this into Definition 5.1 gives the formula for $C_F(\alpha)$. □

Part (b) needs a reversible face. Proposition 5.11 gives a formula of the same kind for every irreducible face.

Corollary 5.3 (symmetric faces)**.** *Let F satisfy* (H) *with $k \geq 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j\in F\setminus\{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F, and*

$$K_F = \mu(F)\,k^{(k+1)/2}\sqrt{\operatorname{tree}(q_F)}\,(4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ *is* $\operatorname{tree}(c)$ *for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.*

Proof. The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 5.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)}\operatorname{tree}(q_F)$, and $\prod_i\alpha_i = k^{-k}$. □

5.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition gets the factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of a lattice cell. A field $v \in \mathbb{R}^F$ enters the continuum density as an exact factor. By Lemma 2.4, $f_T(\alpha)e^{Tv\cdot\alpha}$ is the density f_T of the killed matrix $Q_F + \operatorname{diag} v$, which has the same A_F and μ and the diagonal $q - v$ in place of q. On the lattice the factor is $e^{Tv\cdot n/N} = e^{h\langle v,n\rangle}$.

Theorem 5.4 (local law)**.** *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$, $T \geq 1$ and $v \in \mathbb{R}^F$,*

$$f_T(\alpha)e^{Tv\cdot\alpha} = T^{(k-1)/2}C_F(\alpha)e^{-T(J_F(\alpha)-v\cdot\alpha)}\big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

The field cancels from the two sides, so the case $v = 0$ is the whole content, and the error does not depend on v. The density of the visit fractions on the event that the chain stays in F is a Gaussian factor $T^{(k-1)/2}$ times the large-deviation factor $e^{-TJ_F(\alpha)}$ times the constant $C_F(\alpha)$, with a relative error $O(1/T)$ that is uniform away from the boundary of the face. Good [48, pp. 42–43] proposed a saddle-point analysis of his formula and did not carry it out. Theorem 5.4, with its explicit constant and its uniform relative error, completes that analysis.

Idea of the proof. The proof is in Appendix A. The generating function of the words in H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point comes from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α. The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 5.5 (discrete and continuum)**.** *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for* $\operatorname{supp} n = F$ *with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

The probability of a composition is the continuum density times the volume $N^{-(k-1)}$ of a lattice cell. The proof is in Appendix A. A field multiplies both sides by the same factor $e^{Tv\cdot n/N}$. Together with the restriction identity the two results give one display for all faces.

Theorem 5.6 (the stratified local law)**.** *Fix Q and a start μ. For every face F satisfying (H) with $|F| = k \geq 2$ and every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for* $\operatorname{supp} n = F$*, $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/3}$,*

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2}\log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big),$$

and for a singleton $-\log P_N(Ne_i) = Tq_i - \log \mu_i + O(T^2/N)$. With $T = \tau L$ the exponent of N is $(k-1) + \tau J_V(n/N)$, the lattice entropy of the face plus τ times the Donsker and Varadhan cost of the point.

Proof. This is Theorem 5.4 with $v = 0$, Lemma 5.5 and Proposition 3.1, which replaces J_F by J_V on Δ_F. The singleton case is (2). □

The local law does not give the behaviour near the boundary of a face, where one coordinate of n/N tends to zero. Section 6 proves the layer for faces with at most three states and for larger faces under one hypothesis, and Section 8 treats the complete graph when the small coordinates are of order one.

For a field $v \in \mathbb{R}^d$ we write $\lambda_F(v)$, $\alpha_F^*(v)$ and $K_F(v)$ for the quantities of v_F, and

$$M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\},$$

so that $M_F(T, 0) = M_F$.

Theorem 5.7 (face maxima)**.** *Let F satisfy (H), and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. If $k \geq 2$, then uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,*

$$M_F(T, v) = N^{-(k-1)} T^{(k-1)/2} K_F(v) e^{-T\lambda_F(v)}\big(1 + O_{\mathcal{V}}(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$. For $k = 1$, $M_{\{i\}}(T, v) = e^{Tv_i} M_{\{i\}}$ with $M_{\{i\}}$ given by (2). In both cases, uniformly for $v \in \mathcal{V}$ and $1 \le T \le N^{1/3}$,

$$\log M_F(T, v) = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}(1/T + T^2/N). \tag{10}$$

The best composition on F sits within $O(N/T)$ of $N\alpha_F^*(v)$, and its probability is the Gaussian peak of the local law there. The factor $N^{-(k-1)}$ is the volume of a lattice cell, and $T^{(k-1)/2}$ is the height of a Gaussian density whose spread in each free direction is of order $T^{-1/2}$. The constants depend on $\mathcal{V}$ only through the compact set $\{\alpha_F^*(v) : v \in \mathcal{V}\}$, the bounds of λ_F and K_F on $\mathcal{V}$, and finitely many vectors of V_F.

Idea of the proof. Near $\alpha_F^*(v)$ we use the local law and Lemma 5.5, and maximize a Gaussian with a small linear perturbation. Away from $\alpha_F^*(v)$ the rate exceeds $\lambda_F(v)$ by a fixed amount, and Lemma 4.1, applied with finitely many vectors θ, shows that these compositions are exponentially less likely in T. For bounded T positivity and continuity suffice.

Proof. For $k = 1$ the only composition is Ne_i. Let $k \ge 2$, and put $J^v(\alpha) = J_F(\alpha) - v \cdot \alpha - \lambda_F(v)$, which is nonnegative on Δ_F and vanishes only at $\alpha_F^*(v)$ (Proposition 3.3(a)). Put $X(v) = N^{-(k-1)} T^{(k-1)/2} K_F(v) e^{-T\lambda_F(v)}$, the value that the theorem claims, and write $x = \alpha' - \alpha_F^*(v)'$ for the displacement of α from $\alpha_F^*(v)$. Below, constants depend only on Q, μ and $\mathcal{V}$.

The balls. The Perron vectors of $Q_F + \operatorname{diag} v$ are real-analytic in v, so $\mathcal{A} = \{\alpha_F^*(v) : v \in \mathcal{V}\}$ is a compact subset of Δ_F°, and λ_F and $\log K_F$ are bounded on $\mathcal{V}$. By Propositions 3.3 and 5.2(a), J_F and $\log C_F$ are real-analytic on Δ_F° with $\nabla^2 J_F = \Sigma_\alpha^{-1}$ positive definite, and J^v has a critical point at $\alpha_F^*(v)$. Taylor's theorem with bounds on the derivatives of order up to three on a compact neighbourhood of $\mathcal{A}$ gives $\rho_0 > 0$ and positive constants c_U, C_U, C_1 and C_3 with the following properties. The set $\mathcal{K}_{\mathcal{V}}$ of points within ρ_0 of $\mathcal{A}$ lies in Δ_F°, and for $v \in \mathcal{V}$ the ball $U(v) = \{|x| \le \rho_0\}$ satisfies

$$c_U|x|^2 \le J^v \le C_U|x|^2, \qquad \bigl|\log(C_F/K_F(v)) - g(v) \cdot x\bigr| \le C_1|x|^2$$

on $U(v)$, where $g(v) = \nabla \log C_F(\alpha_F^*(v))$ has $|g(v)| \le C_1$. Also $\log C_F$ is C_3-Lipschitz on $\mathcal{K}_{\mathcal{V}}$.

Outside $U(v)$. The set $\mathcal{E}$ of pairs $(v, \beta) \in \mathcal{V} \times \Delta_F$ with $\beta \notin U(v)^\circ$ is compact, and $J^v(\beta)$ is positive and lower semicontinuous on it, since J_F is a supremum of continuous functions. So $J^v \ge 3\eta$ on $\mathcal{E}$ for some $\eta > 0$. For each $(v, \beta) \in \mathcal{E}$ the definition of J_F gives $\theta \in V_F$ with $\langle q - \theta - v, \beta\rangle > \lambda_F(v) + 2\eta$, as in Section 4, and the same θ works near (v, β). By compactness finitely many vectors $\theta_j \in V_F$ suffice. Now let $\operatorname{supp} n = F$ with $(v, n/N) \in \mathcal{E}$, and take j that works at $(v, n/N)$. Lemma 4.1 with $\theta = \theta_j$, multiplied by $e^{h\langle v, n\rangle}$, gives

$$P_N(n) e^{Tv\cdot n/N} \le C(T/N)^{k-1} e^{-T(\lambda_F(v)+2\eta)} = X(v) \cdot O\bigl(T^{(k-1)/2} e^{-2\eta T}\bigr).$$

Inside $U(v)$. By Theorem 5.4 and Lemma 5.5 with $\mathcal{K} = \mathcal{K}_{\mathcal{V}}$, for $\alpha = n/N \in U(v)$,

$$P_N(n) e^{Tv\cdot n/N} = X(v)\, \tfrac{C_F(\alpha)}{K_F(v)}\, e^{-TJ^v(\alpha)}(1 + E), \qquad |E| \le C_2(1/T + T^2/N). \tag{11}$$

For $T \ge 2C_1/c_U$ the logarithm of the two middle factors is at most

$$g \cdot x + C_1|x|^2 - c_U T|x|^2 \le g \cdot x - \tfrac{c_U T}{2}|x|^2 \le \frac{|g|^2}{2c_U T},$$

the maximum of a concave quadratic. So $P_N(n) e^{Tv\cdot n/N} \le X(v)(1 + O(1/T + T^2/N))$ on $U(v)$. The bound outside $U(v)$ is smaller for large T, and hence $M_F(T, v) \le X(v)(1 + O(1/T + T^2/N))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_F^*(v)_i| < 1$ for all i, which exists for

$N \ge N_0$. Then $|x| \le k/N$, $C_F(n^0/N) \ge K_F(v)e^{-kC_3/N}$ and $TJ^v(n^0/N) \le C_U k^2 T/N^2$, so (11) gives $P_N(n^0)e^{Tv\cdot n^0/N} \ge X(v)(1 - O(1/T + T^2/N))$.

Bounded T. Fix T_2 such that the two bounds give $M_F(T,v)/X(v) \in [\frac12, \frac32]$ for $T \ge T_2$. For $1 \le T \le T_2$ it is enough to show $M_F(T,v)/X(v) \in [c, C]$ with $0 < c < C$. There $N^{k-1}X(v)$ lies between two positive constants, and so does $f_T(\alpha)e^{Tv\cdot\alpha}$ on $[1, T_2] \times \mathcal{V} \times \mathcal{K}_{\mathcal{V}}$, as it is positive and continuous. So Lemma 5.5 gives both bounds, and the bound outside $U(v)$ is $O(N^{-(k-1)})$. This proves the first display, and (10) is its logarithm.

The maximizer. Let $\hat{n}$ be a maximizer with displacement $\hat{x}$. For $T \ge T_3$ the bound outside $U(v)$ is below $X(v)/2 \le P_N(n^0)e^{Tv\cdot n^0/N}$, so $\hat{n}/N \in U(v)$, and $|E| \le \frac12$ in (11). Taking logarithms in the comparison of $\hat{n}$ with n^0, and using $\log(C_F/K_F(v)) \le C_3|\hat{x}|$ and $TJ^v \ge c_U T|\hat{x}|^2$, we get

$$c_U T|\hat{x}|^2 \le C_3|\hat{x}| + C_4\Big(\frac{1}{T} + \frac{T^2}{N}\Big) \le C_3|\hat{x}| + \frac{2C_4}{T}.$$

The last step uses $T \le N^{1/3}$, which gives $T^2/N \le 1/T$. This is the step that needs the exponent $\frac13$. Solving the quadratic inequality gives $|\hat{x}| \le \big(C_3 + \sqrt{C_3^2 + 8c_U C_4}\big)/(2c_U T)$. For $1 \le T < T_3$ the claim holds since $|\hat{x}|$ is bounded. □

5.3 Crossings and tie windows

By Theorem 5.7, the difference $\log M_F - \log M_G$ is an explicit increasing function of T up to an error $O(1/T + T^2/N)$, and its zero is the crossing.

Theorem 5.8 (the crossing equation)**.** *Let F and G satisfy (H), with $\Delta = k_F - k_G \ge 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) For large N, every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies

$$\delta T_N + \frac{\Delta}{2}\log T_N = \Delta L - \log\frac{K_F}{K_G} + O\Big(\frac{1}{L}\Big). \tag{12}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac12\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac12\log\frac{\Delta}{\delta} - \frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

Proof. We subtract (10) for G from (10) for F. Uniformly on $[1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \log\frac{K_F}{K_G} + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi' \ge \delta$.

(i) At $T = \frac{\Delta}{2\delta}L$ the display gives $D_N = -\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it gives $D_N = \Delta L + O(\log L) > 0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. So D_N is continuous and has a zero in between.

(ii) First we locate a zero $T_N \in [1, N^{1/3}]$ roughly. The error term is $O(1)$ on $[1, N^{1/3}]$, so $\Xi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\Xi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$, so $T_N > \frac{\Delta}{2\delta}L$

because Ξ is increasing. Also $\delta T_N \le \Xi(T_N)$, so $T_N \le \frac{2\Delta}{\delta} L$. With T_N of order L the error term is $O(1/T_N + T_N^2/N) = O(1/L)$, and $D_N(T_N) = 0$ is (12).

(iii) Let T° be the solution of $\Xi(T^\circ) = \Delta L - \log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N) = \Xi(T^\circ) + O(1/L)$, and $\Xi' \ge \delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, note that $T^\circ = \frac{\Delta}{\delta} L + O(\log L)$, so $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$. Putting this into $\delta T^\circ = \Delta L - \frac{\Delta}{2} \log T^\circ - \log(K_F/K_G)$ and dividing by δ gives the explicit form. □

In words, the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$. The crossing balances the three. On the 4-cycle the explicit form in (iii) is still off by 0.06 at $N = 10^{14}$, so computations should solve (12). We do not prove that the zero of D_N is unique in general.

At a switch value τ^* of Theorem 4.4 several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph (Corollary 7.3). Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 5.9 (tie windows)**.** *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 4.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F - 1}{2} \log \tau^* - \tau^* \lambda_F t$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^* L + \frac{m^*}{2} \log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \le \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \le N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \le t \le \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \le t \le -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

Consequently the following holds for every $\eta > 0$ and large N. If $|t| \le \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \le \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j runs over the kinks of E, and it changes near each kink. Coincident lines do not distinguish their faces at this order.

In words, inside the window the logarithm of every tied face maximum is the same leading term plus the line $\ell_F(t)$, faces that do not tie are smaller by a power of N, and far out in the window the face with the least (for $t > 0$) or largest (for $t < 0$) exit rate wins. In between, the maximizing line changes only near the kinks of E. Faces with the same line can require a finer comparison.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). We first put T into (10). Write $T = \tau^* L(1 + y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \le \frac{1}{2}$ for $|t| \le L/4$ and large N. Then $\log T = \log \tau^* + \log L + \log(1 + y)$ and $T\lambda_F = \tau^* \lambda_F (L - \frac{1}{2}\log L + t)$, and the error term of (10) is $O(1/L)$. Since $\tau^* \lambda_F + k_F - 1 = m^*$, we get, uniformly for $|t| \le L/4$,

$$\log M_F = -m^* L + \tfrac{m^*}{2} \log L + \ell_F(t) + \tfrac{k_F - 1}{2} \log(1 + y) + O(1/L) \qquad (F \in \mathcal{F}^*). \tag{13}$$

(i) For t in a compact set, $y = O(\log L/L)$, so $\log(1+y) = O(\log L/L)$ in (13).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. (If there is no such face, (ii) says nothing, and we put $g = 1$.) Put $\varepsilon_1 = \min\bigl(\tau^*/2,\, g/(4q_{\max}+4)\bigr)$. (Any constant below $g/(2q_{\max})$ would do in the second entry. We use that $0 \le \lambda_F \le q_{\max}$ for every face, since the Perron root of Q_F lies between its largest diagonal entry and 0.) The entry $\tau^*/2$ keeps T of order L. Namely, if $|T/L - \tau^*| \le \varepsilon_1$, then $\frac{1}{2}\tau^* L \le T \le \frac{3}{2}\tau^* L$. So for large N we have $T_0 \le T \le N^{1/3}$, which is the range of Theorem 4.2(ii), and $\log T = O(\log L)$. Take an admissible $F \notin \mathcal{F}^*$ and any $G \in \mathcal{F}^*$. The difference $\Phi_\tau(F) - \Phi_\tau(G)$ is at least g at $\tau = \tau^*$, and its slope in τ is $\lambda_F - \lambda_G$, which is at most $q_{\max}$ in absolute value. So $\Phi_\tau(F) - \Phi_\tau(G) \ge g/2$ for $|\tau - \tau^*| \le \varepsilon_1$. With $\tau = T/L$ the upper bound of Theorem 4.2(ii) for F and the lower bound for G give $\log M_F - \log M_G \le -(\Phi_\tau(F) - \Phi_\tau(G))L + O(\log L) \le -\frac{g}{4}L$ for large N. So (ii) holds with $\varepsilon_0 = g/4$.

(iii) Take $\varepsilon = \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$. Then for $|t| \le \varepsilon L$ and large N we have $|T/L - \tau^*| = \tau^*|y| \le \varepsilon_1$, so (ii) applies and no global mode lies on a face outside $\mathcal{F}^*$. It remains to compare the tied faces. Let $t_0 \ge 1$ be larger than every $|t_j|$, so that E is a single line on $[t_0, \infty)$. Let $F \in \mathcal{F}^*$ have this line, and let $G \in \mathcal{F}^*$ have another line. We show that $M_F > M_G$ for $t_0 \le t \le \varepsilon L$ if t_0 and N are large. Since $\ell_F \ge \ell_G$ on $[t_0, \infty)$, the slopes satisfy $\lambda_F \le \lambda_G$. As both faces are tied,

$$\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \ge 0.$$

There are two cases.

First let $\Delta = 0$. Then the two lines are parallel, and $\ell_F - \ell_G = \log(K_F/K_G) > 0$. The terms with $\log(1+y)$ in (13) are the same for F and G, so $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L) > 0$ for large N.

Now let $\Delta > 0$. Then $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$, and (13) gives

$$\log \frac{M_F}{M_G} = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O\Bigl(\frac{1}{L}\Bigr).$$

For large N and $t \ge 1$ we have $|\log(1+y)| \le 2|y| \le (2t + \log L)/L \le t/2$. So the right side is at least $\frac{3}{4}\Delta t - C$, which is positive for $t \ge t_0$ once t_0 is large.

So for $t_0 \le t \le \varepsilon L$ every global mode lies on a face whose line is E on $[t_0, \infty)$. The slope of ℓ_F is $-\tau^*\lambda_F$, and E follows the line of largest slope on the far right, so such a face has the least λ_F in $\mathcal{F}^*$. The case $t \le -t_0$ is the same with the signs reversed.

The consequence. Fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$, a finite union of closed intervals without kinks. Take $G \in \mathcal{F}^*$. The function $E - \ell_G$ is convex, piecewise linear and nonnegative, and its slope changes only at kinks of E. So its zero set is empty, or a single kink of E, or an interval whose finite endpoints are kinks of E. Hence each interval of S_η lies inside the zero set or is disjoint from it. In the first case $\ell_G = E$ on the interval, and in the second $E - \ell_G \ge g_\eta$ there for some $g_\eta > 0$. Now let $t \in S_\eta$ and let $F \in \mathcal{F}^*$ have $\ell_F(t) = E(t)$. By (i), every $G \in \mathcal{F}^*$ whose line is not E near t has $\log(M_F/M_G) \ge g_\eta - O(\log L/L) > 0$, and by (ii) the faces outside $\mathcal{F}^*$ are smaller still. This is the claim for $t \in S_\eta$. For $t_0 \le |t| \le \varepsilon L$ it is (iii). Since η is arbitrary, the line of the face that carries the mode can change only at $t = t_j + o(1)$. It does change near each kink, because E is a different line on the two sides of a kink. □

Remark 5.10 (kinks are crossings). At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^*\delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 5.8. The coefficient $\frac{1}{2}$ of $\log\log N$ in both theorems comes from the power $T^{(k-1)/2}$ in Theorem 5.7, so it needs only (H). Example 5.14 shows that it can fail on a reducible face.

5.4 A tree formula for every face

The next proposition extends Proposition 5.2(b) from reversible faces to every irreducible face. It uses two counts of spanning trees in place of one, namely tree as before and the count arb of spanning trees oriented toward a root, which is defined just before Lemma A.3. Its proof rests on Lemma A.3 in Appendix A.

Proposition 5.11 (the constant on a general face)**.** *Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ put*

$$f_{ij}(\alpha) = l_{\alpha,i}\, q_{ij}\, r_{\alpha,j} \qquad (i \neq j \text{ in } F),$$

which does not change when r_α is scaled.

(a) The weights $f(\alpha)$ are the stationary flow of the tilted chain, that is, $f_{ij}(\alpha) = \alpha_i(\widehat{Q}_\alpha)_{ij}$, and $\sum_{j\neq i} f_{ij}(\alpha) = \sum_{j\neq i} f_{ji}(\alpha) = \alpha_i\theta_i(\alpha)$ for every i. The number $\mathrm{arb}_i(f(\alpha))$ does not depend on the root i, and

$$\det \Sigma_\alpha = \frac{(\prod_i \alpha_i)^2\ \mathrm{tree}\big(f(\alpha) + f(\alpha)^\top\big)}{\mathrm{arb}(f(\alpha))^2}$$

and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\mathrm{arb}(f(\alpha))}{\prod_i \alpha_i \sqrt{\mathrm{tree}(f(\alpha) + f(\alpha)^\top)}}.$$

(b) Also $\mathrm{arb}(f(\alpha))/\prod_i \alpha_i = \sum_{i\in F} \det\big(\mathrm{diag}\,\theta(\alpha) - A_F\big)_{(i)}$. At $\alpha = \alpha_F^*$ this is $\prod_\rho(\rho - \lambda_F)$, the product over the other $k-1$ eigenvalues ρ of $-Q_F$, counted with multiplicity. So with $l_F \cdot r_F = 1$ and $f_{ij} = l_{F,i}\, q_{ij}\, r_{F,j}$,

$$K_F = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\mathrm{arb}(f)}{\prod_i \alpha^*_{F,i} \sqrt{\mathrm{tree}(f + f^\top)}} = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\prod_\rho(\rho - \lambda_F)}{\sqrt{\mathrm{tree}(f + f^\top)}}.$$

(c) Let $F = \{1, \dots, k\}$ be a directed cycle, that is, $q_{i,i+1} = a_i > 0$ with indices modulo k, and $q_{ij} = 0$ for all other $i \neq j$ in F. Then for every $\alpha \in \Delta_F^\circ$

$$\det \Sigma_\alpha = \frac{k\,(\prod_i \alpha_i)^2}{\big(\prod_i \alpha_i a_i\big)^{(k-1)/k}}.$$

If moreover $q_i = a_i$ for all i, then $\alpha^*_{F,i} = \varphi^*/a_i$ with $\varphi^* = (\sum_i 1/a_i)^{-1}$, and

$$\det \Sigma_{\alpha_F^*} = \frac{k\,(\varphi^*)^{k+1}}{\prod_i a_i^2}, \qquad K_F = \frac{\mu(F)\,\prod_i a_i}{(2\pi)^{(k-1)/2}\sqrt{k}\,(\varphi^*)^{(k+1)/2}}.$$

In the reversible case of Proposition 5.2(b), $f(\alpha) = c(\alpha)$ is symmetric, $\mathrm{arb}(f) = \mathrm{tree}(c)$ and $\mathrm{tree}(2c) = 2^{k-1}\,\mathrm{tree}(c)$, so (a) reduces to that proposition.

Proof. (a) By Definition 5.1, $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_{\alpha,i}q_{ij}r_{\alpha,j} = f_{ij}(\alpha)$ for $i \neq j$, and the diagonal of $\widehat{Q}_\alpha$ is $-\theta(\alpha)$. So $\widehat{Q}_\alpha\mathbf{1} = 0$ and $\alpha^\top\widehat{Q}_\alpha = 0$ give the two flow identities. Lemma A.3 for $G = \widehat{Q}_\alpha$ and $\pi = \alpha$, for which $\Sigma = \Sigma_\alpha$, gives $\det\Sigma_\alpha$, and Definition 5.1 gives $C_F(\alpha)$.

(b) Since $-\widehat{Q}_\alpha = R^{-1}(\mathrm{diag}\,\theta(\alpha) - A_F)R$ and a diagonal conjugation keeps principal minors, Lemma A.3(i) gives the first claim. At α_F^*, $\mathrm{diag}\,\theta - A_F = -Q_F - \lambda_F I$, whose eigenvalues are the numbers $\rho - \lambda_F$, exactly one of them 0. The sum of its principal minors of order $k-1$ is the

elementary symmetric polynomial of degree $k-1$ in these eigenvalues, which is the product of the other $k-1$. With $l_{\alpha_F^*} \cdot r_{\alpha_F^*} = 1$, (a) gives both forms of K_F.

(c) The flow out of i equals the flow into i, so all $f_{i,i+1}(\alpha)$ equal one number φ, and their product $\prod_i l_{\alpha,i} a_i r_{\alpha,i+1} = \prod_i \alpha_i a_i$ gives $\varphi = (\prod_i \alpha_i a_i)^{1/k}$. The only spanning tree oriented toward i with nonzero weight is the cycle without the edge out of i, so $\operatorname{arb}(f(\alpha)) = \varphi^{k-1}$, and the symmetrized flow has conductance φ on each edge of the undirected cycle (for $k = 2$ one edge of conductance 2φ), so $\operatorname{tree}(f + f^\top) = k\varphi^{k-1}$. Now (a) gives $\det \Sigma_\alpha$. If $q_i = a_i$, then Q_F is a generator, $\lambda_F = 0$, $r_F = \mathbf{1}$ and $l_F = \alpha_F^*$ has constant flow $l_{F,i} a_i$, so $\alpha^*_{F,i} = \varphi^*/a_i$, $\prod_i \alpha^*_{F,i} = (\varphi^*)^k / \prod_i a_i$ and $(\mu \cdot r_F)(l_F \cdot \mathbf{1}) = \mu(F)$. □

On the complete graph the tree sum of Proposition 5.2(b) has a closed form. This is classical, and we include a short proof.

Lemma 5.12 (weighted Cayley formula)**.** *For $x_1, \dots, x_k > 0$ and the conductances $c_{ij} = x_i x_j$,*

$$\operatorname{tree}(c) = x_1 x_2 \cdots x_k \, (x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ this is Cayley's count k^{k-2} of the spanning trees of the complete graph K_k.

Proof. Put $X = \sum_i x_i$. The Laplacian of the conductances is $\operatorname{diag}(Xx_i) - xx^\top$, so without its first row and column it is $\operatorname{diag}(Xx_2, \dots, Xx_k) - x'x'^\top$ with $x' = (x_2, \dots, x_k)$. The matrix determinant lemma gives the determinant $X^{k-1} \prod_{i \geq 2} x_i \, (1 - \sum_{i \geq 2} x_i / X) = X^{k-2} \prod_i x_i$, and this is $\operatorname{tree}(c)$ by the matrix-tree theorem [5]. (For integer conductances this is the multigraph case, and both sides are polynomials in the c_{ij}.) □

Example 5.13 (the complete face)**.** Let F be a face of $k \geq 2$ states with $q_{ij} = 1$ for all $i \neq j$ in F, and any exit rates, for example a face of $Q = J - dI$, where $q_i = d - 1$. Then F is reversible with $\nu = \mathbf{1}$, so by Proposition 5.2(b) we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$ (entrywise), and the conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}$ are products. Put $A = \sum_{i \in F} \sqrt{\alpha_i}$. Lemma 5.12 gives $\operatorname{tree}(c(\alpha)) = (\prod_i \alpha_i)^{1/2} A^{k-2}$, so for every $\alpha \in \Delta_F^\circ$

$$C_F(\alpha) = \frac{(\mu \cdot \sqrt{\alpha}) \, A^{k/2}}{(4\pi)^{(k-1)/2} \, (\prod_i \alpha_i)^{3/4}} = \frac{(\mu \cdot \sqrt{\alpha}) \, A}{\sqrt{\det(2\pi \Sigma_\alpha)}}, \qquad J_F(\alpha) = \sum_{i \in F} q_i \alpha_i - (A^2 - 1).$$

The second form of $C_F(\alpha)$ uses $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$. For $Q = J - dI$ the rate is $J_F(\alpha) = d - A^2$, the rate of Dupuis and Liu [18, Eq. (3.2)]. At the center $\alpha = \mathbf{1}/k$ we have $A^2 = k$ and $C_F = \mu(F) k^{k-1/2} (4\pi)^{-(k-1)/2}$, which is Corollary 5.3 with $\operatorname{tree}(q_F) = k^{k-2}$. With the uniform start, $dC_F(\alpha)$ is the constant $D(\alpha)$ of [29, Theorem 3.3 and Proposition 3.6], written there as a mass over a Gaussian volume. Section 7 uses this example.

5.5 A reducible face

The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the coefficient $\frac{1}{2}$ of Remark 5.10 can fail, and in the next example it is 1 and then 0.

Example 5.14 (a directed 3-cycle)**.** Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$. From state 1 the chain moves only along $1 \to 2 \to 3 \to 1$, so the admissible faces are $\{1\}$, $\{1, 2\}$ (reducible, $\lambda = 1$) and $\{1, 2, 3\}$ (irreducible, $\lambda = 0$).

(i) Exactly $M_{\{1\}} = (1-bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1+2b)$ and $K_{\{1,2,3\}} = (1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$, and $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log\frac{b-1}{b}] + O(\log L/L)$. The coefficient of $\log\log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log\frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0.

Proof. (i) A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$. Moving one visit from state 2 to state 1 multiplies it by $(1-bh)/(1-h) < 1$, so the maximum is at $n_1 = 1$. On the full face Q is a generator, so $\lambda = 0$, $r = \mathbf{1}$ and α^* is the stationary law $(1, b, b)/(1+2b)$. Proposition 5.11(c) with $(a_1, a_2, a_3) = (b, 1, 1)$ and $q_i = a_i$ gives $\varphi^* = b/(1+2b)$ and the value of $K_{\{1,2,3\}}$.

(ii) The function $\log(M_{\{1,2\}}/M_{\{1\}}) = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$ has the derivative $1/h - (N-2)/(1-h) + (N-1)b/(1-bh) > 0$ in h, since $b > 1$ and $1 - bh < 1 - h$. It tends to $-\infty$ as $h \to 0$ and to $+\infty$ as $bh \to 1$, so it has exactly one zero. Its expansion $\log b + \log T - L + (b-1)T + O(T^2/N)$ gives $(b-1)T + \log T = L - \log b + O(T^2/N)$ at the zero, and with $\log T = \log L - \log(b-1) + O(\log L/L)$ this gives the formula.

(iii) For the full face $k = 3$, so Theorem 5.7 gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/L)$ on $[L/2, 2L]$, with the same power T^1 as $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$. So $\log T$ cancels, and equating the two gives $T = L + \log b - \log K_{\{1,2,3\}} + O(1/L)$. □

The face $\{1,2\}$ has the power T^1 in its maximum where an irreducible pair would have $T^{1/2}$, and this changes the coefficient of $\log\log N$ at both crossings. The exact parts of (i) and the uniqueness of the root in (ii) are checked in Lean.

5.6 A weak field

Give each visit to state i the extra weight $e^{\xi_i/N}$, for a fixed vector $\xi \in \mathbb{R}^d$. Over N steps these weights multiply to $e^{\langle \xi, n\rangle/N}$, a factor of order 1, which is the order of the constants K_F. So such a field cannot change the first-order rule, but it can change the winner inside a tie window. For a composition n put

$$w_\xi(n) = e^{\langle \xi, n\rangle/N} P_N(n), \qquad Z_{N,\xi} = \sum_n w_\xi(n), \qquad \widetilde{P}_{N,\xi}(n) = \frac{w_\xi(n)}{Z_{N,\xi}},$$

and for a face F let $M_F^\xi = \max\{w_\xi(n) : \operatorname{supp} n = F\}$, which is $M_F(T, \xi/T)$ in the notation of Theorem 5.7. Write $\|\xi\| = \max_i |\xi_i|$. On the complete graph with the uniform start, $\widetilde{P}_{N,\xi}$ is the law of the counts of the Potts chain in a field of size ξ_i/N at each site in state i [29].

Proposition 5.15 (a weak field)**.** *Fix Q, μ and a compact set $\mathcal{X} \subset \mathbb{R}^d$.*

(a) $\widetilde{P}_{N,\xi}$ is the law of the composition when the law of the path $X_1, \dots, X_N$ is multiplied by $\exp(N^{-1}\sum_{s=1}^N \xi_{X_s})$ and normalized. For every face F, $e^{-\|\xi\|} M_F \le M_F^\xi \le e^{\|\xi\|} M_F$.

(b) Let F satisfy (H)*. Uniformly for $\xi \in \mathcal{X}$ and $1 \le T \le N^{1/3}$,*

$$M_F^\xi = e^{\langle \xi, \alpha_F^*\rangle} M_F\big(1 + O(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ of w_ξ on F satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$.

(c) *Fix ξ and $\tau^* > 0$, and let m^*, $\mathcal{F}^*$ and ℓ_F be as in Theorem 5.9, with every face in $\mathcal{F}^*$ irreducible. Then Theorem 5.9 holds for the modes of $\widetilde{P}_{N,\xi}$, with M_F replaced by M_F^ξ and ℓ_F by $\ell_F^\xi = \ell_F + \langle \xi, \alpha_F^* \rangle$. Thus it is the constants $K_F e^{\langle \xi, \alpha_F^* \rangle}$ that determine whether two field-shifted lines coincide.*

On a face satisfying (H) a weak field shifts the line by the field averaged over the quasi-ergodic law. On the complete graph α_F^* is uniform on F, and the tilt is the average of ξ over F (Corollary 7.11). By (a) the first-order rule of Theorem 4.2(iii) holds for $\widetilde{P}_{N,\xi}$ without change, and by (b) so does the location in Theorem 4.2(iv).

Proof. (a) A path $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (I + hQ)_{x_s x_{s+1}}$, and $\sum_s \xi_{x_s} = \langle \xi, n \rangle$ depends on the path only through its composition n. So the multiplied law, summed over the paths with composition n, is $w_\xi(n)$. Since n/N is a probability vector, $|\langle \xi, n \rangle|/N \le \|\xi\|$.

(b) For a singleton the claim is exact. Let $k \ge 2$. For $T \ge 1$ the field ξ/T lies in the compact set $\mathcal{V} = \{s\xi : 0 \le s \le 1,\ \xi \in \mathcal{X}\}$, so Theorem 5.7 applies to $M_F^\xi = M_F(T, \xi/T)$ uniformly. The map $v \mapsto \lambda_F(v)$ is real-analytic with $\nabla \lambda_F(0) = -\alpha_F^*$, the derivative of a simple eigenvalue, so $T\lambda_F(\xi/T) = T\lambda_F - \langle \xi, \alpha_F^* \rangle + O(1/T)$. Also $\alpha_F^*(\xi/T) = \alpha_F^* + O(1/T)$ and $K_F(\xi/T) = K_F + O(1/T)$, since Perron vectors and C_F are real-analytic. Dividing by the face maximum of Theorem 5.7 at $v = 0$ gives (b).

(c) The proof of Theorem 5.9 uses (10) for the faces in $\mathcal{F}^*$ and the bounds of Theorem 4.2(ii) for the other faces. By (b), (10) holds for M_F^ξ with $\log K_F$ replaced by $\log K_F + \langle \xi, \alpha_F^* \rangle$, so (13) and part (i) of that theorem hold with ℓ_F^ξ. By (a), the bounds of Theorem 4.2(ii) hold for M_F^ξ with κ replaced by $\kappa + \|\xi\|$, so part (ii) holds. Part (iii) and the consequence use only (13) and part (ii). Finally the modes of $\widetilde{P}_{N,\xi}$ are the maximizers of w_ξ, and $\max_n w_\xi(n) = \max_F M_F^\xi$. □

6 The boundary layer at a facet

The local law of Theorem 5.4 holds on compact subsets of the open face. This section follows the law of the occupation times from the inside of a face F to one of its facets $G = F \setminus \{i\}$, where the time spent in i is small. We prove an exact formula for the law near the facet, which is the run expansion with the runs in i pulled out (Lemmas 6.1 and 6.2). It reduces the layer to one statement about a weighted chain on G, a local law with a small complex parameter (Hypothesis L). We prove it when G has one or two states, so the layer theorem (Theorem 6.7) is unconditional for faces with at most three states. For larger faces we deduce the theorem from Hypothesis L, which we state but have not proved. We compute the constants of the layer for every irreducible Q (Proposition 6.4) and check that they match the local law (Proposition 6.5). The near-vertex theorem of Section 8 is the pair-face case of this picture on the lattice (Section 6.7).

6.1 Setting

Let F be an irreducible face with $k = |F| \ge 2$, let $i \in F$, and put $G = F \setminus \{i\}$. We assume that G satisfies (H), so G is irreducible and $\mu(G) > 0$. The start mass μ_i may be zero. For $j \in G$ put

$$c_j^- = q_{ji}, \qquad c_j^+ = q_{ij},$$

the rates into i and out of i. Both vectors are nonnegative and nonzero, since F is irreducible. For $z \in \mathbb{C}$ let

$$Q_G(z) = Q_G + z\, c^- (c^+)^\top, \qquad \mu^{(z)} = \mu_G + z\mu_i c^+, \qquad e^{(z)} = \mathbf{1} + zc^-,$$

and write $Q_G(z) = A(z) - \operatorname{diag} d(z)$ with $A(z)$ the off-diagonal part and $d(z) = q_G - z\, c^- \circ c^+$. A jump $j \to j'$ of $Q_G(z)$ of rate $z c_j^- c_{j'}^+$ stands for a visit to i that is entered from j and left to j'.

Occupation coordinates. Let E_F be the event of Lemma 2.4. On E_F let $s \in (0,\infty)^G$ be the times spent in the states of G, $S = \sum_{j\in G} s_j$, and $x = T - S$ the time spent in i. We write

$$f_F(s,x) = e^{-\langle q_G, s\rangle - q_i x} H_F(s,x), \qquad f_G(s) = e^{-\langle q_G, s\rangle} H_G(s),$$

where H_F and H_G are the sums H of Lemma 2.4 for the faces F and G. By that lemma f_F is the density of the occupation times on E_F in any $k-1$ of the coordinates, and $f_G(s)$ is the same density for G at total time S. In this notation Theorem 5.4 for the face G reads

$$f_G(S\beta) = S^{-(k-2)/2} C_G(\beta) e^{-S J_G(\beta)} \big(1 + O_{\mathcal{K}}(1/S)\big) \tag{14}$$

uniformly for β in a compact $\mathcal{K} \subset \Delta_G^\circ$ and $S \geq 1$, with $C_G(\beta) = \mu_j$ and $J_G = q_j$ when $G = \{j\}$. The profile on the facet is $\beta = s/S \in \Delta_G^\circ$.

The weighted chain. For real $z \geq 0$ the matrix $Q_G(z)$ is Metzler and irreducible. We let $J_G(\beta, z)$ be its rate, the supremum in Proposition 3.3 with $Q_G(z)$ in place of Q_F, and $C_G(\beta, z)$ its local-law constant, which is Definition 5.1 for the off-diagonal part $A(z)$ with the start $\mu^{(z)}$ in place of μ and the end vector $e^{(z)}$ in place of $\mathbf{1}$, namely

$$C_G(\beta, z) = \frac{(\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})}{(2\pi)^{(k-2)/2} \sqrt{\det \Sigma_\beta(z)}}.$$

So $J_G(\beta, 0) = J_G(\beta)$ and $C_G(\beta, 0) = C_G(\beta)$.

The facet constants. Let $r_\beta = r_\beta(0)$ and $l_\beta = \beta / r_\beta$ be the vectors of Definition 5.1 for the face G, so $l_\beta \cdot r_\beta = 1$. Put

$$a_\beta = c^- \cdot l_\beta, \qquad b_\beta = c^+ \cdot r_\beta, \qquad \omega_\beta = a_\beta b_\beta, \qquad \pi^{\mathrm{in}}_{\beta,j} = \frac{c_j^- l_{\beta,j}}{a_\beta}, \qquad \pi^{\mathrm{out}}_{\beta,j} = \frac{c_j^+ r_{\beta,j}}{b_\beta}.$$

Only ω_β is intrinsic, since a_β and b_β move in opposite ways when r_β is scaled. The law π^{in}_β is the law of the state from which the tilted chain enters i, and π^{out}_β the law of the state into which it leaves i. Let $\mathcal{L}_\beta$ be the Laplacian on G with the conductances $\kappa_{jj'} = l_{\beta,j} q_{jj'} r_{\beta,j'} + l_{\beta,j'} q_{j'j} r_{\beta,j}$ for $j \neq j'$, which are those of the additive reversibilization of the tilted generator $\widehat{Q}_\beta$. Its kernel is $\mathbb{R}\mathbf{1}$, since G is irreducible. Let $\mathcal{L}_\beta^+$ be its pseudo-inverse and

$$\gamma_\beta = q_i + \frac{\omega_\beta}{2} (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta)^\top \mathcal{L}_\beta^+ (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta). \tag{15}$$

The quadratic form is the effective resistance of the network κ between the laws π^{in}_β and π^{out}_β. So $\gamma_\beta \geq q_i$, with equality exactly when the two laws agree. Finally put

$$\zeta = 2\sqrt{\omega_\beta S x}, \qquad z_* = \sqrt{\frac{x}{\omega_\beta S}}, \qquad \mathbf{B}_\beta(x, S) = e^{-\gamma_\beta x} \sqrt{\frac{\omega_\beta S}{x}}\, I_1(\zeta),$$

with the modified Bessel function I_1. We call $\mathbf{B}_\beta$ the Bessel kernel.

6.2 The exact representation

Lemma 6.1 (the runs in i pulled out)**.** *For $z \in \mathbb{C}$ and $s \in (0,\infty)^G$ let*

$$f_G^{(z)}(s) = e^{-\langle d(z),s\rangle} \sum_{m \in \mathbb{Z}_{\geq 1}^G} W_z(m) \prod_{j \in G} \frac{s_j^{m_j-1}}{(m_j-1)!},$$

$$W_z(m) = \sum_w \mu_{w_1}^{(z)} \prod_{l<|w|} A(z)_{w_l w_{l+1}}\, e_{w_{|w|}}^{(z)},$$

the inner sum over the words w in G with no two equal neighbours and m_j letters j. Then $z \mapsto f_G^{(z)}(s)$ is entire, $f_G^{(0)} = f_G$, and for $x > 0$ and every $\rho > 0$

$$f_F(s,x) = e^{-q_i x} \sum_{K \geq 1} \frac{x^{K-1}}{(K-1)!}\, [z^K] f_G^{(z)}(s) \tag{16}$$

$$= e^{-q_i x}\, \frac{1}{2\pi \mathrm{i}} \oint_{|z|=\rho} e^{x/z} f_G^{(z)}(s)\, \frac{dz}{z^2}. \tag{17}$$

So $f_G^{(z)}$ is the killed density on the facet of the chain in which each visit to i is replaced by an instantaneous jump $j \to j'$ of rate $z c_j^- c_{j'}^+$, and z counts the visits. The K visits have total length x, with density $q_i^K x^{K-1} e^{-q_i x}/(K-1)!$, and each return carries a factor $c_{j'}^+/q_i$. So the powers of q_i cancel.

Proof. Fix a word c in F with no equal neighbours, every letter of F, and $K = m_i(c) \geq 1$ letters i. Its term in $H_F(s,x)$ is $\mu_{c_1} \prod q_{c_l c_{l+1}}$ times $x^{K-1}/(K-1)!$ times $\prod_{j \in G} s_j^{m_j-1}/(m_j-1)!$. The letters i are isolated in c. We delete them and mark the places where they were. The result is a marked word c' in G, in which two equal neighbours occur exactly where c had $j\,i\,j$. A mark between letters j and j' carries $q_{ji} q_{ij'} = c_j^- c_{j'}^+$. A mark before the first letter j' carries $\mu_i c_{j'}^+$, and a mark after the last letter j carries c_j^-. Two neighbours without a mark are different and carry $q_{jj'}$. So the weight of c is the product of these weights. Between two different letters of c' there is at most one mark, and between two equal letters exactly one.

We group the marked words by their reduced word w, obtained by merging each block of equal letters of c' into one letter, and by the numbers $n_1, n_2, \ldots$ of marks inside the blocks. If j forms ϱ blocks with n inner marks in all, it has $\varrho + n$ letters in c', and there are $\binom{n+\varrho-1}{\varrho-1}$ ways to spread the marks. Since $\binom{n+\varrho-1}{\varrho-1}/(n+\varrho-1)! = 1/(n!(\varrho-1)!)$,

$$\sum_{n \geq 0} \binom{n+\varrho-1}{\varrho-1} \frac{y^n s^{n+\varrho-1}}{(n+\varrho-1)!} = e^{ys} \frac{s^{\varrho-1}}{(\varrho-1)!}.$$

With $y = z c_j^- c_j^+$, one power of z for each mark, the inner marks give the factor $e^{z c_j^- c_j^+ s_j}$, which is the diagonal part of $d(z)$. The marks between different letters give $A(z)$, and the first and last marks give $\mu^{(z)}$ and $e^{(z)}$. Collecting the powers of z gives (16). The entries of $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ are at most $q_{\max}(1+|z|q_{\max})$, $1+|z|q_{\max}$ and $1+|z|q_{\max}$ in modulus. So the bound $W(m) \leq k^M q_{\max}^{M-1}$ of Section 2 holds for W_z with $q_{\max}$ replaced by $q_{\max}(1+|z|q_{\max})$, up to a factor $(1+|z|q_{\max})^2$, and the series for $f_G^{(z)}(s)$ converges locally uniformly in z. Hence it is entire. For $z = 0$ no word carries a mark, and $f_G^{(0)}$ is the sum of Lemma 2.4 for G. Finally we expand $e^{x/z} = \sum_{n \geq 0} x^n z^{-n}/n!$ and integrate term by term on the circle. This gives $\sum_{n \geq 0} (x^n/n!)[z^{n+1}] f_G^{(z)}(s)$, which is (16). □

The discrete law has the same structure. The only change is that the visits to i come in blocks of steps, whose lengths form a composition of n_i.

Lemma 6.2 (the discrete form)**.** *Let $hq_{\max} \le 1$, $\operatorname{supp} n = F$, $N' = N - n_i$, and let $\hat{n}$ be n restricted to G. For $z \in \mathbb{C}$ let*

$$P^{(z)}_{N'}(\hat{n}) = \sum_x \mu^{h,(z)}_{x_1} \prod_{t<N'} \big(I + hQ_G + zh^2 c^-(c^+)^\top\big)_{x_t x_{t+1}} e^{h,(z)}_{x_{N'}},$$

the sum over the words $x_1 \cdots x_{N'}$ in G with composition $\hat{n}$, where $\mu^{h,(z)} = \mu_G + zh\mu_i c^+$ and $e^{h,(z)} = \mathbf{1} + zhc^-$. Then $P^{(0)}_{N'}(\hat{n})$ is the probability of the composition $\hat{n}$ for the chain run for N' steps, and

$$P_N(n) = \sum_{K \ge 1} \binom{n_i - 1}{K - 1} (1 - hq_i)^{n_i - K}\, [z^K] P^{(z)}_{N'}(\hat{n}).$$

Proof. We delete the maximal blocks of i from a path with composition n. What is left is a path $x_1 \cdots x_{N'}$ in G with composition $\hat{n}$. A block of length ℓ between $x_t = j$ and $x_{t+1} = j'$ replaces the factor $(I + hQ)_{jj'}$ by $hq_{ji}(1 - hq_i)^{\ell - 1} hq_{ij'}$. A block at the start replaces μ_{x_1} by $\mu_i (1 - hq_i)^{\ell - 1} hq_{ix_1}$, and a block at the end adds the factor $hq_{x_{N'} i}(1 - hq_i)^{\ell - 1}$. We count the blocks by z. If there are K blocks, the factors $(1 - hq_i)^{\ell - 1}$ give $(1 - hq_i)^{n_i - K}$, and the block lengths form one of the $\binom{n_i - 1}{K - 1}$ compositions of n_i into K positive parts. □

6.3 Pair faces

For $k = 2$ the facet is one state and the weighted chain has no words to sum. The contour formula then gives the law in closed form.

Proposition 6.3 (pair faces)**.** *Let $F = \{j, i\}$ with $q_{ji}, q_{ij} > 0$ and $\mu_j > 0$, and put $\omega = q_{ji} q_{ij}$ and $\zeta = 2\sqrt{\omega s x}$. For all $s, x > 0$*

$$f_F(s, x) = \mu_j e^{-q_j s - q_i x} \Big[\sqrt{\frac{\omega s}{x}}\, I_1(\zeta) + \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij} \Big) I_0(\zeta) + \frac{\mu_i}{\mu_j} \sqrt{\frac{\omega x}{s}}\, I_1(\zeta) \Big].$$

Here $\omega_\beta = \omega$, $\gamma_\beta = q_i$ and $f_G(s) = \mu_j e^{-q_j s}$, and for all $s, x > 0$

$$0 \le \frac{f_F(s, x)}{f_G(s) \mathbf{B}_\beta(x, s)} - 1 \le \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij} \Big) \Big(\frac{1}{\omega s} + \sqrt{\frac{x}{\omega s}} \Big) + \frac{\mu_i}{\mu_j}\, \frac{x}{s}.$$

This is the facet version of the formula of Pedler for the occupation times of a two-state chain [47], and Kovchegov, Meredith and Nir give Bessel densities of the same kind [46]. What is new is the reading as a layer with an explicit two-sided bound. For $x \le T/2$ the bound is $O(1/T + \sqrt{x/T})$, and the term $\sqrt{x/T}$ is attained, since the middle term of the formula has exactly this size.

Proof. For $G = \{j\}$ the only word is j, so $f^{(z)}_G(s) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji}) e^{-(q_j - z\omega)s}$. With $z = z_* t$ and $z_* = \sqrt{x/(\omega s)}$ we have $x/z + \omega s z = \frac{\zeta}{2}(t + 1/t)$, and the generating function $e^{(\zeta/2)(t + 1/t)} = \sum_{n \in \mathbb{Z}} I_n(\zeta) t^n$ [27, Eq. 10.35.1] gives

$$\frac{1}{2\pi \mathrm{i}} \oint e^{x/z + \omega s z} z^{n-2}\, dz = z_*^{n-1} I_{n-1}(\zeta) \qquad (n \in \mathbb{Z}). \tag{18}$$

We put this into (17) for $n = 0, 1, 2$ and use $I_{-1} = I_1$. This gives the formula. Each term is nonnegative, which gives the lower bound. For the upper bound we divide the last two terms by the first. This gives $(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}) z_* I_0(\zeta)/I_1(\zeta) + \frac{\mu_i}{\mu_j} x/s$. The lower bound $I_1(\zeta)/I_0(\zeta) \geq \zeta/(1+\sqrt{1+\zeta^2})$ of Amos [45] gives $I_0/I_1 \leq (1+\sqrt{1+\zeta^2})/\zeta \leq 1 + 2/\zeta$, and $z_*(1+2/\zeta) = \sqrt{x/(\omega s)} + 1/(\omega s)$. For a pair $\beta = (1)$, the vectors of the facet are $r_\beta = l_\beta = 1$, so $\omega_\beta = \omega$ and $\pi^{\mathrm{in}}_\beta = \pi^{\mathrm{out}}_\beta$, which gives $\gamma_\beta = q_i$. □

6.4 The constants for every irreducible chain

The rate on F is homogeneous of degree one in the occupation times, and we write $J_F(s,x) = T J_F((s,x)/T)$. For $u \in (0,\infty)^G$ put

$$\Psi_\beta(u) = \sum_{j \in G} \beta_j \Big(q_j - \sum_{j' \in G \setminus \{j\}} q_{jj'} \frac{u_{j'}}{u_j} \Big), \qquad P_\beta(u) = (c^+ \cdot u) \sum_{j \in G} \frac{c_j^- \beta_j}{u_j}.$$

Since $\sum_j \beta_j (-(Q_G(z)u)_j/u_j) = \Psi_\beta(u) - z P_\beta(u)$, the rate of the weighted chain is $J_G(\beta, z) = \sup_u [\Psi_\beta(u) - z P_\beta(u)]$ for real $z \geq 0$.

Proposition 6.4 (the facet expansion of the rate)**.** *Let F be irreducible, $i \in F$, and $G = F \setminus \{i\}$ irreducible.*

(i) For $s = S\beta$ with $\beta \in \Delta_G^\circ$ and $x > 0$,

$$J_F(s,x) = q_i x + \sup_{z>0} \big[S\, J_G(\beta, z) - x/z \big].$$

(ii) The map $z \mapsto J_G(\beta, z)$ is real-analytic near $z = 0$, jointly with β, and

$$J_G(\beta, 0) = J_G(\beta), \qquad \partial_z J_G(\beta, 0) = -\omega_\beta, \qquad \partial_z^2 J_G(\beta, 0) = 2\omega_\beta(\gamma_\beta - q_i).$$

(iii) There are $\delta > 0$ and C such that, uniformly for β in a compact subset of Δ_G° and $0 < x \leq \delta S$,

$$\Big| J_F(s,x) - \big(S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x} + (\gamma_\beta - q_i) x \big) \Big| \leq C\, x^{3/2} S^{-1/2}.$$

(iv) If $\nu_j q_{jj'} = \nu_{j'} q_{j'j}$ on F for some $\nu \in (0,\infty)^F$, then $\pi^{\mathrm{in}}_\beta = \pi^{\mathrm{out}}_\beta$, $\gamma_\beta = q_i$, $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ for all real $z \geq 0$, $\omega_\beta = \big(\sum_{j \in G} \sqrt{q_{ij} q_{ji} \beta_j} \big)^2$, and for all $s = S\beta$ and $x > 0$

$$J_F(s,x) = S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}.$$

So in the visit fraction α_i the rate has a square-root singularity $-2\sqrt{\omega_\beta \alpha_i}$ at the facet, and the time in i is charged at the rate γ_β, not q_i.

Proof. (i) In the supremum that defines $J_F(s,x)$ we write $u = (u_G, \epsilon)$ with $\epsilon = u_i$. The terms with ϵ are $-\epsilon \sum_j s_j c_j^- / u_j - (x/\epsilon)(c^+ \cdot u_G)$, and both coefficients are positive. Their supremum over $\epsilon > 0$ is $-2\sqrt{x S\, P_\beta(u_G)}$. So $J_F(s,x) = q_i x + \sup_u [S\Psi_\beta(u) - 2\sqrt{x S P_\beta(u)}]$. Now $2\sqrt{xSP} = \inf_{z>0} [SzP + x/z]$, and we exchange the two suprema.

(ii) Write $u = e^\psi$. For real $z \geq 0$ the function $\psi \mapsto \Psi_\beta(e^\psi) - z P_\beta(e^\psi)$ is concave, since $P_\beta(e^\psi)$ is a positive combination of the convex functions $e^{\psi_m - \psi_j}$, and it does not change when a constant

is added to ψ. At $z=0$ its maximizer is $\psi_0 = \log r_\beta$, unique up to a constant (Proposition 3.3(b)). The Hessian of $\sum_{j\neq j'} \beta_j q_{jj'} e^{\psi_{j'}-\psi_j}$ at ψ_0 is the Laplacian with the conductances $\kappa_{jj'}$, because $\beta_j q_{jj'} r_{\beta,j'}/r_{\beta,j} = l_{\beta,j} q_{jj'} r_{\beta,j'}$. So the Hessian of Ψ_β at ψ_0 is $-\mathcal{L}_\beta$, which is negative definite on the complement of the constants. By the implicit function theorem on that complement, the critical point $\psi(z)$ is real-analytic in (β, z) near $(\beta, 0)$, and for real $z \geq 0$ it is the maximizer by concavity. So $J_G(\beta, z) = \Psi_\beta(\psi(z)) - zP_\beta(\psi(z))$ is real-analytic. The envelope theorem gives $\partial_z J_G = -P_\beta(\psi(z))$, so $\partial_z J_G(\beta, 0) = -(c^+ \cdot r_\beta)(c^- \cdot l_\beta) = -\omega_\beta$. Differentiating the critical equation gives $-\mathcal{L}_\beta \psi'(0) = \nabla P$, with ∇P the gradient of $P_\beta(e^\psi)$ at ψ_0. Its entries are

$$\partial_{\psi_m} P = c^+_m r_{\beta,m}\,(c^- \cdot l_\beta) - (c^+ \cdot r_\beta)\, c^-_m l_{\beta,m} = \omega_\beta(\pi^{\text{out}}_{\beta,m} - \pi^{\text{in}}_{\beta,m}),$$

which sum to zero. So $\partial_z^2 J_G(\beta, 0) = -\nabla P \cdot \psi'(0) = \nabla P^\top \mathcal{L}_\beta^+ \nabla P$, and this is $2\omega_\beta(\gamma_\beta - q_i)$ by (15).

(iii) By the proof of (i), $J_F(s, x) - q_i x = S \sup_\psi [\Psi_\beta(\psi) - 2\varepsilon P_\beta(\psi)^{1/2}]$ with $\varepsilon = \sqrt{x/S}$. The maximizer lies in a bounded set of ψ modulo constants, uniformly for β in a compact set, as in the proof of Proposition 3.3(b). Comparing its value with the value at ψ_0 gives $\Psi_\beta(\psi_0) - \Psi_\beta(\psi^*) \leq 2\varepsilon \sup P_\beta^{1/2}$ on that set, and the strict concavity near ψ_0 then gives $\psi^* - \psi_0 = O(\varepsilon^{1/2})$. The critical equation $\nabla\Psi_\beta(\psi^*) = \varepsilon \nabla P_\beta(\psi^*)/\sqrt{P_\beta(\psi^*)}$ has a right side of size $O(\varepsilon)$, and $\nabla^2 \Psi_\beta$ is invertible near ψ_0 on the complement of the constants, so $\psi^* - \psi_0 = O(\varepsilon)$. Now we expand to second order around ψ_0. With $g = \nabla(P^{1/2}) = \nabla P/(2\sqrt{\omega_\beta})$, the maximizer is $\psi^* = \psi_0 - 2\varepsilon \mathcal{L}_\beta^+ g + O(\varepsilon^2)$, and the value is $J_G(\beta) - 2\varepsilon\sqrt{\omega_\beta} + 2\varepsilon^2 g^\top \mathcal{L}_\beta^+ g + O(\varepsilon^3)$. Here $2g^\top \mathcal{L}_\beta^+ g = \nabla P^\top \mathcal{L}_\beta^+ \nabla P/(2\omega_\beta) = \gamma_\beta - q_i$ by (ii). Multiplying by S gives (iii), since $S\varepsilon^3 = x^{3/2} S^{-1/2}$.

(iv) By Proposition 5.2(b) for the face G we may take $r_{\beta,j} = \sqrt{\beta_j/\nu_j}$ and $l_{\beta,j} = \sqrt{\beta_j \nu_j}$. Then $c^-_j l_{\beta,j} = q_{ji}\sqrt{\beta_j \nu_j} = \nu_i q_{ij} \sqrt{\beta_j/\nu_j} = \nu_i c^+_j r_{\beta,j}$, so $\pi^{\text{in}}_\beta = \pi^{\text{out}}_\beta$ and $\gamma_\beta = q_i$. More is true. For every real $z \geq 0$ the matrix $Q_G(z)$ is reversible with respect to ν_G, since $\nu_j c^-_j c^+_{j'} = \nu_i q_{ij} q_{ij'}$ is symmetric in j, j'. So by Proposition 5.2(b) its optimal vector is r_β for every z, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Next $\omega_\beta = \nu_i \big(\sum_j q_{ij}\sqrt{\beta_j/\nu_j}\big)^2 = \big(\sum_j \sqrt{q_{ij} q_{ji} \beta_j}\big)^2$, using $q_{ji} = q_{ij}\nu_i/\nu_j$. Finally (i) gives $\sup_{z>0}[-Sz\omega_\beta - x/z] = -2\sqrt{\omega_\beta S x}$. □

For $1/T \ll x \ll T$ the Bessel kernel is $(4\pi)^{-1/2}(\omega_\beta S)^{1/4} x^{-3/4} \exp(2\sqrt{\omega_\beta S x} - \gamma_\beta x)(1 + O(1/\zeta))$, by $I_1(\zeta) = e^\zeta (2\pi\zeta)^{-1/2}(1 + O(1/\zeta))$ [27, Eq. 10.40.1]. So Proposition 6.4(iii) says that $f_G(s)\mathbf{B}_\beta(x, S)$ and the local law of F, which in occupation coordinates is $T^{-(k-1)/2} C_F(\alpha) e^{-TJ_F(\alpha)}$, have the same exponent up to $O(x^{3/2} S^{-1/2})$. For their powers of x and their constants to agree as well we need $C_F((1-\epsilon)\beta, \epsilon) \sim (4\pi)^{-1/2} \omega_\beta^{1/4} \epsilon^{-3/4} C_G(\beta)$ as $\epsilon \to 0$. The next proposition proves this for reversible Q.

6.5 Prefactor matching

Proposition 6.5 (prefactor matching)**.** *Let Q be reversible on F with respect to ν, and let G satisfy (H), with $|G| \geq 1$. Uniformly for β in a compact subset of Δ_G°, as $\epsilon \to 0$,*

$$C_F((1-\epsilon)\beta, \epsilon) = (4\pi)^{-1/2}\, \omega_\beta^{1/4}\, \epsilon^{-3/4}\, C_G(\beta)\big(1 + O(\sqrt{\epsilon})\big),$$

with $C_G(\beta) = \mu_j$ when $G = \{j\}$.

Proof. We use Proposition 5.2(b) at $\alpha = ((1-\epsilon)\beta, \epsilon)$. On G the vectors $r_\alpha = \sqrt{\alpha/\nu}$ and $l_\alpha = \sqrt{\alpha\nu}$ are $\sqrt{1-\epsilon}$ times r_β and l_β, and at i they are $O(\sqrt{\epsilon})$. So $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = (\mu_G \cdot r_\beta)(l_\beta \cdot \mathbf{1})(1 + O(\sqrt{\epsilon}))$, where $\mu_G \cdot r_\beta > 0$ by (H). Next $\prod_{j\in F} \alpha_j = \epsilon(1-\epsilon)^{k-1} \prod_{j\in G} \beta_j$. The conductances are $c_{jj'}(\alpha) =$

$(1-\epsilon)c_{jj'}(\beta)$ on G, and at i they are $c_{ij}(\alpha)=\sqrt{\epsilon(1-\epsilon)}\sqrt{q_{ij}q_{ji}\beta_j}$. A spanning tree of F with m edges at i has weight $O(\epsilon^{m/2})$. The trees with one edge at i are a spanning tree of G and one edge $\{i,j\}$. So $\mathrm{tree}(c(\alpha))=\sqrt{\epsilon}\,\sqrt{\omega_\beta}\,\mathrm{tree}(c_G(\beta))(1+O(\sqrt{\epsilon}))$, by $\sum_j\sqrt{q_{ij}q_{ji}\beta_j}=\sqrt{\omega_\beta}$ from Proposition 6.4(iv). We put these into the formula for C_F and divide by the same formula for C_G, which has one factor $(4\pi)^{1/2}$ fewer. The powers of ϵ are $\epsilon^{1/4}$ from the tree and ϵ^{-1} from the product, which gives $\epsilon^{-3/4}$. For $G=\{j\}$ we use $C_{\{j\}}=\mu_j$ and tree $=1$ on one vertex. □

The first amplitude correction. Put $c_1(\beta)=\partial_z\log C_G(\beta,z)$ at $z=0$. For reversible Q, Proposition 5.2(b) applied to $Q_G(z)$, whose vector $r_\beta(z)=r_\beta$ does not move with z, gives

$$c_1(\beta)=\frac{\mu_i b_\beta}{\mu_G\cdot r_\beta}+\frac{a_\beta}{l_\beta\cdot\mathbf{1}}+\frac{\omega_\beta}{2}\sum_{\{j,j'\}\subseteq G}R_{jj'}\pi_{\beta,j}\pi_{\beta,j'},\tag{19}$$

with $\pi_\beta=\pi_\beta^{\mathrm{in}}=\pi_\beta^{\mathrm{out}}$ and $R_{jj'}$ the effective resistance for the conductances $l_{\beta,j}q_{jj'}r_{\beta,j'}$. The three terms come from a start in i, an end in i, and the rule $\partial\log\mathrm{tree}(c)/\partial c_e=R_e$ of Kirchhoff, since the conductance of $\{j,j'\}$ grows by $z\omega_\beta\pi_{\beta,j}\pi_{\beta,j'}$.

6.6 The facet theorem

For $|G|\ge 2$ the layer needs the local law of the weighted chain $Q_G(z)$, uniformly for complex z near 0. We state it as a hypothesis. We prove it for $|G|=2$ (Lemma 6.6) and not for $|G|\ge 3$.

Hypothesis L (local law of the weighted facet chain)**.** *Let G satisfy* (H) *with $|G|=k-1\ge 2$, let $i\notin G$ with $F=G\cup\{i\}$ irreducible, and let $\mathcal{K}\subset\Delta_G^\circ$ be compact. There are $z_0>0$, C, and functions $J_G(\beta,z)$ and $C_G(\beta,z)$ on $\mathcal{K}\times\{|z|<2z_0\}$, holomorphic in z, jointly continuous, real for real z, equal to the rate and the local-law constant of Section 6.1 for real $0\le z<2z_0$, and with $C_G(\beta,0)=C_G(\beta)>0$, such that uniformly for $\beta\in\mathcal{K}$, $S\ge 1$ and complex $|z|\le z_0$*

$$f_G^{(z)}(S\beta)=S^{-(k-2)/2}\,C_G(\beta,z)\,e^{-S\,J_G(\beta,z)}\,(1+\varepsilon),\qquad |\varepsilon|\le C/S.$$

At $z=0$ this is the local law (14), that is Theorem 5.4 for the face G. For $G=\{j\}$ the same statement holds with $\varepsilon=0$, since $f_G^{(z)}(s)=(\mu_j+z\mu_i q_{ij})(1+zq_{ji})e^{-(q_j-z\omega)s}$ by the proof of Proposition 6.3. For real $0\le z<2z_0$ the weights are nonnegative, and the proof of Theorem 5.4 needs only one change, the end vector $e^{(z)}$ in place of $\mathbf{1}$, which turns the factor $b(z')$ of Lemma A.2 into $(\mu_1^{(z)}+\mu^{(z)\prime\top}X(z')b_1)(e_1^{(z)}+a_1^\top X(z')e^{(z)\prime})$. The open part is complex z. For two-state facets the word sum is a closed Bessel expression, and the complex case follows from the large-argument expansion of I_0 and I_1.

Lemma 6.6 (two-state facets)**.** *Hypothesis L holds when $|G|=2$.*

Proof. Let $G=\{1,2\}$, and write $a_{12}(z)=q_{12}+zc_1^-c_2^+$ and $a_{21}(z)=q_{21}+zc_2^-c_1^+$ for the entries of $A(z)$, $\tau(z)=a_{12}(z)a_{21}(z)$, $p=\mu^{(z)}$ and $e=e^{(z)}$. A word in G with no equal neighbours alternates. The words that start with 1 are $1(21)^n$, with weight $p_1\tau^n e_1$ and $(n+1,n)$ letters, and $(12)^n$, with weight $p_1\tau^{n-1}a_{12}e_2$ and (n,n) letters, for $n\ge 1$. With $\xi=2\sqrt{\tau s_1 s_2}$ the power series of I_1 and I_0 give

$$\sum_{n\ge1}\frac{\tau^n s_1^n s_2^{n-1}}{n!\,(n-1)!}=\sqrt{\frac{\tau s_1}{s_2}}\,I_1(\xi),\qquad \sum_{n\ge1}\frac{(\tau s_1 s_2)^{n-1}}{(n-1)!^2}=I_0(\xi).$$

We do the same for the words that start with 2 and obtain

$$f_G^{(z)}(s) = e^{-\langle d(z),s\rangle}\Big[\Big(p_1e_1\sqrt{\frac{\tau s_1}{s_2}} + p_2e_2\sqrt{\frac{\tau s_2}{s_1}}\Big)I_1(\xi) + \big(p_1a_{12}e_2 + p_2a_{21}e_1\big)I_0(\xi)\Big].$$

Both sides are entire in z, so the branch of the square root does not matter. Since G is irreducible, $\tau(0) = q_{12}q_{21} > 0$. We choose z_0 so that $|\tau(z) - \tau(0)| \le \tau(0)/2$ for $|z| \le 2z_0$ and take the principal branch of $\sqrt{\tau}$. Then $\xi = 2S\sqrt{\beta_1\beta_2\tau}$ at $s = S\beta$ satisfies $|\arg\xi| \le \pi/12$ and $|\xi| \ge cS$, uniformly on $\mathcal{K}$. The expansion $I_\nu(\xi) = e^\xi(2\pi\xi)^{-1/2}(1 + O(1/|\xi|))$ holds for $\nu = 0, 1$, uniformly in the sector $|\arg\xi| \le \pi/4$ and $|\xi| \ge 1$ [27, Eq. 10.40.1]. Since $(2\pi\xi)^{-1/2} = S^{-1/2}(4\pi)^{-1/2}(\tau\beta_1\beta_2)^{-1/4}$, this gives Hypothesis L with

$$J_G(\beta, z) = \langle d(z), \beta\rangle - 2\sqrt{\beta_1\beta_2\tau}, \qquad C_G(\beta, z) = \frac{U + V}{\sqrt{4\pi}\,(\tau\beta_1\beta_2)^{1/4}},$$

where $U = p_1e_1\sqrt{\tau\beta_1/\beta_2} + p_2e_2\sqrt{\tau\beta_2/\beta_1}$ and $V = p_1a_{12}e_2 + p_2a_{21}e_1$, and with $\varepsilon = (U\varepsilon_1 + V\varepsilon_0)/(U + V)$, where $|\varepsilon_\nu| \le C/S$ are the errors of the two expansions. Both functions are holomorphic in z, jointly continuous and real for real z. At $z = 0$ we have $p = \mu_G$ and $e = \mathbf{1}$, so $U + V > 0$ because $\mu(G) > 0$. After shrinking z_0, $|U + V|$ is bounded below and $|U| + |V|$ above on $\mathcal{K} \times \{|z| \le 2z_0\}$, which gives $|\varepsilon| \le C/S$ once $cS \ge 1$. For $1 \le S \le 1/c$ the error ε is continuous on a compact set, so it is bounded, and we enlarge C. For real $z \ge 0$ the rate of the two-state matrix $Q_G(z)$ is $\sup_u[\beta_1(d_1 - a_{12}u_2/u_1) + \beta_2(d_2 - a_{21}u_1/u_2)] = \langle d, \beta\rangle - 2\sqrt{\beta_1\beta_2\tau}$, by the inequality of the arithmetic and geometric means. The constant agrees with that of Section 6.1, because for real $z \ge 0$ both are the limit of $S^{1/2}e^{SJ_G(\beta,z)}f_G^{(z)}(S\beta)$ by the paragraph above. □

When Hypothesis L holds, so always for $k \le 3$, we define for $\beta \in \mathcal{K}$ and $x, S > 0$ the generalized kernel

$$\mathbf{K}_\beta(x, S) = e^{-q_ix}\,\frac{1}{2\pi\mathrm{i}}\oint_{|z|=z_*} e^{x/z}\,\frac{C_G(\beta, z)}{C_G(\beta, 0)}\,e^{S(J_G(\beta)-J_G(\beta,z))}\,\frac{dz}{z^2},$$

for $z_* < 2z_0$. The integral does not depend on the radius below $2z_0$.

Theorem 6.7 (the facet layer)**.** *Let F be irreducible with $k = |F| \ge 2$, let $i \in F$, let $G = F \setminus \{i\}$ satisfy* (H)*, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. If $k \le 3$ we assume nothing more. If $k \ge 4$, assume that Hypothesis L holds for G, i and $\mathcal{K}$. Then there are $\delta > 0$, T_0 and C such that the following hold uniformly for $\beta \in \mathcal{K}$, $T \ge T_0$ and $0 < x \le \delta T$, with $S = T - x$ and $s = S\beta$.*

(a) $f_F(s, x) = f_G(s)\,\mathbf{K}_\beta(x, S)\,(1 + E)$ *with* $|E| \le C/T$.

(b) $f_F(s, x) = f_G(s)\,\mathbf{B}_\beta(x, S)\,e^E$ *with* $|E| \le C\big(1/T + \sqrt{x/T}\,(1 + x)\big)$.

(c) If Q is reversible on F, then $\gamma_\beta = q_i$, (b) holds with $|E| \le C(1/T + \sqrt{x/T})$, and

$$f_F(s, x) = f_G(s)\,\mathbf{B}_\beta(x, S)\Big(1 + c_1(\beta)\,z_*\frac{I_0(\zeta)}{I_1(\zeta)} + E'\Big), \qquad |E'| \le C\,\frac{1 + x}{T}.$$

For $k = 2$, (a) holds with $E = 0$, and Proposition 6.3 gives (b) for all $x > 0$ with an explicit bound.

So the theorem is unconditional for every face with at most three states, and for $k \ge 4$ it is conditional on Hypothesis L. In the reversible case the Bessel form has relative error $O(1/T + \sqrt{x/T})$ on the whole layer $x \le \delta T$. When Q is not reversible our bound gives the relative form $f_F = f_G\mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}(1 + x)))$ only for $x \le CT^{1/3}$, and beyond that only the logarithmic form

(b) and the kernel of (a). The cause is the cubic term $S\,\partial_z^3 J_G(\beta,0)z^3/6$ at $z \approx z_*$, of size $x^{3/2}T^{-1/2}$, which the generalized kernel keeps and the Bessel kernel does not. For the non-reversible cycle $q_{12}=q_{23}=q_{31}=2$, $q_{21}=q_{32}=q_{13}=1$ with $G=\{1,2\}$, $\beta=(\frac12,\frac12)$ and $x=T/20$, the ratio $f_F/(f_G\mathbf{B}_\beta)$ is 19.6, 265 and 48268 at $T=800$, 1600 and 3200, so the relative form does fail there.

Proof of (a). Hypothesis L holds with $\varepsilon=0$ for $k=2$, by Lemma 6.6 for $k=3$, and by assumption for $k\ge 4$. Let $k\ge 2$ and take $\delta\le\frac12$, so $S\ge T/2$. Write $\Lambda(z)=S^{-(k-2)/2}C_G(\beta,z)e^{-SJ_G(\beta,z)}$. By Hypothesis L, $f_G(s)=\Lambda(0)(1+O(1/S))$, and by (17), for every $\rho\le z_0$,

$$e^{q_i x}f_F(s,x)=\mathcal{A}+\mathcal{D},\qquad \mathcal{A}=\frac{1}{2\pi\mathrm{i}}\oint_{|z|=\rho}e^{x/z}\Lambda(z)\frac{dz}{z^2},\qquad |\mathcal{D}|\le\frac{C}{S}\,\mathcal{A}^{\mathrm{abs}},$$

where $\mathcal{A}^{\mathrm{abs}}$ is the integral of $|e^{x/z}\Lambda(z)||z|^{-2}$ against $|dz|/(2\pi)$. By definition $\mathbf{K}_\beta(x,S)=e^{-q_i x}\mathcal{A}/\Lambda(0)$. So it suffices to show that $\mathcal{A}>0$ and $\mathcal{A}^{\mathrm{abs}}\le C'\mathcal{A}$. Put

$$\Xi(z)=\frac{C_G(\beta,z)}{C_G(\beta,0)}\,e^{S(J_G(\beta)-J_G(\beta,z)-\omega_\beta z)},$$

so that $\Lambda(z)/\Lambda(0)=e^{S\omega_\beta z}\Xi(z)$. By Proposition 6.4(ii), the holomorphy of C_G, and Cauchy estimates on the compact set $\mathcal{K}\times\{|z|\le\frac32 z_0\}$, the function $R=\log\Xi$ is holomorphic on $|z|\le z_0$ after shrinking z_0, and

$$R(z)=c_1 z-S\omega_\beta(\gamma_\beta-q_i)z^2+O(|z|^2+S|z|^3),\qquad |R''(z)|\le C(1+S).$$

We take δ so small that $z_*\le\sqrt{2\delta/\omega_{\min}}\le z_0/2$, where $\omega_{\min}=\min_{\mathcal{K}}\omega_\beta>0$.

Few visits, $\zeta\le Z_0$ for a fixed large Z_0. Take $\rho=\max(z_*,1/S)$, which is at most C/S. On the circle $|x/z|\le Z_0/2$, $|S\omega_\beta z|\le C$ and $|\Xi(z)-1|\le C/S$. Let $\mathcal{A}_0$ be $\mathcal{A}$ with $\Xi=1$. By (18) it equals $\Lambda(0)\sqrt{\omega_\beta S/x}\,I_1(\zeta)$, which is at least $\Lambda(0)\omega_\beta S$, because $\sqrt{\omega S/x}\,I_1(\zeta)=\omega S\sum_{n\ge0}(\zeta/2)^{2n}/(n!(n+1)!)$. Also $|\mathcal{A}-\mathcal{A}_0|\le C\Lambda(0)\rho^{-1}/S\le C\Lambda(0)$ and $\mathcal{A}^{\mathrm{abs}}\le C\Lambda(0)\rho^{-1}\le C\Lambda(0)S$. So $\mathcal{A}\ge\Lambda(0)(\omega_\beta S-C)>0$ and $\mathcal{A}^{\mathrm{abs}}\le C'\mathcal{A}$ for S large.

Many visits, $\zeta\ge Z_0$. Let $\lambda(z)=x/z+S\omega_\beta z+R(z)$, so that $e^{x/z}\Lambda(z)=\Lambda(0)e^{\lambda(z)}$. It is real on the real axis, and $\lambda'(z)=-x/z^2+S\omega_\beta+O(1+S|z|)$. By the implicit function theorem it has a real critical point $\rho_s=z_*(1+O(\sqrt{x/S}+1/\zeta))$, and we take $\rho=\rho_s$. For $z=\rho e^{\mathrm{i}\phi}$,

$$\operatorname{Re}\lambda(z)-\lambda(\rho)=(x/\rho+S\omega_\beta\rho)(\cos\phi-1)+\operatorname{Re}[R(z)-R(\rho)].$$

Since R is real on the real axis, $\operatorname{Re}[R'(\rho)(z-\rho)]=R'(\rho)\rho(\cos\phi-1)$, and $|R''|\le C(1+S)$ bounds the rest by $2C(1+S)\rho^2(1-\cos\phi)$. Also $|R'(\rho)|\rho\le C(\rho+S\rho^2)$. Now $S\rho^2\le Cx$, $x/\rho+S\omega_\beta\rho\ge\zeta$, and $x/\zeta\le C\sqrt{\delta}$. So for δ small $\operatorname{Re}\lambda(z)-\lambda(\rho)\le-\frac{\zeta}{2}(1-\cos\phi)$, and

$$\mathcal{A}^{\mathrm{abs}}\le\Lambda(0)\rho^{-1}e^{\lambda(\rho)}\frac{1}{2\pi}\int_{-\pi}^{\pi}e^{-\frac{\zeta}{2}(1-\cos\phi)}d\phi\le C\Lambda(0)\rho^{-1}e^{\lambda(\rho)}\zeta^{-1/2}.$$

For the lower bound we write $\mathcal{A}=\Lambda(0)\rho^{-1}\frac{1}{2\pi}\int e^{\zeta\ell(\phi)}e^{-\mathrm{i}\phi}d\phi$ with $\ell(\phi)=\lambda(\rho e^{\mathrm{i}\phi})/\zeta$. Here $\ell(0)$ is real, $\ell'(0)=0$, $-\ell''(0)=\rho^2\lambda''(\rho)/\zeta$ is real and lies in $[\frac12,2]$, the derivatives of ℓ up to order four are bounded uniformly in all parameters, and the gap above holds. The parameters (β,x,S) do not range over a compact set, so we use the uniform form of Lemma B.1 in Remark B.2, with $\sigma_{\min}=\frac12$ and the gap $\frac12(1-\cos\varepsilon)$. With $m=1$, $\Psi=\ell$, $a=e^{-\mathrm{i}\phi}$ and ζ in place of T it gives $\mathcal{A}=\Lambda(0)\rho^{-1}e^{\lambda(\rho)}(2\pi\zeta(-\ell''(0)))^{-1/2}(1+O(1/\zeta))$, which is positive and at least $c\,\mathcal{A}^{\mathrm{abs}}$. □

Proof of (b) *and* (c). We use two facts on Bessel functions besides (18). For integers $n \geq 1$, $I_n(\zeta) = \frac{1}{\pi}\int_0^\pi e^{\zeta\cos t}\cos(nt)\,dt$ [27, Eq. 10.32.3]. So $I_n \leq I_1$ for all $\zeta > 0$, and $|1-\cos nt-(1-\cos t)| \leq n^2t^2/2$ and a Laplace estimate give $|I_1(\zeta) - I_n(\zeta)| \leq Cn^2 I_1(\zeta)/\zeta$ for $\zeta \geq 1$. Also $I_0/I_1 \leq 1 + 2/\zeta$, as in the proof of Proposition 6.3.

The reversible case. By Proposition 6.4(iv) and holomorphy, $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ on the disc, so $\Xi(z) = C_G(\beta, z)/C_G(\beta, 0) = 1 + c_1 z + \sum_{n\geq 2} c_n z^n$ with $|c_n| \leq Cz_0^{-n}$. By (18) the term $c_n z^n$ contributes $c_n z_*^n I_{n-1}(\zeta)/I_1(\zeta)$ relative to the main term $z_*^{-1}I_1(\zeta)$. For $n \geq 2$ this is at most $|c_n| z_*^n$, and the sum over $n \geq 2$ is $O(z_*^2) = O(x/T)$, since $z_* \leq z_0/2$. The term $n = 1$ is $c_1 z_* I_0(\zeta)/I_1(\zeta) = O(z_* + 1/(\omega_\beta S)) = O(\sqrt{x/T} + 1/T)$. With (a) this proves both claims of (c), for all $0 < x \leq \delta T$.

The general case, $x \leq 1$. Write $\Xi(z) = e^{-S\omega_\beta(\gamma_\beta - q_i)z^2}(1 + c_1 z + R_2(z))$ with $|R_2(z)| \leq C(|z|^2 + S|z|^3)$. On $|z| = z_*$ the Gaussian factor has modulus at most $e^{(\gamma_\beta - q_i)x} \leq C$. We expand it in powers of z^2 and use (18). With $y = (\gamma_\beta - q_i)x$ the term 1 gives the relative factor $\sum_{m\geq 0}(-y)^m I_{2m-1}(\zeta)/(m!\, I_1(\zeta))$. For $\zeta \geq 1$ it differs from e^{-y} by at most $C\sum_{m\geq 1} y^m m^2/(m!\,\zeta) \leq Cx/\zeta = O(\sqrt{x/S})$. For $\zeta < 1$ we have $x < 1/(4\omega_\beta S)$, and $|I_{2m-1}/I_1 - 1| \leq 1$ bounds the difference by $e^y - 1 = O(1/S)$. In the same way the term $c_1 z$ gives $O(z_*(1 + 1/\zeta)) = O(\sqrt{x/S} + 1/S)$. The term R_2 is bounded through the modulus of the integrand, and $\frac{1}{2\pi}\int e^{\zeta\cos\phi}d\phi = I_0(\zeta)$ gives $O\big((z_*^2 + Sz_*^3)I_0(\zeta)/I_1(\zeta)\big) = O(1/S + \sqrt{x/S})$. So $\mathbf{K}_\beta = \mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}))$ for $x \leq 1$.

The general case, $1 \leq x \leq \delta T$. Here $\zeta \geq 2\sqrt{\omega_{\min} S} \geq Z_0$. We use the evaluation of $\mathcal{A}$ in the proof of (a), and the same evaluation with $\Xi = 1$, whose integral is exactly $\Lambda(0)z_*^{-1}I_1(\zeta)$. For $\Xi = 1$ the critical point is z_*, the critical value is ζ, and $-\ell''(0) = 1$. The ratio of the two evaluations is $e^{\lambda(\rho_s)-\zeta}(1 + O(\sqrt{x/S} + 1/\zeta))$, since $\rho_s/z_* - 1$ and the relative change of $-\ell''(0)$ are $O(\sqrt{x/S} + 1/\zeta)$. Now $x/z + S\omega_\beta z$ takes the value ζ at its critical point z_*, and the shift from z_* to ρ_s changes λ only by $O(\lambda''(\rho_s)(\rho_s - z_*)^2) = O(\zeta x/S + 1/\zeta)$. So

$$\lambda(\rho_s) - \zeta = R(z_*) + O(\zeta x/S + 1/\zeta) = -(\gamma_\beta - q_i)x + O\big(\sqrt{x/S}\,(1 + x)\big),$$

because $c_1 z_* = O(\sqrt{x/S})$, $S\omega_\beta z_*^2 = x$, $Sz_*^3 = O(x^{3/2}S^{-1/2})$, $\zeta x/S = O(x^{3/2}S^{-1/2})$ and $1/\zeta \leq C\sqrt{x/S}$ for $x \geq 1$. Hence $\mathbf{K}_\beta = \mathbf{B}_\beta \exp(O(\sqrt{x/T}(1 + x)))$, and with (a) this proves (b). □

At $x = \alpha_i T$ with α_i small but fixed, the saddle point of $\mathbf{K}_\beta$ reproduces the exponent $-TJ_F(\alpha)$ exactly, by the Legendre identity of Proposition 6.4(i). So part (a) and Theorem 5.4 overlap, and together they cover the face up to its facet, unconditionally for $k \leq 3$ and under Hypothesis L for $k \geq 4$.

6.7 The lattice and the near-vertex case

Take $Q = J - dI$, the uniform start and $\ell = 1$ in Section 8, with $F = \{1, 2\}$, $G = \{1\}$, $i = 2$ and $n = (b, a)$. Then $q_{12} = q_{21} = 1$, $p = 1 - hq_1$ and $u = h/p$, and Lemma 6.2 gives exactly

$$P_N(b, a) = \sum_{K\geq 1} \binom{a-1}{K-1}(1 - hq_2)^{a-K}\,[z^K]\,(\mu_1 + zh\mu_2)(1 + zh)(1 - hq_1 + zh^2)^{b-1}.$$

Its main part is $\mu_1 p^{N-1}\sum_K \binom{a-1}{K-1}\binom{b-1}{K}u^{2K}$, which becomes $B_a(bu^2)$ of (23) when $\binom{b-1}{K}$ is replaced by $b^K/K!$. So for $\ell = 1$ the near-vertex law is the discrete pair-face case, with $\omega = 1$, $S = hb$, $x = ha$ and $\gamma = q_2$. In Theorem 8.2 the parameter $w = bu^2 = \omega Sh$ is of order one, Lemma 8.1 pins $B_a(w)/(w\mathcal{I}(aw))$ between $1 - w/2$ and 1, and the continuum layer does not apply. When

$T^2/N \to 0$ the parameter w tends to 0, and the same lemma shows that the lattice adds no separate regime.

Open problems. Hypothesis L for $|G| \geq 3$ would follow from a complex-phase version of Lemma B.1. The torus representation of Lemma A.2 holds for the complex weights of $Q_G(z)$, since they are dominated in modulus by those of $Q_G(|z|)$, and the holomorphic implicit function theorem gives a complex critical point near the real one. What is missing is the Laplace lemma for a complex symmetric Hessian with positive definite real part, written with its constants. A discrete form of Theorem 6.7, with the Laguerre polynomial B_{n_i} in place of the Bessel kernel, would need a discrete Hypothesis L and the saddle bounds of the proof above on the lattice, and we do not state it.

The scripts of Section 11 check the statements of this section on three chains with three states, including the cycle above, for which $\omega_\beta = 5.1820$ and $\gamma_\beta = 3.4179$ against $q_3 = 3$. The generalized kernel matches f_F/f_G to $O(1/T)$ for x from $0.01/T$ to $T/4$ on all three, and no statement is deduced from these numbers.

7 The two-state walk and the complete graph

This section applies the second-order theory to the two simplest chains, the two-state walk and the complete graph, which is the Potts chain. On the complete graph every face ties at first order, and the theory of Section 5 with a few exact facts resolves the tie, also with a start law and a weak field.

7.1 The two-state walk

We use the polynomials $S_{a,b}$ and S_N, the root u_N, the crossing p_N, $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$ of Section 2.1. The first result shows that, at every N, the comparison of a symmetric pair with one of its states is exactly the two-state walk.

Proposition 7.1 (an exact pair identity)**.** *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N-1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u), \qquad u = \frac{ah}{1-qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F, $M_F = M_{\{i\}} S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof. We group the paths by their number of switches. A path with this composition is a word in i, j with m letters i. Each stay has probability $1-qh$ and each switch ah, so a word with s switches has probability $\mu_i(1-qh)^{N-1-s}(ah)^s = \mu_i(1-qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1-qh)^{N-1}$. Sum over these words and divide. This gives $\sum_w u^{s(w)} = S_{m,N-m}(u)$, which is the identity. By [28, Proposition 4.1] the polynomials $S_{m,N-m}$ grow coefficientwise as m moves toward the middle, and $S_{m,N-m} = S_{N-m,m}$. So the balanced m give the largest $S_{m,N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). □

Lean checks the identity for a chain with two states. For $d \geq 3$ the paths with this composition never leave F, so the same computation applies, but this case is not in Lean.

Corollary 7.2 (the two-state walk)**.** *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [28] with $T = N(1-p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1,2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2}\log z_N = L + c_0 + O(1/L)$.*

Proof. Corollary 5.3 with $k = 2$, $\mathrm{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 7.1 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only solution of $M_{\{1,2\}} = M_{\{1\}}$ is $T = z_N$, and by Theorem 5.8(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N. Theorem 5.8(ii) with $\Delta = \delta = 1$ gives the equation. □

The persistent-walk paper [28] proves the next term $1/(8z_N)$ of this equation and bounds its remainder at every N.

7.2 The complete graph

From now on in this section $Q = J - dI$. The chain switches to each other state with probability h (the r of [29]) and keeps its state with probability $p = 1 - (d-1)h$. The following symbols of [29] are local to this section and to Appendix C. Put

$$\sigma = p - h, \qquad u = \frac{h}{p}, \qquad v = \frac{h}{\sigma} = \frac{u}{1-u}, \qquad V_N = \frac{p^{N-1}}{d}.$$

So u is the odds of a switch to one given state against no switch, and V_N is the probability of a vertex Ne_i for the uniform start, that is, $M_{\{i\}}$. For a composition n we write $S_n(u)$ for the sum of w^j over the words with the counts n, where j is the number of switches of the word, and $W_n(j)$ for the number of these words with j switches. For a composition with some zero coordinates, S_n is the polynomial of its positive coordinates, which does not depend on d. By (3), the uniform start has

$$P_N(n) = V_N S_n(u) \qquad (0 < p \leq 1), \tag{20}$$

and so does every start that is constant on $\operatorname{supp} n$, with V_N replaced by $\mu_a p^{N-1}$ for $a \in \operatorname{supp} n$. A composition is *balanced* if its positive coordinates differ by at most 1. For the uniform start all faces of k states have the same face maximum M_F, and all balanced compositions with k positive coordinates have the same probability. For $p \geq 1/d$ this probability is M_F, by Theorem 7.6 below. (The preprint [29] writes $C_{N,k}$ for it.) Put

$$a_k = \frac{1}{2}\log(4\pi) - \frac{k+\frac{1}{2}}{k-1}\log k, \qquad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}} \qquad (k \geq 2),$$

and $D_1 = 1$. Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Corollary 7.3 (the complete graph)**.** *A face F of k states has $\lambda_F = d - k$ and α_F^* uniform, and if $\mu(F) > 0$ then $K_F = \frac{\mu(F)}{k} D_k$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. With the uniform start, the crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k$.*

Proof. Each Q_F is symmetric with row sums $-(d-k)$, and the complete graph on F has k^{k-2} spanning trees by Lemma 5.12. So Corollary 5.3 gives λ_F, α_F^* and

$$K_F = \mu(F) k^{(k+1)/2} k^{(k-2)/2} (4\pi)^{-(k-1)/2} = \frac{\mu(F)}{k} D_k.$$

A vertex has $K = \frac{1}{d}$ and $\lambda = d-1$ for the uniform start, so $b_{FG} = -\frac{1}{k-1}\log(dK_F) = -\frac{1}{k-1}\log D_k = a_k$. The arithmetic that gives b_{FG} from K_F here and in Corollary 7.2 is checked in Lean. □

Remark 7.4 (first-order winners on other graphs). Theorem 4.4 reads the first-order winners of any chain off the exit rates of its admissible faces. Two cases have short proofs. On the complete graph a face of k states has $\Phi_\tau(F) = \tau(d-k) + k - 1 = (d-1)\tau + (k-1)(1-\tau)$ by Corollary 7.3. So the vertices win for $\tau < 1$, the whole set $[d]$ wins for $\tau > 1$, and all faces tie at $\tau = 1$. Now let Q be symmetric and fix a state x. Rayleigh–Ritz with the unit vector f that equals $(d-1)^{-1/2}$ off x and 0 at x gives $\lambda_{[d]\setminus\{x\}} \le q_x/(d-1)$, with equality exactly when f is an eigenvector of $-Q$ restricted to $[d]\setminus\{x\}$. That matrix maps f to the vector with entries $q_{yx}(d-1)^{-1/2}$, so equality holds exactly when $q_{yx} = q_x/(d-1)$ for all $y \neq x$. Take unit rates on a connected graph and a start of full support, and let x have the least degree δ. The vertices of degree δ and the face $[d]$ tie at $\tau_0 = (d-1)/\delta$. The restricted matrix is block diagonal, so a component C of $[d]\setminus\{x\}$ of least exit rate has $\lambda_C = \lambda_{[d]\setminus\{x\}}$ and $\Phi_{\tau_0}(C) \le \tau_0\lambda_{[d]\setminus\{x\}} + d - 2 \le d - 1$. Equality needs $q_{yx} = \delta/(d-1)$ for every $y \neq x$, which for unit rates means that the graph is complete. So the winners go from the vertices straight to $[d]$ exactly when the graph is complete. On every other connected graph a face that avoids x beats both at τ_0, and the mode can move to a community that does not contain its first vertex.

By Example 5.13 at the center, $D_k = dK_F = k/\sqrt{\det(2\pi\Sigma_k)}$ with $\Sigma_k = \Sigma_{\mathbf{1}/k} = (2/k^2)(I - J/k)$ in the coordinates α'. So D_k is the mass k of a face, in units of a vertex, over the Gaussian volume of the fluctuations of its fractions. The preprint [29, Proposition 3.7] obtains it also by summing the exact law over the whole face. Here Σ_α is the occupation covariance defined in Definition 5.1. Example 5.13, Theorem 5.4 and Lemma 5.5 give every composition whose fractions are bounded away from the boundary of its face. Let $\alpha = n/N$ have k positive coordinates, all at least ε, and put $A = \sum_i \sqrt{\alpha_i}$. Uniformly for such compositions and for $1 \le T \le N^{1/3}$,

$$\frac{P_N(n)}{V_N} = D(\alpha)\,\frac{T^{(k-1)/2}e^{(A^2-1)T}}{N^{k-1}}\Big(1 + O_\varepsilon\Big(\frac{1}{T} + \frac{T^2}{N}\Big)\Big),$$
$$D(\alpha) = \frac{A^{k/2+1}}{(4\pi)^{(k-1)/2}(\prod_i \alpha_i)^{3/4}}, \tag{21}$$

for the uniform start. The relation between $D(\alpha)$ and $C_F(\alpha)$ is given in Example 5.13, and $V_N = \frac{1}{d}e^{-(d-1)T}(1 + O(T^2/N))$. This is [29, Theorem 3.3], and at the center $D(\mathbf{1}/k) = D_k$. By the Cauchy–Schwarz inequality $A^2 \le k$, with equality only at the center.

The next theorem holds at every N in its exact parts, and its asymptotic parts come from Section 5.

Theorem 7.5 (balanced crossings). *Let μ be uniform and let F be a face of $2 \le k \le d$ states.*

(i) For every $N \ge k$ there is exactly one $p = p_{N,k} \in (0,1)$ with $M_F = V_N$. The face maximum M_F is larger than V_N below $p_{N,k}$ and smaller above it, and so is the probability of a balanced composition of F. The crossing $p_{N,k}$ is strictly increasing in N, $p_{k,k} = 1/(1 + (d-1)(k!)^{-1/(k-1)})$, and every crossing has $u \le (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$.

(ii) For large N, uniformly for $1 \le T \le N^{1/3}$,

$$\log\frac{M_F}{V_N} = (k-1)\Big[T - L + \frac{1}{2}\log T - a_k\Big] + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

With $T = L - \frac{1}{2}\log L + t$, the ratio M_F/V_N tends to $e^{(k-1)(t-a_k)}$, uniformly for t in compact sets.

(iii) The value $T_{N,k} = N(1-p_{N,k})/(d-1)$ *of* T *at the crossing satisfies*

$$T_{N,k} + \frac{1}{2}\log T_{N,k} = L + a_k + O(1/L),$$

and

$$T_{N,k} = L - \frac{1}{2}\log L + a_k + \frac{\log L}{4L} + O\Big(\frac{1}{L}\Big). \tag{22}$$

(iv) $a_2 > a_3 > a_4 > \cdots$.

For $d = 2$, $T_{N,2}$ is the crossing z_N of Corollary 7.2, and $a_2 = c_0$. The proof is in Appendix C. The exact part uses that S_n has nonnegative coefficients and the first nonzero coefficient $W_n(k-1) = k!$, from the words that use each state in a single run. The next coefficient $W_n(k) = k!(k-1)(N-k)/2$ grows with N, which moves the root. The asymptotic part is Theorems 5.7 and 5.8. The equation in (iii) has the next term $1/(8T_{N,k})$ on its right side. The persistent-walk paper [28] proves it for $k = 2$, and the preprint [29, Theorem 3.2], which this paper incorporates, proves it for every $k \geq 2$ through a Bessel majorant.

Once the number of runs of each state is fixed, splitting the visits evenly gives the most ways to choose the run lengths. So the face maxima are the balanced compositions.

Theorem 7.6 (exact balancing)**.** *Let* $1/d \leq p < 1$*, and let* μ *be constant and positive on a face* F*. Moving one unit from a coordinate* b *of a composition with support* F *to another coordinate* a*, where* $n_b \geq n_a + 2$ *and* $n_a \geq 1$*, strictly increases the probability of the composition. So the most likely compositions with support* F *are exactly the balanced ones, and they all have the same probability.*

For an integer $n \geq 1$ put

$$B_n(w) = \sum_{m=1}^{n} \binom{n-1}{m-1} \frac{w^m}{m!} = \frac{w}{n} L_{n-1}^{(1)}(-w), \tag{23}$$

where $L_{n-1}^{(1)}$ is the Laguerre polynomial [27, Eq. 18.5.12]. The proof in Appendix C rests on the log-concavity of $B_n(w)$ in n for $w > 0$. The condition $p \geq 1/d$ holds for $T \leq N/d$, and so at every crossing by Theorem 7.5(i).

Corollary 7.7 (exactly one mode change)**.** *Let* μ *be uniform, and fix* $d \geq 2$ *and* $C > 1$*. There is* N_0 *such that for* $N \geq N_0$ *and every* p *with* $0 < T \leq CL$ *the following hold.*

(i) If $p > p_{N,d}$*, the global modes are exactly the* d *vertices.*

(ii) If $p < p_{N,d}$*, they are exactly the balanced compositions with* d *positive coordinates.*

(iii) If $p = p_{N,d}$*, they are these two classes and nothing else.*

Moreover $p_{N,2} < p_{N,3} < \cdots < p_{N,d}$ *and* $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$*. So all the crossings of balanced compositions with the vertices lie in a* p*-interval of length* $(d-1)(a_2 - a_d + o(1))/N$*, and as* p *decreases the balanced compositions with* d *positive coordinates cross first.*

So for large N the mode jumps once, from the vertices straight to the center of the simplex, and no proper face holds it in between. On the triangle the middle composition $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the vertices while the middles of the edges are still below them by the factor $e^{-(a_2 - a_3)} = 0.827$. The proof is in Appendix C. In the tie window at $\tau^* = 1$ this is Theorem 5.9, whose envelope E has a single kink, at $t = a_d$.

Remark 7.8 (an intermediate face can be a mode at finite N). Corollary 7.7 does not hold for every N. Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$. For an edge center and a full center, counting the words gives

$$\frac{P_4(2,2,0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \qquad \frac{P_4(2,1,1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Here $p > 1/3$, so by Theorem 7.6 every global mode is a vertex or a balanced composition, and the three edge centers are the only global modes. Also $0 < T = 14/17 < \log 4$, so this p lies in the range of Corollary 7.7 for every $C > 1$. Hence $N_0 > 4$ when $d = 3$. These values are checked in Lean.

7.3 A start

By Corollary 7.3 the start enters K_F only through $\mu(F)$. It changes the constants by an amount of order 1, the order at which the faces are separated, so a start can decide which face wins.

Corollary 7.9 (crossings with a start)**.** *Let F be a face of $k \geq 2$ states with $\mu(F) > 0$, and let i be a state with $\mu_i > 0$. Put $a_k(\mu) = a_k + \frac{1}{k-1} \log \frac{k\mu_i}{\mu(F)}$. For large N the equation $M_F = M_{\{i\}}$ has a root in $[1, N^{1/3}]$, and every such root satisfies $T + \frac{1}{2} \log T = L + a_k(\mu) + O(1/L)$. With $T = L - \frac{1}{2} \log L + t$, $\log(M_F/M_{\{i\}}) = (k-1)(t - a_k(\mu)) + o(1)$, uniformly for t in compact sets.*

Proof. By Corollary 7.3, $\log(K_F/K_{\{i\}}) = \log D_k + \log \frac{\mu(F)}{k\mu_i} = -(k-1)a_k(\mu)$, and $\Delta = \delta = k - 1$. So Theorem 5.8 gives the root and its equation. For the last claim, Theorem 5.9(i) at $\tau^* = 1$ gives $\log(M_F/M_{\{i\}}) = \ell_F(t) - \ell_{\{i\}}(t) + O(\log L/L)$, and $\ell_F(t) - \ell_{\{i\}}(t) = \log(K_F/\mu_i) + (k-1)t$. □

So a start changes only the constant of each crossing [29, Theorem 4.4], and it can change which face wins.

Example 7.10 (the start decides which face wins)**.** Let $d = 3$ and $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$, and put $t_{\mathrm{ef}} = \log \frac{3}{2} + 2a_3 - a_2 = \frac{1}{2} \log \frac{32\pi}{243}$. Fix t, put $T = L - \frac{1}{2} \log L + t$ and let N be large. If $t < a_2$, the global modes are $(N, 0, 0)$ and $(0, N, 0)$. If $a_2 < t < t_{\mathrm{ef}}$, they are exactly the balanced compositions of the edge $\{1, 2\}$. If $t > t_{\mathrm{ef}}$, every global mode has three positive coordinates. The window $(a_2, t_{\mathrm{ef}}) = (-0.4673558, -0.4412978)$ has width $\log \frac{16}{9\sqrt{3}} = 0.0260580$, while with the uniform start no proper face wins (Corollary 7.7).

Proof. By Lemma 2.2(b) and (c) the admissible faces are the faces that meet $\{1, 2\}$, and they satisfy (H). By Corollary 7.3 they all tie at $\tau^* = 1$, so Theorem 5.9 applies with $m^* = 2$. For a face F of k states its line is $\ell_F(t) = \log K_F - (3-k)t$, and $\ell_F(t) + 2t$ equals $\log \mu_i$ for $F = \{i\}$ and $\log(\mu(F)/k) + (k-1)(t - a_k)$ for $k \geq 2$. So $\ell_F + 2t$ is $\ell_1 = \log \frac{1}{2}$ for the vertices 1 and 2, $\ell_2 = \log \frac{1}{2} + t - a_2$ for the edge $\{1, 2\}$, $\ell_2 - \log 2$ for the other two edges and $\ell_3 = \log \frac{1}{3} + 2(t - a_3)$ for the full face. Hence $E + 2t$ is the upper envelope of ℓ_1, ℓ_2 and ℓ_3. Here $\ell_2 > \ell_1$ exactly when $t > a_2$, $\ell_2 > \ell_3$ exactly when $t < t_{\mathrm{ef}}$, and $\ell_3 > \ell_1$ exactly when $t > t^* = a_3 + \frac{1}{2} \log \frac{3}{2}$. By the definition of a_k, $t_{\mathrm{ef}} - a_2 = \log \frac{3}{2} + 2(a_3 - a_2) = 4 \log 2 - \frac{5}{2} \log 3 = \log \frac{16}{9\sqrt{3}}$, which is positive since $16^2 = 256 > 243 = (9\sqrt{3})^2$. Also $t^* - a_2 = \frac{1}{2}(t_{\mathrm{ef}} - a_2)$, so $a_2 < t^* < t_{\mathrm{ef}}$. Hence the kinks of E are a_2 and t_{ef}, and E is the line of the two vertices for $t < a_2$, of the edge $\{1, 2\}$ for $a_2 < t < t_{\mathrm{ef}}$, and of the full face for $t > t_{\mathrm{ef}}$. For fixed t that is not a kink, the consequence in Theorem 5.9 puts every global mode on a face with that line. The vertices Ne_1 and Ne_2 have the same probability $\frac{1}{2}p^{N-1}$. On the edge, μ is constant and $p > 1/3$ for large N, so by Theorem 7.6 the most likely compositions with support $\{1, 2\}$ are exactly the balanced ones, and they all have the same probability. □

The comparison of the lines and the value of the width are checked in Lean.

7.4 A weak field

Proposition 5.15 applies to the complete graph with the uniform start. There α_F^* is uniform, so the field shifts the line of a face S by the average $\bar{\xi}_S = \frac{1}{|S|}\sum_{i\in S}\xi_i$ of the field over S. In the Potts chain the field ξ_i/N acts at each site in state i.

Corollary 7.11 (weak-field modal phase diagram)**.** *Let μ be uniform, put $T = L - \frac{1}{2}\log L + t$ and define the face scores*

$$\ell_S(t,\xi) = \begin{cases} \xi_i, & S = \{i\}, \\ (|S|-1)(t - a_{|S|}) + \bar{\xi}_S, & |S| \geq 2. \end{cases}$$

Uniformly for (t,ξ) in compact sets,

$$\log \max_n \frac{e^{\langle \xi, n\rangle/N} P_N(n)}{V_N} \longrightarrow \max_{\varnothing \neq S \subseteq [d]} \ell_S(t,\xi). \tag{24}$$

For fixed (t,ξ) and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has a support that maximizes the limiting score. If the maximizing support is unique, all global modes have exactly that support, and if it has $|S| \geq 2$ states they satisfy $n/N = \mathbf{1}_S/|S| + O(1/L)$. Equal limiting scores do not imply that both supports hold modes at finite N.

If the fields are sorted as $\xi_1 \geq \cdots \geq \xi_d$, the maximum of the scores is the upper envelope of the d lines

$$\ell_1(t) = \xi_1, \qquad \ell_k(t) = (k-1)(t - a_k) + \frac{1}{k}\sum_{i=1}^{k} \xi_i, \quad 2 \leq k \leq d. \tag{25}$$

Away from ties, the winning support size is nondecreasing in t. If the fields are strictly ordered, the winning supports are the nested sets $\{1,\dots,k\}$. Every support size $k \in \{2,\dots,d-1\}$ is the unique winning size on an open interval of t for suitable fixed fields. So proper faces can hold the mode under a field of order $1/N$ per step.

Proof. Step 1. The best composition on a support. Let S have $k \geq 2$ states. By Theorem 7.5(ii), $\log(M_S/V_N) = (k-1)(t - a_k) + o(1)$, uniformly for t in a compact set. Proposition 5.15(b), with α_S^* uniform, gives $\log(M_S^\xi/V_N) = \ell_S(t,\xi) + o(1)$, uniformly for (t,ξ) in compact sets. A vertex has $M_{\{i\}}^\xi = e^{\xi_i} V_N$ exactly. There are finitely many supports, which proves (24).

Step 2. The modes. A support whose score is below the maximum by a fixed amount has M_S^ξ smaller than the largest face maximum for large N, so it holds no global mode. A global mode with support S maximizes w_ξ on S, and Proposition 5.15(b) gives its location.

Step 3. The lines. Among the supports of size k the largest $\bar{\xi}_S$ is the average of the k largest fields, which gives (25). The line ℓ_k has slope $k-1$. If size k_1 wins at t_1 and size k_2 at $t_2 > t_1$, adding $\ell_{k_1}(t_1) \geq \ell_{k_2}(t_1)$ and $\ell_{k_2}(t_2) \geq \ell_{k_1}(t_2)$ gives $(k_2 - k_1)(t_2 - t_1) \geq 0$, so $k_2 \geq k_1$, as in Theorem 4.4(b). If the fields are strictly ordered, the unique best support of size k is $\{1,\dots,k\}$.

Step 4. Every intermediate size. To realize a given $2 \leq k < d$, set $\xi_1 = \cdots = \xi_k = 0$ and $\xi_{k+1} = \cdots = \xi_d = -H$, and take $t > a_k$ so close to a_k that $t < a_{k-1}$ if $k > 2$. Then the score of $\{1,\dots,k\}$ is positive, the largest vertex score is 0, and every face with $2 \leq j < k$ states scores at most $(j-1)(t - a_j) < 0$, by Theorem 7.5(iv). A face with $j > k$ states scores at most $(j-1)(t - a_j) - H(j-k)/j$, and another face with k states at most $(k-1)(t - a_k) - H/k$, so both lose once H is large. The comparisons are strict, so they hold on an open interval of t, and small changes make the fields strictly ordered without removing the phase. □

One field can make every support size win in turn, from one state to all d. This is the sequence of phases in Figure 1.

Corollary 7.12 (one field realizes the full face hierarchy)**.** *Let μ be uniform, fix $d \geq 2$ and $H > 3/4$, and let $\xi_i = -H(i-1)^2$ for $1 \leq i \leq d$. In particular $\xi_i = -(i-1)^2$ works in every fixed dimension. Put*

$$t_k = \log D_k - \log D_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and $t_0 = -\infty$, $t_d = +\infty$. Then $t_1 < \cdots < t_{d-1}$, and for every $k \in \{1, \ldots, d\}$ the support $\{1, \ldots, k\}$ is the unique winning support on the nonempty interval $t_{k-1} < t < t_k$. So for each fixed t in this interval and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has exactly those k positive coordinates.

Proof. Since $\frac{1}{k}\sum_{i=1}^{k}(i-1)^2 = (k-1)(2k-1)/6$ and $-(k-1)a_k = \log D_k$, the lines (25) are $\ell_k(t) = (k-1)t + \log D_k - H(k-1)(2k-1)/6$, also for $k = 1$, and $\ell_{k+1}(t) - \ell_k(t) = t - t_k$. Extend $\log D_k$ to the smooth function $c(x) = (x + \frac{1}{2})\log x - \frac{x-1}{2}\log(4\pi)$ for $x \geq 1$. Then $0 < c''(x) = (x - \frac{1}{2})/x^2 \leq \frac{1}{2}$, and integrating twice over unit intervals gives $\log D_{k+2} - 2\log D_{k+1} + \log D_k \leq \frac{1}{2}$. Hence

$$t_{k+1} - t_k = \frac{2H}{3} - (\log D_{k+2} - 2\log D_{k+1} + \log D_k) > 0.$$

For $t_{k-1} < t < t_k$, summing the differences $\ell_{j+1}(t) - \ell_j(t) = t - t_j$ shows that line k strictly exceeds every other line. The fields are strictly decreasing, so the unique best support of size k is $\{1, \ldots, k\}$, and Corollary 7.11 gives the conclusion for finite N. Most of this arithmetic is checked in Lean. □

8 Near a vertex

This section is the worked case of the boundary layer of Section 6 (Theorem 6.7 and Proposition 6.3) on the complete graph near a vertex, and Section 6.7 shows that with one rare count it is the discrete pair-face layer. The vertex law covers the regime in which T is of order $\sqrt{N}$, where the lattice parameter is of order one and the Laguerre kernel cannot be replaced by the Bessel kernel of the continuum layer.

We keep the notation of Section 7.2, namely $Q = J - dI$, the uniform start, p, u, v, and $S_n(u) = P_N(n)/V_N$ by (20). Let one count b be large and let the other positive counts $a_1, \ldots, a_\ell$ be small compared with b, with $\ell \geq 1$ fixed. We put $|a| = a_1 + \cdots + a_\ell$ and write $S_{(b,a)}$ for S_n with $n = (b, a_1, \ldots, a_\ell)$. These rare counts are not the constants a_k of Section 7.2. Since S_n does not depend on the start, $P_N(n) \leq (\max_i \mu_i)p^{N-1}S_n(u)$, so upper bounds on S_n hold for every start.

A rare run away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the rare counts enter through bu^2, and for fixed rare counts the crossing moves to odds u of order $b^{-1/2}$, that is, to T of order $\sqrt{N}$. Use B_a from (23) and put

$$\mathcal{I}(x) = \sum_{j \geq 0} \frac{x^j}{j!\,(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}},$$

where I_1 is the modified Bessel function [27, Eq. 10.25.2], and $\mathcal{I}(0) = 1$. By the Laguerre form in (23), the next lemma is a quantitative form of the limit of Laguerre polynomials to Bessel functions.

Lemma 8.1 (a uniform Laguerre–Bessel inequality)**.** *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\,\mathcal{I}(aw)} \leq \frac{w}{2}. \tag{26}$$

The proof is in Appendix D. Put

$$\mathcal{B}(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \qquad \kappa(y) = \frac{y\,\mathcal{B}'(y)}{\mathcal{B}(y)}.$$

The polynomial $\mathcal{B}$ has positive coefficients, $\mathcal{B}(0) = 0$ and $\mathcal{B}(1) \geq 1$, so $\mathcal{B}(y_a) = 1$ has exactly one positive root y_a, and $y_a \leq 1$. The polynomial $S_{(b,a)}$ has nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ, so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$, the crossing odds of the composition $(b, a_1, \ldots, a_\ell)$ with a vertex.

Theorem 8.2 (the vertex law)**.** *Fix $\ell \geq 1$ and put*

$$c_\ell = \ell + \sqrt{\ell} + 1, \qquad \varepsilon = \sqrt{\frac{|a|+1}{b}}, \qquad \delta = 16 c_\ell \varepsilon.$$

For all positive counts $b, a_1, \ldots, a_\ell$ with $\delta \leq 1/2$,

$$(1-\delta)\sqrt{y_a/b} \;\leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}, \qquad \mathcal{B}(y_a) = 1. \tag{27}$$

In particular

$$\left| \frac{u_{b,a}}{\sqrt{y_a/b}} - 1 \right| \leq 16 c_\ell \sqrt{\frac{|a|+1}{b}} \qquad \text{whenever } \sqrt{\frac{|a|+1}{b}} \leq \frac{1}{32 c_\ell}, \tag{28}$$

so $u_{b,a} = \sqrt{y_a/b}\,(1 + o(1))$ as $(|a|+1)/b \to 0$. The constants are not optimized.

For fixed rare counts Lemma D.2 gives $S_{(b,a)}(\sqrt{y/b}) \to \mathcal{B}(y)$ as $b \to \infty$, and the theorem shows that the crossing stays near $\sqrt{y_a/b}$ uniformly as long as $|a| = o(b)$. For $d = 2$ and $\ell = 1$ its leading term is that of the fixed-edge expansion of [28, Theorem 4.10]. The proof is in Appendix D. It rests on two facts. First, $\log \mathcal{B}$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell |a| y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})]$ exactly, where $\mathcal{X}$ is a gamma variable plus a compound binomial sum with mean $2 + (b-1)u$ (Lemma D.2). Jensen's inequality then gives a lower bound that exceeds 1 at $u = (1+\delta)\sqrt{y_a/b}$, and the tangent line of $\log \mathcal{B}$ at bu^2 with the moment generating function of $\mathcal{X}$ gives an upper bound that is below 1 at $u = (1-\delta)\sqrt{y_a/b}$.

Remark 8.3 (the boundary of the simplex). Away from a vertex, the preprint [29] gives a Bessel majorant for every positive count vector [29, Theorem 6.2] and shows that, when at least two coordinates are of order N, a small coordinate starts to matter at size of order $N/(\log N)^2$ [29, Remark 6.3 and Theorem C.1]. We do not use these results.

9 What the occupation law determines

The forward theory says which face carries the mode for a given chain. This section asks which coarse functionals of the law of the occupation counts determine the generator, and how much they lose. We compare three of them, from fine to coarse. The first is the law of the counts

itself, at every N and T. The second is its logarithmic scale on single compositions, which by the stratified local law (Theorem 5.6) is the Donsker and Varadhan functional J_V. The third is a finite label, the modal support of the chain killed at known rates (Section 9.1). The organizing result is that the second and the third carry the same information, namely the principal minors of $-Q$ (Corollary 9.3). Theorem 9.13 describes this information. It is Q up to time reversal when Q has no cut, and a finite fiber of generators in general. For a reversible chain on a known graph it is all of Q, and finitely many boundary coordinates of a design fixed in advance recover it (Section 9.2).

9.1 The killed chain and its modal support

Fix a field $v \in \mathbb{R}^d$ and a time scale $\tau > 0$, put $T = \tau L$ and $h = T/N$, and weight each composition by $e^{Tv\cdot n/N}$. Along a path the weight is $\prod_t e^{hv_{X_t}}$. If $v \le 0$, this is the probability that the chain survives when it is killed at each visit to i with probability $1 - e^{hv_i}$. So $P_N(n)e^{Tv\cdot n/N}$ is the probability of surviving with composition n, and its maximizers are the modes of the composition of the surviving chain, since a normalization does not move a mode. On a face the extra killing shifts the diagonal of the killed generator, $Q_F + \operatorname{diag} v_F$ in place of Q_F. For a face F let

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v_F\big), \qquad \Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v),$$

so $\lambda_F(0) = \lambda_F$ and $\Phi_{\tau,0} = \Phi_\tau$. If F is reducible, Q_F is block triangular in the order of its strongly connected components, and $\lambda_F(v)$ is the least of the numbers $\lambda_C(v)$ over these components C.

The shift $v \mapsto v + c\mathbf{1}$ multiplies the weighted law by the constant e^{Tc}, and since $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$ it moves every score $\Phi_{\tau,v}(F)$ by the same $-\tau c$. So it leaves every winner unchanged, and every field may be taken nonpositive. A nonpositive field is a known killing rate, and $\lambda_F(v)$ is the exit rate of F for the killed chain. The observation is then the support of the most likely composition of the surviving trajectory, for a chain with known extra killing watched up to the horizon $T = \tau L$. Throughout this section the field and the time scale are calibrated, that is, v and τ are known in the units of Q.

The survival probability itself, $\sum_n P_N(n)e^{Tv\cdot n/N}$, is a finer observation. For irreducible Q it equals $e^{-T\lambda_V(v)}$ up to a bounded factor, so it gives λ_V on all fields, and by Lemma 9.10 below it already determines every principal minor of $-Q$, with no descent through the faces. The modal-support label is coarser. It does not require measuring a probability, only the support of the most likely composition, and it is the datum that the forward theory describes. We show that it loses nothing.

The controlled score law below says that the face that maximizes the weighted law has the least score $\Phi_{\tau,v}$, up to $O(\log L)$ in the exponent. A face F wins strictly at (τ, v) if it is the unique minimizer of $\Phi_{\tau,v}$ over the admissible faces. The modal-support map is $(\tau, v) \mapsto F$ on the strict winning regions, and the data are the boundaries between adjacent strict regions, the controlled boundaries. We do not assume that a face tied in Φ is observed as a maximizer on the boundary. For two states a factor $\sqrt{L}$ decides a tie, and we do not use it.

Proposition 9.1 (the controlled score law)**.** *Fix Q, a start μ with full support, a field v and a compact interval of τ in $(0, \infty)$. For every admissible face F,*

$$\log \max_{\operatorname{supp} n = F} \big\{P_N(n)e^{Tv\cdot n/N}\big\} \le -\Phi_{\tau,v}(F)\, L + O(\log L),$$

with equality up to $O(\log L)$ if F is irreducible, and for a singleton the maximum is exactly $\mu_i(1 - hq_i)^{N-1}e^{Tv_i}$. A reducible admissible face F has $\Phi_{\tau,v}(F) \ge \Phi_{\tau,v}(C) + 1$ for one of its strongly connected components C. Hence if F is the unique minimizer of $\Phi_{\tau,v}$ over the admissible faces, then F is irreducible and the support of every maximizer of the weighted law is F for all large N.

Proof. The proposition is the field form of Theorem 4.2(ii) and (iii). Let $k = |F|$. For large N the whole interval of τ gives $T_0 \le T = \tau L \le N^{1/3}$, and $\log T = O(\log L)$. The only composition with support $\{i\}$ is Ne_i, so by (2) the maximum is exactly $\mu_i(1 - hq_i)^{N-1}e^{Tv_i}$, whose logarithm is $-T(q_i - v_i) + O(1) = -\Phi_{\tau,v}(\{i\})L + O(1)$. For $k \ge 2$, Remark 4.3 gives the upper bound of Theorem 4.2(ii) for $M_F(T, v)$ with $\lambda_F(v)$ in place of λ_F. It uses (7) on each block of a path, so it holds for reducible faces too, and it reads $\log M_F(T, v) \le -(k-1)L - T\lambda_F(v) + (k-1)\log T + \kappa = -\Phi_{\tau,v}(F)L + O(\log L)$. If F is irreducible, it satisfies (H) because μ has full support, and (10) of Theorem 5.7 with $\mathcal{V} = \{v\}$ gives the matching lower bound.

Let F be reducible and admissible. Then $\lambda_F(v)$ is the least $\lambda_C(v)$ over the components C of F, and a component C where it is attained has $|C| \le k - 1$. It is admissible by Lemma 2.2(b), and $\Phi_{\tau,v}(F) = \Phi_{\tau,v}(C) + k - |C| \ge \Phi_{\tau,v}(C) + 1$. So a unique minimizer F of $\Phi_{\tau,v}$ is irreducible. Let $\varepsilon > 0$ be the least gap $\Phi_{\tau,v}(G) - \Phi_{\tau,v}(F)$ over the other admissible faces G. Their weighted maxima are at most $N^{-\Phi_{\tau,v}(F)-\varepsilon+o(1)}$, that of F is at least $N^{-\Phi_{\tau,v}(F)-o(1)}$, and a face that is not admissible carries no mass. There are finitely many faces, so for large N every maximizer has support F. □

The upper bound holds up to the boundary of the face, and this is what lets a fixed field act on all compositions at once.

Definition 9.2 (controlled boundaries)**.** Two generators have the same controlled boundaries if, for every start of full support and every field used below, they have the same strict winner at every point (τ, v) outside a closed nowhere dense set. The fields used are the pair fields of Section 9.3, with a field δ on one state, 0 on another and $-\bar{v}$ elsewhere, and the fields $(v_F, -\bar{v})$ of Lemma 9.12, equal to v_F on a face F and to $-\bar{v}$ off it, with $\bar{v}$ large.

The ties form a closed nowhere dense set. A minimizer of $\Phi_{\tau,v}$ is irreducible, and for irreducible faces $F \ne H$ the difference $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(H)$ vanishes for at most one τ at fixed v if $|F| \ne |H|$. If $|F| = |H|$ and $j \in F \setminus H$, it vanishes for at most one v_j when the other coordinates are fixed, since $\lambda_H(v)$ does not depend on v_j while $\partial\lambda_F/\partial v_j = -(l \circ r)_j < 0$ for the Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$. So the set where the least score is attained twice is closed and nowhere dense. Off it Proposition 9.1 identifies the observed support with the strict winner for all large N, and the ties themselves are never used.

The organizing result of the section compares the second and the third functional of the introduction.

Corollary 9.3 (the Donsker and Varadhan functional and the modal supports)**.** *For two irreducible generators Q and Q' the following are equivalent.*

(i) They have the same Donsker and Varadhan functional J_V.

(ii) They have the same controlled boundaries.

(iii) They have the same principal minors, that is, they lie in the same fiber of Theorem 9.13(b).

The proof is given after Theorem 9.13. The equivalence of (i) and (iii) uses only the variational formula, Lemma 9.9 and Lemma 9.10, and (ii) is the strengthening to a single label. The law of the counts itself does better for a generic start. From the stationary start the law of the composition is the same under Q and under its time reversal Q^*, for every N and T with $Tq_{\max} \le N$ (Theorem 9.13(f)), so from that start no count statistic separates them. A start outside a finite union of proper subspaces separates every member of the fiber through the amplitudes $K_V(v)$.

9.2 The reversible case

For a reversible chain the information is all of Q, and an explicit finite experiment reads it. Let Q be reversible and irreducible with stationary law π, on a known connected graph $G = (V, E)$ with $m = |E|$ edges, where $\{i, j\} \in E$ exactly when $q_{ij} > 0$. Put $w_{ij} = q_{ij} q_{ji}$ for the edge products and

$$S = D_{\sqrt{\pi}}(-Q) D_{\sqrt{\pi}}^{-1}, \qquad S_{ii} = q_i, \qquad S_{ij} = -\sqrt{w_{ij}} \ \ (\{i, j\} \in E),$$

the symmetrized killed matrix. We assume known constants $R, b > 0$ with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on every edge. The field and the time are calibrated, that is, v and τ are known in the units of Q.

Two facts are used throughout. First, S is symmetric and similar to $-Q$, whose eigenvalues have nonnegative real parts, so S is positive semidefinite. Hence $0 \le q_k \le R$ for every k, $\sqrt{w_{kl}} \le R$ on every edge, and $|q_k - q_l| \le R$. Second, for a face F on which G induces a connected graph, $-(Q_F + \operatorname{diag} v_F)$ is similar to $S_F - \operatorname{diag} v_F$ by the restriction of $D_{\sqrt{\pi}}$, so

$$\lambda_F(v) = \lambda_{\min}\big(S_F - \operatorname{diag} v_F\big), \tag{29}$$

as in Proposition 3.4. Since Q has symmetric support, the admissible faces are exactly these connected faces (Lemma 2.2(c)).

The design uses, for a retained pair (i, j), the field

$$v_i = \delta, \qquad v_j = 0, \qquad v_k = -\bar{v} \ \ (k \ne i, j),$$

which suppresses every other state, with the constants

$$\bar{\delta} = 2R, \qquad \tau_0 = \frac{1}{2R}, \qquad U = \frac{8R}{b^2}, \qquad \bar{v} = 4R + 4UR^2,$$

and the ranges $|\delta| \le 3R$ and $\frac{1}{4R} \le \tau \le U$.

Lemma 9.4 (suppression)**.** *Under the design for the pair (i, j), every irreducible face that contains a suppressed state has score at least $\frac{3}{4}$ larger than the best face inside $\{i, j\}$, uniformly on the ranges of δ and τ. Hence the strict winner is one of $\{i\}$, $\{j\}$ and $\{i, j\}$, and the last only if $\{i, j\}$ is an edge.*

Proof. Let F be connected and let $J = F \setminus \{i, j\}$ be nonempty.

Wholly suppressed faces. If $F \cap \{i, j\} = \emptyset$, then $S_F - \operatorname{diag} v_F = S_F + \bar{v} I \ge \bar{v} I$, so $\lambda_F(v) \ge \bar{v}$ by (29). The singleton $\{j\}$ has score $\tau q_j \le \tau R$. Hence $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(\{j\}) \ge \tau(\bar{v} - R) \ge (3R + 4UR^2)/(4R) \ge \frac{3}{4}$.

Mixed faces. Let $G_0 = F \cap \{i, j\}$ be nonempty. In the order G_0, J,

$$S_F - \operatorname{diag} v_F = \begin{pmatrix} B & E \\ E^{\top} & S_J + \bar{v} I \end{pmatrix}, \qquad B = S_{G_0} - \operatorname{diag} v_{G_0}, \quad E = S[G_0, J].$$

Here $\|E\|_2 \le R$ and $S_J \ge 0$. Let $e = \lambda_{\min}(B)$, so $|e| \le \|S_{G_0}\|_2 + |\delta| \le 4R$. The test vector $(x, 0)$, with x a unit eigenvector of B for e, gives $\lambda_F(v) \le e$. For a unit vector (x, y),

$$\left\langle \begin{pmatrix} x \\ y \end{pmatrix}, (S_F - \operatorname{diag} v_F) \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle \ge e|x|^2 - 2R|x||y| + \bar{v}|y|^2 \ge \lambda_{\min} \begin{pmatrix} e & -R \\ -R & \bar{v} \end{pmatrix} \ge e - \frac{R^2}{\bar{v} - e},$$

by $\sqrt{s^2 + R^2} \le s + R^2/(2s)$ with $s = (\bar{v} - e)/2 > 0$. Since $\bar{v} - e \ge \bar{v} - 4R = 4UR^2$, we get $e - 1/(4U) \le \lambda_F(v) \le e$. Now $e = \lambda_{G_*}(v)$ for an irreducible face $G_* \subseteq \{i, j\}$. If G_0 is connected,

take $G_* = G_0$ and use (29). If $G_0 = \{i, j\}$ is not an edge, B is diagonal and G_* is the singleton with the smaller diagonal entry. In both cases $|G_*| \le |F| - 1$, so $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(G_*) \ge 1 - \tau/(4U) \ge \frac{3}{4}$ as $\tau \le U$.

So the unique minimizer of $\Phi_{\tau,v}$, if there is one, is an irreducible face inside $\{i, j\}$, and Proposition 9.1 identifies it with the strict winner. □

Theorem 9.5 (reversible reconstruction)**.** *Let Q be reversible and irreducible on the known connected graph G with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on edges, and let the start have full support.*

(a) (Exit differences.) For each $i \ne 1$, under the design for the pair $(i, 1)$ at $\tau = \tau_0$, the strict winner changes from $\{1\}$ to $\{i\}$ at $\delta = x_i = q_i - q_1$, and $|x_i| \le R$. Put $d_i = x_i$ and $d_1 = 0$.

(b) (Products.) For each edge $\{i, j\}$, under the design for (i, j) at $\delta = \bar\delta$, the strict winner changes from a singleton to the pair at

$$\tau = t_{ij} = \frac{\sqrt{(\bar\delta - \Delta_{ij})^2 + 4w_{ij}} + \bar\delta - \Delta_{ij}}{2w_{ij}}, \qquad \Delta_{ij} = d_i - d_j,$$

with $1/R \le t_{ij} < U/2$, and $w_{ij} = (\bar\delta - \Delta_{ij})/t_{ij} + t_{ij}^{-2}$.

(c) (Reconstruction.) Let B be the symmetric matrix with diagonal d_i, entries $-\sqrt{w_{ij}}$ on the edges and zero elsewhere, let $\lambda = \lambda_{\min}(B)$ and let ϕ be its positive unit eigenvector. Then

$$S = B - \lambda I, \qquad Q = -D_\phi^{-1} S D_\phi, \qquad \pi_i = \phi_i^2.$$

So the $m + d - 1$ limiting boundary coordinates $(d_i)_{i \ne 1}$ and $(t_{ij})_{\{i,j\} \in E}$ determine Q, and the design is fixed by R, b and the labels before any boundary is observed.

Proof. By Lemma 9.4 and Proposition 9.1, at each point of the ranges the strict winner is the unique minimizer of $\Phi_{\tau,v}$ among the irreducible faces inside the retained pair, when it is unique. Note that $\tau_0 \in [\frac{1}{4R}, U]$, since $b \le R$.

(a) The candidates are $\{i\}$, $\{1\}$ and, if $\{i, 1\}$ is an edge, the pair. By Weyl's inequality $\lambda_{\{i,1\}}(v) \ge \min(q_i - \delta, q_1) - \sqrt{w_{i1}}$, so at τ_0 the score of the pair exceeds the better singleton score by at least $1 - \tau_0 R = \frac{1}{2}$, and the pair never wins. The singleton $\{i\}$ has the smaller score exactly when $q_i - \delta < q_1$, that is when $\delta > x_i$. Since $|x_i| \le R$, both sides of x_i lie in $[-3R, 3R]$. Hence $\{1\}$ wins strictly on $[-3R, x_i)$, $\{i\}$ wins strictly on $(x_i, 3R]$, and the boundary is $\delta = x_i$.

(b) Put $a = q_i - \bar\delta$, $c = q_j$ and $s = c - a = \bar\delta - \Delta_{ij}$, where $\Delta_{ij} = q_i - q_j$ by (a). Then $R \le s \le 3R$, and $\{i\}$ beats $\{j\}$ by $\tau s > 0$. By (29) the pair has $\lambda_{\{i,j\}}(v) = a - g$, the smaller eigenvalue of $\left(\begin{smallmatrix} a & -\sqrt{w} \\ -\sqrt{w} & c \end{smallmatrix}\right)$ with $w = w_{ij}$, where

$$g = \frac{\sqrt{s^2 + 4w} - s}{2} = \frac{2w}{\sqrt{s^2 + 4w} + s} \in (0, \sqrt{w}\,].$$

So $\Phi(\{i, j\}) - \Phi(\{i\}) = 1 - \tau g$. Hence $\{i\}$ wins strictly below $t_{ij} = 1/g$, the pair wins strictly above it, and $1/g$ is the displayed formula. Also $g \le \sqrt{w} \le R$ gives $t_{ij} \ge 1/R > \frac{1}{4R}$. The formula increases in s and decreases in w, and $s \le 3R$, $w \ge b^2$, $b \le R$, so $t_{ij} \le (\sqrt{13} + 3)R/(2b^2) < U/2$. At $\tau = t_{ij}$ the pair eigenvalue $a - 1/t_{ij}$ is a root of $(a - x)(c - x) = w$, which gives $t_{ij}^{-1}(s + t_{ij}^{-1}) = w$, the inversion formula.

(c) $B = S - q_1 I$, since S has diagonal q_k, entries $-\sqrt{w_{kl}}$ on edges and zero off the graph. The positive unit vector $\sqrt{\pi}$ satisfies $S\sqrt{\pi} = D_{\sqrt{\pi}}(-Q)\mathbf{1} = 0$. As $S \ge 0$, the eigenvalue 0 is $\lambda_{\min}(S)$,

and it is simple with eigenvector $\sqrt{\pi}$ by the theorem of Perron and Frobenius for the irreducible nonnegative matrix $RI - S$. So $\lambda = -q_1$, $S = B - \lambda I$, $\phi = \sqrt{\pi}$ and $\pi_i = \phi_i^2$, and $Q = -D_\phi^{-1} S D_\phi$ is the definition of S. The coordinates used are $d - 1$ values x_i and m values t_{ij}, and the fields and time scales of the design depend only on R, b and the labels. □

The reconstruction of a reversible generator from its exit rates and edge products is classical. Kim and Landim [24, Lemma 5.1 and Corollary 5.3] recover the same quantities from the values of J_V, and with the stationary law from $J_V^{-1}(0)$ (their Lemma 5.2) deduce that J_V determines a reversible generator. Here the coordinates are read off the boundaries of strict support regions of the weighted composition law, with a design fixed in advance.

Theorem 9.6 (minimality)**.** *The reversible generators on the fixed connected graph G form a real-analytic manifold of dimension $m + d - 1$, with global coordinates the edge products (w_{ij}) and the exit differences $(q_i - q_1)$. Let W be a nonempty open subset of it, for example the set where $\|S\|_2 < R$ and $\sqrt{w_{ij}} > b$ when that set is nonempty. Let $k < m + d - 1$, let $f \colon W \to \mathbb{R}^k$ be C^1 and let $\ell \colon W \to \Lambda$ be locally constant with Λ finite. Then the observation (f, ℓ) is not injective on any nonempty open subset of W. So no family of fewer than $m + d - 1$ locally C^1 scalar observations with locally constant finite labels is locally identifying.*

Proof. Let $\mathcal{Q}_G$ be the set of reversible generators with $q_{ij} > 0$ exactly on the edges of G, and $\kappa(Q) = ((w_{ij})_E, (q_i - q_1)_{i \neq 1})$. Given $w \in (0, \infty)^m$ and $d \in \mathbb{R}^{d-1}$ (with $d_1 = 0$), form B as in Theorem 9.5(c), let λ and ϕ be its least eigenvalue and positive unit eigenvector, which is simple since $cI - B$ is irreducible and nonnegative for large c, and put $\Psi(w, d) = -D_\phi^{-1}(B - \lambda I) D_\phi$. This is a reversible generator on G, with respect to ϕ^2, with products w_{ij} and exit rates $d_i - \lambda$, so $\kappa \circ \Psi = \mathrm{id}$, and Theorem 9.5(c), whose proof uses only reversibility and irreducibility, gives $\Psi \circ \kappa = \mathrm{id}$. The map Ψ is real-analytic by the implicit function theorem at a simple eigenvalue, and κ is polynomial. So κ is a global real-analytic chart of $\mathcal{Q}_G$, of dimension $m + d - 1$.

For the lower bound let $W' \subseteq W$ be nonempty and open. Shrinking it, we may assume that ℓ is constant on W' and that W' is a ball in the chart κ, so f is a C^1 map on a ball of $\mathbb{R}^p$, $p = m + d - 1$. Let ρ be the largest rank of df on W', attained at x_0. The rank is lower semicontinuous, so it equals ρ near x_0, and by the constant rank theorem f is constant on a C^1 submanifold through x_0 of dimension $p - \rho \geq p - k \geq 1$. So (f, ℓ) is not injective on W'. □

The chart is the continuous-time version of that of Wolfer and Watanabe [44, Theorems 3, 5 and 6] for reversible kernels, and the lower bound is the classical rank argument. Regularity is needed, since interleaving the digits of the $m + d - 1$ coordinates gives one injective scalar.

Remark 9.7 (partial data). Suppose that the products are known on $J \subseteq E$ and the exit differences along a comparison graph K on V. In the chart κ the generators with these data are those with the given w_e on J, any $w_e > 0$ on $E \setminus J$, and d equal to the given one plus a vector that is constant on each component of K and zero on that of the state 1. So they are parametrized by $(0, \infty)^{m - |J|} \times \mathbb{R}^{c(K) - 1}$, with $c(K)$ the number of components of K, and the data identify Q exactly when $J = E$ and K is connected. For instance, for $\ell > \sqrt{2}$ and $z = \ell - 2/\ell$ the generators

$$Q_\ell = \begin{pmatrix} -\ell & 2/\ell & z \\ \ell/2 & -\ell & \ell/2 \\ z & 2/\ell & -\ell \end{pmatrix}$$

satisfy detailed balance with $\nu = (1, 4/\ell^2, 1)$, have all exit rates equal to ℓ and $w_{12} = w_{23} = 1$, and have $w_{13} = z^2$. So the data of a spanning tree miss a parameter.

Theorem 9.8 (stability)**.** *In the setting of Theorem 9.5, let the observed dimensionless coordinates d_i/R and Rt_{ij} have sup-norm errors at most ε, and let $\widehat{Q}$ be the reconstruction from the observed values. There are ε_0 and C depending only on d and b/R such that for $\varepsilon \le \varepsilon_0$ the reconstruction $\widehat{Q}$ is a reversible generator on G and*

$$\|\widehat{Q} - Q\|_2 \le C\, R\, \varepsilon.$$

Among the constants are the bound $\phi_{\min} \ge \phi_* = (b/R)^{d-1}/\sqrt{d}$ for the ground vector $\phi = \sqrt{\pi}$ and the bound $\lambda_2(S) \ge g_* = 2b\phi_*^2/(d-1)$ for the gap of S.

Proof. The reconstruction commutes with the scaling $Q \mapsto Q/R$, so we take $R = 1$, $\bar{\delta} = 2$ and $\eta = b \le 1$. Then $\sigma_{ij} = t_{ij} \ge 1$ and $a_{ij} = 2 - d_i + d_j \in [1, 3]$, by Theorem 9.5. Hats denote reconstructed values, and $\widehat{d}_1 = d_1 = 0$.

Products. Let $\varepsilon \le \eta^2/20$. Then $\widehat{\sigma} \ge \frac{3}{4}$ and $|\widehat{\sigma}^{-1} - \sigma^{-1}| \le \frac{4}{3}\varepsilon$, and $w = a/\sigma + \sigma^{-2}$ gives $|\widehat{w} - w| \le (\frac{20}{3} + \frac{28}{9})\varepsilon < 10\varepsilon$. So $\widehat{w} \ge \eta^2/2$ and $|\sqrt{\widehat{w}} - \sqrt{w}| \le 10\varepsilon/\eta$, and the largest absolute row sum gives $\|\widehat{B} - B\|_2 \le \beta := 10d\varepsilon/\eta$. By Weyl's inequality $|\widehat{\lambda} - \lambda| \le \beta$, so $\|\widehat{S} - S\|_2 \le 2\beta$. The matrix $\widehat{B}$ has negative entries exactly on the edges of G, so the chart Ψ of the proof of Theorem 9.6 makes $\widehat{Q}$ a reversible generator with the graph G.

The constants. Row l of $S\phi = 0$ reads $q_l\phi_l = \sum_{k\sim l} \sqrt{w_{lk}}\phi_k$, so $\phi_l \ge \eta\phi_k$ on every edge. Some ϕ_{k_0} is at least $1/\sqrt{d}$, and every state is joined to k_0 by at most $d-1$ edges, so $\phi_{\min} \ge \phi_*$. For $x \perp \phi$ write $x = \phi \circ f$. Then $\langle x, Sx\rangle = \sum_{\{k,l\}\in E} c_{kl}(f_k - f_l)^2$ with $c_{kl} = \sqrt{w_{kl}}\phi_k\phi_l \ge \eta\phi_*^2$, while $\|x\|^2 = \sum_k \pi_k f_k^2$ and $\sum_k \pi_k f_k = 0$. In a spanning tree $\mathcal{T}$ of G the path from k to l has at most $d-1$ edges, so the inequality of Cauchy and Schwarz gives

$$\|x\|^2 = \tfrac{1}{2}\sum_{k,l} \pi_k\pi_l(f_k - f_l)^2 \le \tfrac{d-1}{2}\sum_{e\in\mathcal{T}} (\nabla_e f)^2 \le \frac{d-1}{2\eta\phi_*^2}\langle x, Sx\rangle,$$

and $\lambda_2(S) \ge g_*$.

The generator. Write $\widehat{\phi} = c\phi + z$ with $z \perp \phi$ and $c > 0$. Since $Sz = (S - \widehat{S})\widehat{\phi}$, we get $\|z\| \le 2\beta/g_*$, and since $1 - c \le 1 - c^2 = \|z\|^2$, also $\gamma = \|\widehat{\phi} - \phi\| \le \sqrt{2}\|z\| \le 2\sqrt{2}\beta/g_*$. This is the bound of Davis and Kahan in its simplest form. Let ε be so small that $\gamma \le \phi_*/2$. Then $\widehat{\phi}_{\min} \ge \phi_*/2$, $|\widehat{\phi}_l/\widehat{\phi}_k - \phi_l/\phi_k| \le 4\gamma/\phi_*^2$ and $\widehat{\phi}_l/\widehat{\phi}_k \le 2/\phi_*$, so on an edge

$$|\widehat{q}_{kl} - q_{kl}| = \bigl|\sqrt{\widehat{w}_{kl}}\,\widehat{\phi}_l/\widehat{\phi}_k - \sqrt{w_{kl}}\,\phi_l/\phi_k\bigr| \le \frac{20\varepsilon}{\eta\phi_*} + \frac{4\gamma}{\phi_*^2}.$$

The diagonal satisfies $|\widehat{q}_k - q_k| = |\widehat{S}_{kk} - S_{kk}| \le 2\beta$, and off the graph both matrices vanish. So every entry of $\widehat{Q} - Q$ is at most $C_1(d,\eta)\varepsilon$, and $\|\widehat{Q} - Q\|_2 \le dC_1\varepsilon$. The conditions on ε depend only on d and η. □

The perturbation bounds are standard. The point is that the constants come from R and b alone, with no separate assumption on the spectral gap.

9.3 The non-reversible case

Without reversibility the exit rates and the edge products no longer determine Q, since the time reversal $Q^* = D_\pi^{-1}Q^\top D_\pi$ has the same exit rates, the same products and the same stationary law. The right data are the principal minors of $-Q$. As in Section 3.2, $\lambda_F(v)$ is the smallest real eigenvalue of $-Q_F - \operatorname{diag} v_F$. For $A \subseteq F$ the principal submatrix of $-Q_F$ on A is $-Q_A$, since a

killed generator keeps the full exit rates on its diagonal. So the principal minors of $-Q_F$ are the numbers $\det(-Q_A)$, and they are the coefficients of the multiaffine polynomial

$$R_F(w) = \det\bigl(-Q_F - \operatorname{diag} w\bigr) = \sum_{A\subseteq F} \det(-Q_A) \prod_{i\in F\setminus A} (-w_i), \qquad w \in \mathbb{R}^F,$$

with $\det(-Q_\emptyset) = 1$.

Lemma 9.9 (irreducibility)**.** *If F is irreducible, then R_F is irreducible over $\mathbb{C}$.*

Proof. For $|F| = 1$ the polynomial has degree 1. Let $|F| \geq 2$ and suppose that $R_F = GH$ with G and H nonconstant. The polynomial R_F has degree exactly 1 in each w_i, as its coefficient at $\prod_F(-w_i)$ is 1, and degrees in w_i add, so the sets X and Y of variables of G and H partition F, both are nonempty, and G and H are multiaffine. Scale them so that the coefficients of $\prod_X(-w_i)$ in G and of $\prod_Y(-w_i)$ in H are 1, and write $G = \sum_{A\subseteq X} g_A \prod_{X\setminus A}(-w_i)$ and $H = \sum_{B\subseteq Y} h_B \prod_{Y\setminus B}(-w_i)$. Comparing coefficients gives $\det(-Q_{A\cup B}) = g_A h_B$, so $g_A = \det(-Q_A)$ and $h_B = \det(-Q_B)$ by taking $B = \emptyset$ or $A = \emptyset$. Hence

$$\det(-Q_U) = \det(-Q_{U\cap X}) \det(-Q_{U\cap Y}) \qquad \text{for every } U \subseteq F. \tag{30}$$

Since G_F is strongly connected, it has an arc $x \to y$ with $x \in X$ and $y \in Y$, and a shortest path from y back to x closes it into a simple directed cycle that meets X and Y. Let U be a minimal set, for inclusion, among the vertex sets of such cycles, and let C be such a cycle on U. Expand $\det(-Q_U)$ over the permutations σ of U. The permutations that preserve $U\cap X$ give $\det(-Q_{U\cap X})\det(-Q_{U\cap Y})$. Let σ be another permutation with a nonzero product. Every nontrivial cycle of σ is a directed cycle of G_U, and one of them meets X and Y. By minimality its vertex set is U, so σ is a single cycle through all of U, and its term is $\operatorname{sgn}\sigma \prod_{u\in U}(-q_{u\sigma(u)}) = (-1)^{|U|-1}(-1)^{|U|}\prod_u q_{u\sigma(u)} = -\prod_u q_{u\sigma(u)}$. Hence

$$\det(-Q_U) - \det(-Q_{U\cap X})\det(-Q_{U\cap Y}) = -\sum_\sigma \prod_{u\in U} q_{u\sigma(u)} < 0,$$

the sum over the Hamiltonian cycles of G_U, which include C. This contradicts (30). □

Al Ahmadieh [34, Lemma 5.1 and Corollary 5.6] proves over any field that a matrix is irreducible exactly when $\det(\operatorname{diag} x + A)$ is an irreducible polynomial. The proof above is a short version for the matrices $-Q_F$. The converse direction is immediate here, since a reducible $-Q_F$ makes R_F the product of the polynomials of its diagonal blocks.

Lemma 9.10 (the Perron branch)**.** *Let F be irreducible, let Q' be another generator whose face F is irreducible, and let $c \in \mathbb{R}$. Suppose that $\lambda'_F(v) = \lambda_F(v) + c$ for all v in a nonempty open set $\mathcal{U} \subseteq \mathbb{R}^F$. Then $-Q'_F$ and $-Q_F + cI$ have the same principal minors, and $\lambda'_F = \lambda_F + c$ on all of $\mathbb{R}^F$. In particular, for $c = 0$, $-Q'_F$ and $-Q_F$ have the same principal minors.*

Proof. Let $k = |F|$ and number the states of F as $1, \dots, k$. Put $\widetilde{R}(w) = \det(-Q_F + cI - \operatorname{diag} w) = R_F(w - c\mathbf{1})$, which is irreducible over $\mathbb{C}$ by Lemma 9.9. For $k = 1$ the hypothesis says $q'_1 = q_1 + c$, which is the claim. Let $k \geq 2$, and put $\psi(v) = v + \lambda'_F(v)\mathbf{1}$. For $v \in \mathcal{U}$ the number $\lambda'_F(v) = \lambda_F(v) + c$ is an eigenvalue of $-Q'_F - \operatorname{diag} v$ and of $-Q_F + cI - \operatorname{diag} v$, so $R'_F(\psi(v)) = 0 = \widetilde{R}(\psi(v))$.

Both polynomials are affine in w_1. Write $\widetilde{R} = p_1 w_1 + p_0$ and $R'_F = p'_1 w_1 + p'_0$ with p_1, p_0, p'_1, p'_0 polynomials in $w' = (w_2, \dots, w_k)$. The coefficient of $\prod_{i\geq 2}(-w_i)$ in p_1 is -1, so $p_1 \neq 0$. The polynomial $\mathcal{D} = p_1 R'_F - p'_1 \widetilde{R} = p_1 p'_0 - p'_1 p_0$ does not involve w_1 and vanishes on $\psi(\mathcal{U})$. We

claim that the projection of $\psi(\mathcal{U})$ to w' contains a nonempty open set. Fix $v^0 \in \mathcal{U}$ and consider $y \mapsto (\psi(v^0 + (0,y)))_{i\ge 2}$ for small $y \in \mathbb{R}^{k-1}$. The Perron root of $Q'_F + \operatorname{diag} v$ is simple, so λ'_F is real-analytic with $\partial\lambda'_F/\partial v_i = -\beta_i$, where $\beta = l \circ r > 0$ for the Perron vectors with $l \cdot r = 1$, and $\sum_i \beta_i = 1$. The derivative of the map at $y = 0$ is $I - \mathbf{1}\beta'^{\top}$ with $\beta' = (\beta_i)_{i\ge 2}$, of determinant $\beta_1 > 0$, and the inverse function theorem proves the claim. So $\mathcal{D} = 0$ and $p_1 R'_F = p'_1 \widetilde{R}$. The irreducible $\widetilde{R}$ divides $p_1 R'_F$, and it involves w_1, so it does not divide $p_1 \neq 0$, which is free of w_1. Hence it divides R'_F. Both have total degree k and the coefficient 1 at $\prod_i(-w_i)$, so $R'_F = \widetilde{R}$. Finally $\lambda'_F(v)$ is the smallest real root of $t \mapsto R'_F(v + t\mathbf{1}) = \widetilde{R}(v + t\mathbf{1})$, which is $\lambda_F(v) + c$. □

Delmas, Dronnier and Zitt [38, Theorem 2.1] prove a multiplicative analogue, for the functions $\eta \mapsto \varrho(K \operatorname{diag} \eta)$ of nonnegative matrices on $\{0,1\}^n$. The additive version above needs agreement only on an open set, and uses the irreducibility of R_F in place of their induction over the cube. Cohen [49] proved that $v \mapsto \varrho(A + \operatorname{diag} v)$ is convex for a Metzler matrix A, which is the concavity of λ_F in Proposition 3.4, and Friedland [50, Theorem 4.1] derived it from a variational formula of Donsker and Varadhan type and showed that for irreducible A it is affine only along the direction of $\mathbf{1}$. These are statements about the map $v \mapsto \varrho(A + \operatorname{diag} v)$. Lemma 9.10 is the inverse statement, that this map on an open set determines the principal minors of A, and we did not find it there.

The next lemma replaces the Schur estimate of Lemma 9.4, which needs a symmetric matrix, by the bound of Collatz and Wielandt. For a Metzler matrix B and a positive vector u, the Perron root of B is at most $\max_i (Bu)_i/u_i$ [1].

Lemma 9.11 (suppression without reversibility)**.** *Let Q be a generator, F a nonempty face, $\mathcal{K} \subset \mathbb{R}^F$ compact and $\eta > 0$. There is $\bar v_0$ such that for $\bar v \ge \bar v_0$, every field v with $v_F \in \mathcal{K}$ and $v_j = -\bar v$ off F, and every face $G \not\subseteq F$, we have $\lambda_G(v) \ge \lambda_{G\cap F}(v) - \eta$ if $G \cap F \neq \emptyset$, and $\lambda_G(v) \ge \bar v$ if $G \cap F = \emptyset$.*

Proof. Put $P = G \cap F$, $S = G \setminus F$ and $B = Q_G + \operatorname{diag} v_G$, whose Perron root is $-\lambda_G(v)$. If $P = \emptyset$, every row of B has off-diagonal sum at most q_s and diagonal $-q_s - \bar v$, so its row sums are at most $-\bar v$, and the bound of Collatz and Wielandt with $u = \mathbf{1}$ gives $\lambda_G(v) \ge \bar v$.

Let $P \neq \emptyset$ and $\varrho_P(v) = -\lambda_P(v)$, and let $\mathbf{J}_P$ have ones off the diagonal of P. For $\zeta > 0$ the Perron root of the irreducible Metzler matrix $Q_P + \operatorname{diag} v_P + \zeta \mathbf{J}_P$ is continuous in (ζ, v_P) and decreases to $\varrho_P(v)$ as $\zeta \downarrow 0$, so by Dini's theorem some ζ makes it at most $\varrho_P(v) + \eta/2$ for all $v_F \in \mathcal{K}$. Let x be its Perron vector with $\max x = 1$, and $x_* > 0$ the least entry of x over $\mathcal{K}$. Test B with $u = x$ on P and $u = \xi$ on S. For $p \in P$, since $\mathbf{J}_P x \ge 0$,

$$(Bu)_p \le \big((Q_P + \operatorname{diag} v_P)x\big)_p + \xi q_{\max} \le (\varrho_P + \tfrac{\eta}{2})x_p + \xi q_{\max} \le (\varrho_P + \eta)x_p$$

once $\xi \le \eta x_*/(2q_{\max})$. For $s \in S$,

$$(Bu)_s \le \sum_{p\in P} q_{sp}x_p + \xi\Big(\sum_{s'\in S\setminus\{s\}} q_{ss'} - q_s - \bar v\Big) \le q_{\max} - \xi\bar v \le (\varrho_P + \eta)\xi$$

once $\bar v \ge q_{\max}/\xi + |\varrho_P| + \eta$, which holds uniformly on $\mathcal{K}$ for large $\bar v$. There are finitely many P and S. So the Perron root of B is at most $\varrho_P(v) + \eta$, that is $\lambda_G(v) \ge \lambda_P(v) - \eta$. □

Lemma 9.12 (exposure by descent)**.** *Let Q be irreducible with a start of full support, and let F be an irreducible face with $|F| \ge 2$. For $v_F \in \mathbb{R}^F$ and $\bar v > 0$ write $(v_F, -\bar v)$ for the field equal to v_F on F and to $-\bar v$ off F. Outside a closed nowhere dense set $\mathcal{E} \subset \mathbb{R}^F$, every v^0 has a unique partner G, an irreducible face $G \subsetneq F$, with the following property. There are an open neighbourhood $\mathcal{U}$ of v^0, a real-analytic function $\tau_1 > 0$ on $\mathcal{U}$, and numbers $\delta > 0$ and $\bar v_0$, such that for $\bar v \ge \bar v_0$ and the fields $(v_F, -\bar v)$, on $O = \{(\tau, v_F) : v_F \in \mathcal{U},\ |\tau - \tau_1(v_F)| < \delta\}$,*

(i) the points with $\tau > \tau_1(v_F)$ lie in the strict winning region of F, and those with $\tau < \tau_1(v_F)$ in that of G,

(ii) on the hypersurface $\tau = \tau_1(v_F)$ we have $\lambda_F(v) = \lambda_G(v) - (|F| - |G|)/\tau_1(v)$.

The neighbourhood $\mathcal{U}$, the function τ_1 and the numbers δ and $\bar{v}_0$ depend on Q. For $F = V$ there is nothing to suppress. Hence the controlled boundaries determine, for every irreducible face F, the function λ_F on all of $\mathbb{R}^F$, up to one additive constant common to all faces, and $\lambda_V(0) = 0$ fixes the constant. In this comparison of Q with a second generator Q' the ranges of τ, δ and $\bar{v}$ are chosen for both, so they depend on Q', and unlike in the reversible case no design is fixed in advance. The statement is one of identifiability.

Proof. Let $k = |F|$. For an irreducible face $H \subsetneq F$, singletons included, put

$$\rho_H(v) = \frac{k - |H|}{\lambda_H(v) - \lambda_F(v)}, \qquad v \in \mathbb{R}^F.$$

The denominator is positive by Lemma 3.2, and ρ_H is real-analytic, since the Perron roots of irreducible Metzler matrices are simple. Note that $\Phi_{\tau,v}(F) < \Phi_{\tau,v}(H)$ exactly when $\tau > \rho_H(v)$.

Generic partners. Let $H_1 \neq H_2$ be two such faces with some $j \in H_1 \setminus H_2$, and take $v = te_j$ with $t \to \infty$. Then λ_{H_2} stays fixed, $\lambda_F \leq \lambda_{H_1} \leq q_j - t$ since a Perron root is at least every diagonal entry, and $\lambda_F \geq -t$ by the bound of Collatz and Wielandt with $u = \mathbf{1}$. Hence $\rho_{H_2} \to 0$, while $\rho_{H_1} \geq (k - |H_1|)/q_j > 0$, as $q_j > 0$ on the irreducible face F. So $\rho_{H_1} - \rho_{H_2}$ is a nonzero real-analytic function, with a closed nowhere dense zero set. The set $\mathcal{E}$ where the maximum of ρ_H over H is attained twice is closed and lies in the finite union of these zero sets.

The boundary. Let $v^0 \notin \mathcal{E}$, let G be the unique maximizer of $\rho_H(v^0)$ and put $\tau_1 = \rho_G$. Choose $\delta > 0$ with $3\delta < \tau_1(v^0)$ and $\rho_H(v^0) < \tau_1(v^0) - 3\delta$ for every irreducible $H \subsetneq F$ other than G. By continuity there is a neighbourhood $\mathcal{U}$ of v^0 with compact closure on which $\tau_1 > 2\delta$ and $\rho_H < \tau_1 - 2\delta$ for all these H. Then $\tau - \rho_H \geq \delta$ on the closure of O, and $\Phi(H) - \Phi(F) = (\lambda_H - \lambda_F)(\tau - \rho_H) \geq \kappa$ for some $\kappa > 0$. Also $\Phi(F) - \Phi(G) = (\lambda_G - \lambda_F)(\tau_1(v) - \tau)$ is negative above the hypersurface and positive below it, and a reducible face loses to one of its components (Proposition 9.1). So among the faces inside F the least score is attained only by F above and only by G below. Apply Lemma 9.11 to the closure of $\mathcal{U}$, with $\eta = 1/(2\tau_+)$ for an upper bound τ_+ of τ on O. Let $G' \not\subseteq F$ be admissible. If $P = G' \cap F$ is nonempty and C is a component of P with $\lambda_C(v) = \lambda_P(v)$, then $\Phi(G') \geq |P| + \tau\lambda_P(v) - \frac{1}{2} \geq \Phi(C) + \frac{1}{2}$. If P is empty, $\Phi(G') \geq \tau_- \bar{v}$ for a lower bound $\tau_- > 0$ of τ on O, which exceeds the scores inside F by $\frac{1}{2}$ for large $\bar{v}$. This proves (i), and the tie $\Phi(F) = \Phi(G)$ on the hypersurface is (ii).

The induction. Let Q' be a generator with the same controlled boundaries. By Lemma 3.2 the face V is the strict winner for Q on an open set of points (τ, v), so it wins strictly for Q' at some points, and Proposition 9.1 makes Q' irreducible. For $i \neq 1$ take $F = \{1, i\}$ for both generators at once, with the field δ on i and 0 on 1, τ in a compact subinterval of $(0, 1/(2\max(q_{\max}, q'_{\max}))]$, δ in a compact interval containing $q_i - q_1$ and $q'_i - q'_1$ in its interior, and $\bar{v}$ above the thresholds of Lemma 9.11 for Q and Q'. An irreducible pair has eigenvalue at least $\min(q_1, q_i - \delta) - \sqrt{w_{1i}}$ and loses to the better singleton by at least $\frac{1}{2}$, for both generators, and Lemma 9.11 removes the other faces. So the boundaries $\delta = q_i - q_1$ and $\delta = q'_i - q'_1$ agree. Put $c = q'_1 - q_1$, so that $\lambda'_{\{i\}} = \lambda_{\{i\}} + c$ for every i. We show by induction on $|F|$ that every irreducible face F of Q is irreducible for Q' and that $\lambda'_F = \lambda_F + c$ on $\mathbb{R}^F$. Let $|F| \geq 2$, choose $v^0 \notin \mathcal{E}$ with partner G, and take O as above. Outside a closed nowhere dense subset of O, the faces F above the hypersurface

and G below it are strict winners for Q' too, hence irreducible for Q'. By continuity of the scores of Q', $\Phi'(F) = \Phi'(G)$ on the hypersurface. With (ii) and the induction hypothesis for G,

$$\lambda'_F(v) = \lambda'_G(v) - \frac{|F| - |G|}{\tau_1(v)} = \lambda_G(v) + c - \frac{|F| - |G|}{\tau_1(v)} = \lambda_F(v) + c \qquad (v \in \mathcal{U}).$$

Lemma 9.10 extends this to all of $\mathbb{R}^F$. For $F = V$ both generators have $\lambda_V(0) = \lambda'_V(0) = 0$, since $\mathbf{1}$ is a positive null vector of Q and of Q'. So $c = 0$. □

The induction is an algorithm. For each face choose a field at random, decrease τ from a large value, and read the first change of label and its position. The lemma is the place where the boundaries of all faces, and not only those of pairs, are used.

For a face F that is irreducible for Q and a field v, let $\alpha^*_F(v)$ be the minimizer of $J_F - v \cdot \alpha$ over Δ_F (Proposition 3.4) and $K_F(v) = C_F(\alpha^*_F(v))$, the constant of Definition 5.1 at that point. So $K_F(0) = K_F$, and $K_F(v)$ is the constant of the face maxima with a field (Theorem 5.7).

Theorem 9.13 (non-reversible reconstruction)**.** *Let Q be irreducible with a start of full support, and suppose that the controlled boundaries of all irreducible faces are observed.*

(a) The boundaries determine every principal minor of $-Q$.

(b) Every generator Q' with the same principal minors is irreducible, has the same spectrum as Q, and the set of such generators is finite. It contains Q and its time reversal Q^*.

(c) If Q has no cut, that is, for every partition $V = X \cup Y$ with $|X|, |Y| \geq 2$ one of the blocks $Q[X, Y]$, $Q[Y, X]$ has rank at least 2, then the set in (b) is $\{Q, Q^*\}$, which is a single point exactly when Q is reversible.

(d) In general the set in (b) is the set of generator points of the cut-transpose orbit of $-Q$. Each cut-transpose of an irreducible Z-matrix C across a rank-one cut X, with $Y = V \setminus X$,

$$C = \begin{pmatrix} C_{XX} & -ab^\top \\ -cu^\top & C_{YY} \end{pmatrix} \mapsto \begin{pmatrix} C_{XX} & -ac^\top \\ -bu^\top & C^\top_{YY} \end{pmatrix}, \qquad a, b, c, u \geq 0,$$

is again an irreducible Z-matrix with the same principal minors, and exactly one positive diagonal similarity turns it into a generator. Fewer than $2d$ cut-transposes and one optional transposition reach every point of the fiber. If Q is reversible, the fiber is $\{Q\}$.

(e) If $d = 3$, the fiber is $\{Q, Q^*\}$ with no condition.

(f) There is a finite union $\mathcal{Z}$ of proper linear subspaces of $\mathbb{R}^V$ such that, for every start μ of full support outside $\mathcal{Z}$, every member $Q' \neq Q$ of the fiber has $K'_V(v) \neq K_V(v)$ for all v in a dense open set of fields. If Q is not reversible, then $\pi \in \mathcal{Z}$, and with the start π the law of the composition is the same under Q and Q^* for every N and T with $Tq_{\max} \leq N$. If Q' is the generator point of one cut-transpose across X, then for every start $K'_F = K_F$ on every face $F \subseteq X$ and $K'_F = K^*_F$ on every face $F \subseteq Y$.

Part (f) is stated for the amplitudes. To read $K_F(v)$ off the boundary positions one needs the crossing equation of Theorem 5.8 for the weighted law, which we do not prove here.

What is borrowed. The principal-minor fiber of an irreducible matrix is known, up to diagonal equivalence and transposition. Loewy [42, Theorem 1] identifies a matrix of order $n \geq 4$ without a cut from its principal minors, and Hartfiel and Loewy [41, Theorem 3] do the same for $n \leq 3$ with

no condition. Chatterjee, Ghosh, Gurjar and Raj combine the two in their Lemma 2.8 [36]. For $n \geq 4$, Al Ahmadieh [34, Theorem 4.1 and Lemma 4.5] shows that the fiber is a single class exactly when there is no cut or the matrix is diagonally equivalent to a symmetric one, and Chatterjee, Ghosh, Gurjar and Raj [36, Theorem 1.1] describe the whole fiber by cut-transposes. Kim and Landim [24, Section 5] extract the minors of orders one and two from J_V (their Lemma 5.1), deduce with the stationary law that J_V determines a reversible generator (their Corollary 5.3), and note that the uniform 3-cycle and its reversal share J_V (their Example 5.4). Corollary 9.3 answers the question left open by that example. In contrast, their Corollary 5.8 shows that the joint functional of the empirical measure and the empirical current determines the generator when every state is recurrent. So parts (c) and (e) and the reversible case of (d) are corollaries of the matrix theorems together with the normalization (N) in the proof below. What the paper adds is listed in Section 1.3.

Proof. We first record a normalization, which (b) to (e) use.

(N) The generator point. Let C be an irreducible Z-matrix with the same principal minors as $-Q$. The characteristic polynomial $\det(tI - C) = \sum_{s=0}^{d} (-1)^s t^{d-s} E_s(C)$, with E_s the sum of the principal minors of order s, is that of $-Q$. So $-C$ has the spectrum of Q, whose eigenvalue of largest real part is the simple eigenvalue 0. The theorem of Perron and Frobenius for the irreducible Metzler matrix $-C$ gives a vector $x > 0$, unique up to a factor, with $Cx = 0$. With $D = \operatorname{diag} x$ the matrix $-D^{-1}CD$ has off-diagonal entries $-c_{ij}x_j/x_i \geq 0$ and zero row sums, so it is an irreducible generator. If D' is real, diagonal and invertible and $-D'^{-1}CD'$ has zero row sums, then $C(D'\mathbf{1}) = 0$, so D' is a multiple of D and gives the same matrix. We call $-D^{-1}CD$ the generator point of C. For $C = -Q$ it is Q, and for $C = -Q^{\top}$, whose null vector is π, it is Q^*.

(R) Real similarities. If $-Q' = D(-Q)D^{-1}$ or $-Q' = D(-Q)^{\top}D^{-1}$ for a complex invertible diagonal D, then D may be taken real. In the first case $q'_{ij} = d_i q_{ij}/d_j$, so $d_i/d_j = q'_{ij}/q_{ij}$ is real on every edge of the graph of Q. That graph is strongly connected, so every ratio d_i/d_1 is a product of such ratios and is real, and D/d_1 is a real diagonal similarity. The second case is the same with q_{ji} in place of q_{ij}.

(a) Let Q' be a generator with the same controlled boundaries. The induction in the proof of Lemma 9.12 shows that Q' is irreducible and gives $\lambda'_V = \lambda_V$ on $\mathbb{R}^V$, and Lemma 9.10 with $F = V$ and $c = 0$ gives equal principal minors.

(b) Let Q' be a generator with the same principal minors, and suppose that $-Q'$ is reducible. Then there is a proper nonempty $C \subset V$ with $Q'[V \setminus C, C] = 0$, and $\det(-Q') = \det(-Q'_C)\det(-Q'_{V\setminus C})$. Both factors are proper principal minors, so they equal $\det(-Q_C)$ and $\det(-Q_{V\setminus C})$. By Lemma 3.2 every irreducible component of a proper face has a positive exit rate, so every eigenvalue of $-Q_C$ has positive real part and $\det(-Q_C) > 0$, and likewise $\det(-Q_{V\setminus C}) > 0$. This contradicts $\det(-Q') = \det(-Q) = 0$. So Q' is irreducible, and the same argument applies to every real matrix with the principal minors of $-Q$. This is the case $A = -Q$ of Al Ahmadieh [34, Corollary 5.6], which states it for $n \geq 4$ over any field. The spectra agree by the expansion in (N). The time reversal Q^* is a generator, since $Q^*\mathbf{1} = D_\pi^{-1}Q^{\top}\pi = 0$ and its off-diagonal entries are $\pi_j q_{ji}/\pi_i \geq 0$, and $-Q^* = D_\pi^{-1}(-Q)^{\top}D_\pi$ has the principal minors of $-Q$, since transposition and diagonal similarity preserve them. Finiteness follows from (d).

(c) For $d = 2$ a generator is fixed by q_1 and q_2, which are principal minors, and for $d = 3$ the claim is (e). Let $d \geq 4$. Loewy's theorem [42, Theorem 1] applies to the irreducible matrix $-Q$ without a cut, with no condition on $-Q'$. It gives an invertible diagonal D with $-Q' = D(-Q)D^{-1}$ or $-Q'^{\top} = D(-Q)D^{-1}$, and by (R) we may take D real. In the first case (N) with $C = -Q$ gives $Q' = Q$, and in the second case $Q' = D^{-1}Q^{\top}D$ and (N) with $C = -Q^{\top}$ gives $Q' = Q^*$. Finally

$Q^* = Q$ exactly when $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j, that is when Q is reversible.

(d) Let C be an irreducible Z-matrix with the principal minors of $-Q$, and let X be a cut. The block $C[X, Y]$ is nonzero, by irreducibility, and has nonpositive entries and rank 1. Every row is a nonnegative multiple of a fixed nonzero row, so $C[X, Y] = -ab^\top$ with $a, b \geq 0$, and likewise $C[Y, X] = -cu^\top$. The cut-transpose C' is then a Z-matrix. It has the same principal minors. Indeed, for $U \subseteq V$ write $U_X = U \cap X$ and $U_Y = U \cap Y$. If U_X or U_Y is empty, both minors agree since a determinant is invariant under transposition. Otherwise replace C_{XX} by $C_{XX} + tI$ with t large, so that the block of U_X is invertible. The Schur complement gives

$$\det C_U = \det C_{U_X} \det\bigl(C_{U_Y} - \kappa\, c_{U_Y} b_{U_Y}^\top\bigr), \qquad \det C'_U = \det C_{U_X} \det\bigl(C_{U_Y}^\top - \kappa\, b_{U_Y} c_{U_Y}^\top\bigr),$$

with the same scalar $\kappa = u_{U_X}^\top C_{U_X}^{-1} a_{U_X}$. The two right sides are equal by transposition. Both sides are polynomials in t that agree for large t, so they agree at $t = 0$. By (b), C' is irreducible, and by (N) it has exactly one generator point.

Now let Q' be a generator in the fiber, irreducible by (b). The theorem of Chatterjee, Ghosh, Gurjar and Raj [36, Theorem 1.1] gives a sequence $-Q = C_0, C_1, \ldots, C_k$ with $k < 2d$, each C_{s+1} a cut-transpose of C_s across a cut of C_s [36, Definition 2.10], such that $-Q'$ is diagonally similar to C_k or to $C_k^\top$. Their cut-transpose uses a particular rank-one factorization, and another choice changes it by a real diagonal similarity $\operatorname{diag}(\alpha I_X, I_Y)$ [36, Remark 2.11]. A real invertible diagonal similarity of C by $\operatorname{diag}(D_X, D_Y)$ changes the cut-transpose by $\operatorname{diag}(D_X, D_Y^{-1})$, as a short computation with the blocks shows, and does not change the cuts. So, up to real invertible diagonal similarity, which (N) accepts, every C_s is obtained by our cut-transposes, and Q' is the generator point of C_k or of $C_k^\top$. Conversely every such generator point lies in the fiber. There are fewer than 2^d cuts at each step and fewer than $2d$ steps, so the fiber is finite, which completes (b).

If Q is reversible, $S = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}$ is an irreducible symmetric Z-matrix, and a symmetric matrix is its own cut-transpose. Indeed, if $S[X, Y] = -ab^\top$, then $S[Y, X] = -ba^\top$, so we may take $c = b$ and $u = a$. By the rule for diagonal similarities above, the cut-transpose of a matrix diagonally similar to S is again diagonally similar to S, and so is the transpose $D^{-1}SD$ of DSD^{-1}. Since $C_0 = -Q$ is diagonally similar to S, so is every C_s. Hence $-Q'$ is diagonally similar to $-Q$, and (N) with $C = -Q$ gives $Q' = Q$. See also Engel and Schneider [39, Theorem 3.5] and Al Ahmadieh [34, Lemma 6.1 and Theorem 6.2].

(e) Let $d = 3$ and let Q' be in the fiber, so Q' is irreducible by (b). The theorem of Hartfiel and Loewy [41, Theorem 3], as restated in [36, Lemma 2.8], says that for an irreducible matrix A of order 2 or 3 over a field, every matrix with the principal minors of A is diagonally similar to A or to $A^\top$ over that field. So $-Q' = D(-Q)D^{-1}$ or $-Q' = D(-Q)^\top D^{-1}$ with D diagonal and invertible, real by (R), and (N) gives $Q' = Q$ or $Q' = Q^*$, as in (c).

(f) Let Q' be in the fiber and let F be irreducible for Q. The matrices $-Q_F$ and $-Q'_F$ have the same principal minors, so $R'_F = R_F$, which is irreducible by Lemma 9.9. A reducible $-Q'_F$ would make R'_F the product of the polynomials of its diagonal blocks, so F is irreducible for Q'. Then $\lambda'_F(v)$ and $\lambda_F(v)$ are the smallest real roots of the same polynomial $t \mapsto R_F(v + t\mathbf{1})$, so $\lambda'_F = \lambda_F$, and $J'_F = J_F$ on Δ_F° by the Legendre duality of Proposition 3.4. So $\alpha_F^*(v)$ and, by Proposition 5.2(a), $\det \Sigma$ at that point agree for Q and Q'. Let r and l be the right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$, so that $\alpha_F^*(v) = l \circ r$. The computation after Definition 5.1, with $q - v$ in place of q, shows that r attains the supremum defining $J_F(\alpha_F^*(v))$. So $r_{\alpha_F^*(v)}$ is a multiple of r, $l_{\alpha_F^*(v)}$ the matching multiple of l, and

$$K_F(v) = \frac{(\mu_F \cdot r)(l \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_{\alpha_F^*(v)}}}.$$

Hence $K'_F(v)/K_F(v) = a'_F(v)/a_F(v)$ with $a_F(v) = (\mu_F \cdot r)(l \cdot \mathbf{1})$.

Take $F = V$ and put $g(v) = r(v)\,(l(v) \cdot \mathbf{1})$, which does not depend on the scale of r, so that $a_V(v) = \mu \cdot g(v)$. The map g is real-analytic on $\mathbb{R}^V$. Let W be the span of the vectors $g'(v) - g(v)$ over $v \in \mathbb{R}^V$. If $\mu \notin W^\perp$, the analytic function $a'_V - a_V = \mu \cdot (g' - g)$ is not identically zero, so it is nonzero on a dense open set. We show that $W \neq 0$. Suppose $g' = g$. Then $r'(v)$ is a multiple of $r(v)$, and both matrices have the eigenvalue $\lambda_V(v)$, so $(Q' - Q)r(v) = 0$ for every v. Take $v = te_j$ with $t \to \infty$. Then $\lambda_V \leq q_j - t \to -\infty$, and row $i \neq j$ of the eigenvalue equation gives $(q_i - \lambda_V)r_i \leq q_{\max} \max r$, so $r(te_j)/\max r \to e_j$. Hence $(Q' - Q)e_j = 0$ for every j and $Q' = Q$. So for $Q' \neq Q$ the space $W^\perp$ is proper, and $\mathcal{Z}$ is the union of these spaces over the finitely many $Q' \neq Q$ in the fiber.

For the time reversal, $-Q^* - \operatorname{diag} v = D_\pi^{-1}(-Q - \operatorname{diag} v)^\top D_\pi$, so $r^* = D_\pi^{-1} l$ and $l^* = D_\pi r$, with $l^* \cdot r^* = 1$. At $\mu = \pi$ this gives $a^*_V(v) = (\mathbf{1} \cdot l)(\pi \cdot r) = a_V(v)$ for all v, so $\pi \in \mathcal{Z}$ when $Q^* \neq Q$. More is true. The matrix $P = I + hQ$ has stationary law π, and $P^* = D_\pi^{-1} P^\top D_\pi = I + hQ^*$ is stochastic for $hq_{\max} \leq 1$. For a path $x_1 \cdots x_N$ we have $\pi_{x_1} \prod_t P(x_t, x_{t+1}) = \pi_{x_N} \prod_t P^*(x_{t+1}, x_t)$. Reversal is a bijection of the paths that keeps the composition, so the law of the composition under Q from π equals that under Q^* from π, for every N and T with $hq_{\max} \leq 1$.

Finally let Q' be the generator point of the cut-transpose C' of $C = -Q$ across X, and put $s_0 = (b \cdot \mathbf{1})/(c \cdot \pi_Y)$. The rows X of $Q\mathbf{1} = 0$ give $-Q_X \mathbf{1} = (b \cdot \mathbf{1})a$, the columns Y of $\pi^\top Q = 0$ give $-Q_Y^\top \pi_Y = (\pi_X \cdot a)b$, and summing the columns X of $\pi^\top Q = 0$ gives $(\pi_X \cdot a)(b \cdot \mathbf{1}) = (\pi_Y \cdot c)(u \cdot \mathbf{1})$. With these identities, $x = (\mathbf{1}_X, s_0 \pi_Y)$ satisfies $C'x = 0$. So $Q' = -D_x^{-1} C' D_x$, and its killed blocks are $Q'_X = Q_X$ and $Q'_Y = D_{\pi_Y}^{-1} Q_Y^\top D_{\pi_Y} = Q_Y^*$. The constant $K_F(v)$ of a face depends only on Q_F and μ_F, which gives the last claim. □

A cut-transpose need not be Q or Q^*. Take $d = 4$, $X = \{1, 2\}$ and

$$Q = \begin{pmatrix} -4.7 & 0.7 & 1 & 3 \\ 1.3 & -9.3 & 2 & 6 \\ 2 & 2 & -4.4 & 0.4 \\ 1 & 1 & 2 & -4 \end{pmatrix},$$

whose blocks $Q[X, Y] = (1, 2)^\top (1, 3)$ and $Q[Y, X] = (2, 1)^\top (1, 1)$ have rank one. To four decimals, the generator point of the cut-transpose of $-Q$ across X is

$$Q' = \begin{pmatrix} -4.7 & 0.7 & 2.3810 & 1.6190 \\ 1.3 & -9.3 & 4.7619 & 3.2381 \\ 0.84 & 0.84 & -4.4 & 2.72 \\ 1.8529 & 1.8529 & 0.2941 & -4 \end{pmatrix}.$$

Its entries are rational, for instance $q'_{13} = 50/21$, $q'_{41} = 63/34$ and $q'_{43} = 5/17$. It has the fifteen principal minors of $-Q$, with $Q'_X = Q_X$ and $Q'_Y = Q_Y^*$, but $Q' \neq Q^*$, since $q^*_{12} = 351/530 \approx 0.6623$.

Remark 9.14 (three states by hand). For $d = 3$ the minors of orders 1 and 2 give the exit rates and the products w_{ij}, and $\det(-Q) = 0$ gives the sum and the product $w_{12}w_{23}w_{13}$ of the two oriented 3-cycle products $c = q_{12}q_{23}q_{31}$ and $c' = q_{13}q_{32}q_{21}$. So Q' has the cycle products of Q or of Q^* along every simple cycle, hence differs from it by a potential, and (N) gives (e) directly. The uniform 3-cycle and its reversal, Example 5.4 of Kim and Landim [24], are the case $c = 1$, $c' = 0$.

Proof of Corollary 9.3. (i) implies (iii). If $J_V = J'_V$ on Δ_V, the variational formula (Proposition 3.4) gives $\lambda_V(v) = \min_{\Delta_V}(J_V - v \cdot \alpha) = \lambda'_V(v)$ for every v, and Lemma 9.10 with $\mathcal{U} = \mathbb{R}^V$ and $c = 0$ gives equal principal minors.

(iii) implies (i). Equal principal minors give $R'_V = R_V$. For every v, $\lambda_V(v)$ is the smallest real root of $t \mapsto R_V(v + t\mathbf{1})$, so $\lambda'_V = \lambda_V$ on $\mathbb{R}^V$. By Proposition 3.4, J_V is the Legendre transform of $v \mapsto -\lambda_V(v)$ on Δ_V°, so $J'_V = J_V$ there. On the boundary we use that J_V is convex and lower semicontinuous, as a supremum of affine functions of α. For $\alpha \in \Delta_V$ and $\beta \in \Delta_V^\circ$ the points $\alpha_t = (1-t)\alpha + t\beta$ with $t \in (0,1]$ are interior. Convexity gives $\limsup_{t\downarrow 0} J_V(\alpha_t) \le J_V(\alpha)$, and lower semicontinuity the reverse inequality. So $J_V(\alpha) = \lim_{t\downarrow 0} J_V(\alpha_t)$, likewise for J'_V, and $J'_V(\alpha) = J_V(\alpha)$.

(ii) implies (iii). This is Theorem 9.13(a).

(iii) implies (ii). Let F be a face. Equal principal minors give $R'_F = R_F$. By Lemma 9.9 and the remark after it, F is irreducible for Q exactly when R_F is irreducible, so Q and Q' have the same irreducible faces. On such a face $\lambda'_F(v)$ and $\lambda_F(v)$ are the smallest real roots of $t \mapsto R_F(v + t\mathbf{1})$, so they agree, and the scores $\Phi_{\tau,v}$ of the irreducible faces agree. With a start of full support the irreducible faces are admissible, and every reducible admissible face has a larger score than one of its components (Proposition 9.1). So for both generators the strict winner at (τ, v) is the unique minimizer of $\Phi_{\tau,v}$ over the irreducible faces, when it exists, and the strict winners agree at every point. □

9.4 Finite horizons

Theorem 9.5 reads the coordinates off the limiting boundaries. At a finite horizon the observed boundaries are shifted, and the shift has an explicit form, uniformly over a class of chains and starts. Throughout, the connected graph G on $[d]$ and the numbers $0 < b \le R$ and $0 < \mu_* \le 1/d$ are fixed. Let $\mathcal{C}$ be the set of reversible irreducible generators on G with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on every edge, and let $\mathcal{M}$ be the set of starts with $\min_i \mu_i \ge \mu_*$. For a pair (i,j) and $\delta \in \mathbb{R}$ let $v^{ij}(\delta)$ be the design field of Section 9.2, and let $\mathcal{V}$ be the compact set of these fields with $|\delta| \le 3R$. For a face F and a field v put

$$M_F(T,v) = \max\bigl\{P_N(n)\, e^{Tv\cdot n/N} : \operatorname{supp} n = F\bigr\},$$

with $\max \emptyset = 0$, and recall $\alpha^*_F(v)$ and $K_F(v)$ from before Theorem 9.13.

Proposition 9.15 (uniform face maxima on the class)**.** *Let G induce a connected graph on the face F, with $k = |F| \ge 2$. There are N_0 and C, depending only on G, R, b and μ_*, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $v \in \mathcal{V}$ the following hold.*

(a) For all $N \ge 2$ and $T > 0$ with $TR \le N/2$, $M_F(T,v) \le C(T/N)^{k-1} e^{-T\lambda_F(v) + CT^2/N}$.

(b) For $N \ge N_0$ and $1 \le T \le N^{1/3}$, we have $C^{-1} \le K_F(v) \le C$ and

$$M_F(T,v) = K_F(v)\, N^{-(k-1)} T^{(k-1)/2} e^{-T\lambda_F(v)}(1+E), \qquad |E| \le C\Bigl(\frac{1}{T} + \frac{T^2}{N}\Bigr).$$

For a singleton, $M_{\{i\}}(T,v) = \mu_i(1 - Tq_i/N)^{N-1} e^{Tv_i}$ exactly.

Proof. For fixed (Q,μ), part (b) is Theorem 5.7 with the compact set $\mathcal{V}$ of fields, whose constants already do not depend on $v \in \mathcal{V}$. Part (a) is (7), that is, Lemma 4.1 with the Perron tilt $\theta = q - v - \lambda_F(v)\mathbf{1} \in V_F$, and the singleton is the path i^N. It remains to show that no constant depends on (Q,μ) either.

The parameter set is compact. Let $\mathcal{P} = \mathcal{C} \times \mathcal{M} \times \mathcal{V}$. On an edge $q_{ij} \ge b^2/R$, since $q_{ij}q_{ji} \ge b^2$ and $q_{ij} \le R$, and off the edges $q_{ij} = 0$. Reversibility is the closed condition of Kolmogorov on cycle

products, and $\|S\|_2 \le R$ is closed since π is continuous. So $\mathcal{P}$ is compact, and every connected face is irreducible for every $Q \in \mathcal{C}$.

Joint regularity. The Perron root is simple, so $\lambda_F(v)$, the normalized Perron vectors and $\alpha^*_F(v)$ are real-analytic in (Q, v). The proof of Proposition 5.2(a), run with Q as an extra variable, makes $\theta(\alpha)$, r_α, l_α, Σ_α, J_F and C_F real-analytic in (α, Q), and C_F is linear in μ. So $K_F(v)$ is continuous and positive on $\mathcal{P}$, which gives its bounds, and there is $\varepsilon_0 > 0$ with $\alpha^*_{F,i}(v) \ge 2k\varepsilon_0$ on $\mathcal{P}$. Put $\mathcal{K}_0 = \{\alpha \in \Delta_F : \alpha_i \ge \varepsilon_0 \ \forall i\}$.

The constants. They are the following, and each is continuous on $\mathcal{P}$ or on $\mathcal{P} \times \mathcal{K}_0$, or comes from a compactness argument run there. (i) The constant C_θ of Lemma 4.1 is built from the function b of Lemma A.2, which is continuous where the radius condition (34) holds strictly, as it does on V_F. So it is continuous along the Perron tilt. (ii) In Theorem 5.4 the Perron root in (34) at $\theta'(\alpha)$ is continuous and negative on $\mathcal{P} \times \mathcal{K}_0$, so one radius margin serves, and Lemma B.1 applies on the compact space $\mathcal{P} \times \mathcal{K}_0$, since the matrix $\mathcal{J}$ of Appendix A is continuous and positive definite and $\chi(\alpha, 0) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})/\alpha_1$ is continuous and positive. (iii) In Step 3 of the proof of Lemma 5.5 the lower bound for H comes from one word c^0 with at most $2k-1$ letters that walks along edges of G through every state of F. Its weight is at least $\mu_*(b^2/R)^{|c^0|-1}$, which bounds $H(T\alpha)$ below uniformly for $T \ge 1$ and $\alpha \in \mathcal{K}_0$. Step 4 uses only $q_{\max} \le R$ and $n_{\min} \ge \varepsilon_0 N$. (iv) In Theorem 5.7 the bounds for Σ_α^{-1}, for the third derivatives of J_F and for two derivatives of $\log C_F$ are continuous on $\mathcal{P} \times \mathcal{K}_0$, which gives one radius $\rho_0 \le \varepsilon_0$ and uniform c_U, C_U, C_1 and C_3. Away from the ball, $J_F(\beta) - v \cdot \beta - \lambda_F(v)$ is lower semicontinuous and positive on a compact subset of $\mathcal{P} \times \Delta_F$, hence at least 3η, and finitely many vectors $\theta_u = A_F u/u \in V_F$ serve, with constants bounded by (i). For bounded T, f_T is continuous and positive on $[1, T_2] \times \mathcal{P} \times \mathcal{K}_0$. □

Theorem 9.16 (finite-horizon recovery on the class)**.** *Run the design of Theorem 9.5 at horizon N, with $L = \log N$ and $T = \tau L$, and let $\mathcal{W}_N(\tau, v)$ be the set of supports of the maximizers of $n \mapsto P_N(n)e^{Tv\cdot n/N}$. For an ordered pair (i, j) with $\{i, j\} \in E$ put $v = v^{ij}(\bar\delta)$,*

$$\tau_N^- = \sup\Big\{\tau \in [\tfrac{1}{4R}, U] : \mathcal{W}_N(\tau', v) = \{\{i\}\} \text{ for all } \tau' \in [\tfrac{1}{4R}, \tau]\Big\},$$

$$\tau_N^+ = \inf\Big\{\tau \in [\tfrac{1}{4R}, U] : \mathcal{W}_N(\tau', v) = \{\{i, j\}\} \text{ for all } \tau' \in [\tau, U]\Big\},$$

let $t_{ij,N}$ be any number in $[\tau_N^-, \tau_N^+]$, and let $K_{ij} = K_{\{i,j\}}(v)$. Put $\varepsilon_N = \log L/(2L)$. There are N_0 and C, depending only on G, R, b and μ_*, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $N \ge N_0$ the following hold.

(a) (Singleton switches.) Let $i \ne 1$ and $s = \tau_0 L/N$. For every $\delta \in [-3R, 3R]$ other than

$$x_{i,N} = \frac{1}{\tau_0 L}\Big[\log\frac{\mu_1}{\mu_i} + (N-1)\log\frac{1 - sq_1}{1 - sq_i}\Big],$$

$\mathcal{W}_N(\tau_0, v^{i1}(\delta))$ is $\{\{1\}\}$ for $\delta < x_{i,N}$ and $\{\{i\}\}$ for $\delta > x_{i,N}$. Moreover $|x_{i,N} - x_i - 2R\log(\mu_1/\mu_i)/L| \le CL/N$, so $|x_{i,N} - x_i| \le 2R\log(1/\mu_*)/L + CL/N$.

(b) (Pair entries.) We have $\tau_N^+ - \tau_N^- \le C/L^2$ and

$$\Big|t_{ij,N} - t_{ij}(1 - \varepsilon_N) - \frac{c_{ij}}{L}\Big| \le C\,\frac{\log L}{L^2}, \qquad c_{ij} = -t_{ij}\Big(\frac{1}{2}\log t_{ij} + \log\frac{K_{ij}}{\mu_i}\Big), \qquad |c_{ij}| \le C.$$

(c) *(Reconstruction.)* *Let* $\widehat{Q}_N$ *be the reconstruction of Theorem 9.5(c) from the values* $x_{i,N}$ *and* $t_{ij,N}$, *and* $\widetilde{Q}_N$ *the one from* $x_{i,N}$ *and* $t_{ij,N}/(1-\varepsilon_N)$. *Both are reversible generators on* G, *and*

$$\|\widehat{Q}_N - Q\|_2 \le C\,\frac{\log L}{L}, \qquad \|\widetilde{Q}_N - Q\|_2 \le \frac{C}{L}.$$

Proof. All constants below depend only on G, R, b and μ_*. On the ranges of the design $T \in [L/(4R), UL]$, so for $N \ge N_1$ we have $TR \le N/2$, $T \le N^{1/3}$ and $1/T + T^2/N \le 5R/L$, and Proposition 9.15 applies.

Suppression. Fix a pair and a point of the ranges, and let m^* be the least score among the irreducible faces inside the pair. The proof of Lemma 9.4 uses only R, b and $\bar{v}$, so each connected face G' outside the pair has $\Phi_{\tau,v}(G') \ge m^* + \frac{3}{4}$, and Proposition 9.15(a) or the singleton atom gives $M_{G'} \le C T^{|G'|-1} N^{-\Phi_{\tau,v}(G')}$. The best face inside the pair has face maximum at least $c\,N^{-m^*}$, by the singleton atom with $\mu_i \ge \mu_*$ or by Proposition 9.15(b). Faces that are not connected carry no mass. So for $N \ge N_0$ every maximizer has its support inside the pair.

(a) At τ_0 the pair $\{i,1\}$, if it is an edge, has score at least the better singleton score plus $\frac{1}{2}$, by the proof of Theorem 9.5(a), so as above it is smaller than $\max(M_{\{1\}}, M_{\{i\}})$ by a factor $CLN^{-1/2}$. The two singleton atoms are exact, and

$$\log M_{\{i\}} - \log M_{\{1\}} = \log\frac{\mu_i}{\mu_1} + (N-1)\log\frac{1-sq_i}{1-sq_1} + \tau_0 L\,\delta,$$

which is affine and increasing in δ with zero $x_{i,N}$. This gives the claim on $\mathcal{W}_N$. Since $|\log(1-y)+y| \le y^2$ for $0 \le y \le \frac{1}{2}$, the middle term is $-\tau_0 L(q_i - q_1) + O(L/N) + O(R^2L^2/N)$, and dividing by $\tau_0 L = L/(2R)$ gives the expansion. For large N it gives $x_{i,N} \in (-3R, 3R)$, since $|x_i| \le R$.

(b) Let $\{i,j\}$ be an edge and $v = v^{ij}(\bar{\delta})$. The exact atoms give $\log M_{\{j\}} - \log M_{\{i\}} \le \log(1/\mu_*) - TR + 1$ with $TR \ge L/4$, so $\{j\}$ loses to $\{i\}$. By the proof of Theorem 9.5(b), $\lambda_{\{i\}}(v) - \lambda_{\{i,j\}}(v) = 1/t_{ij}$, and the exact atom gives $\log M_{\{i\}} = \log\mu_i - T(q_i - \bar{\delta}) + O(1/L)$. So Proposition 9.15(b) with $k=2$ gives, on the whole range,

$$D_N(T) = \log M_{\{i,j\}} - \log M_{\{i\}} = -L + \Xi(T) + \kappa + E_N(T), \qquad |E_N(T)| \le \frac{C_5}{L},$$

with $\Xi(T) = T/t_{ij} + \frac{1}{2}\log T$ and $\kappa = \log(K_{ij}/\mu_i)$. Here $|\kappa| \le C$, since K_{ij} is bounded above and below and $\mu_i \ge \mu_*$. The function Ξ increases with $\Xi' \ge 1/t_{ij} \ge 2/U$. Let T° solve $\Xi(T^\circ) = L - \kappa$. Then $D_N(T) < 0$ for $T \le T^\circ - C_5U/L$ and $D_N(T) > 0$ for $T \ge T^\circ + C_5U/L$. As $t_{ij} \in [1/R, U/2)$, the window $|T - T^\circ| < C_5U/L$ lies inside the range for large N. With the suppression step, every maximizer has support $\{i\}$ below the window and $\{i,j\}$ above it. Hence $T^\circ/L - C_5U/L^2 \le \tau_N^- \le \tau_N^+ \le T^\circ/L + C_5U/L^2$, which is the first claim, and $t_{ij,N} = T^\circ/L + O(1/L^2)$. To solve for T° put $\tau^\circ = T^\circ/L$. Then $\tau^\circ = t_{ij}\big(1 - (\frac{1}{2}\log T^\circ + \kappa)/L\big)$, so $\tau^\circ/t_{ij} = 1 + O(\log L/L)$ and $\log T^\circ = \log L + \log t_{ij} + O(\log L/L)$. Putting this back gives $\tau^\circ = t_{ij}(1-\varepsilon_N) + c_{ij}/L + O(\log L/L^2)$, uniformly, since $t_{ij} \le U/2$ and $|\kappa| \le C$.

(c) By (a) and (b) the dimensionless coordinates $x_{i,N}/R$ and $Rt_{ij,N}$ differ from x_i/R and Rt_{ij} by at most $C\log L/L$, and after the correction by at most C/L, since $t_{ij,N}/(1-\varepsilon_N) - t_{ij} = (c_{ij}/L + O(\log L/L^2))/(1-\varepsilon_N)$. Theorem 9.8 gives its admissible error size ε_0 and its constant as functions of d and b/R alone. So for $N \ge N_0$ these errors are below ε_0, and that theorem gives both bounds and the generator property. □

Remark 9.17 (a start that defeats every horizon). The bound $\min_i \mu_i \ge \mu_*$ cannot be dropped, since the start enters through the singleton switches as $2R\log(\mu_1/\mu_i)/L$. Let $d = 2$, $q_{12} = q_{21} = 1$,

$\tau = \frac{1}{4}$ and $v = (0, 1)$. The scores are $\Phi(\{2\}) = 0$, $\Phi(\{1\}) = \frac{1}{4}$ and $\Phi(\{1, 2\}) = 1 + \frac{1}{4} \cdot \frac{1-\sqrt{5}}{2} > \frac{3}{4}$, so for a fixed start $\{2\}$ wins for large N. Both singleton atoms carry the factor $(1 - T/N)^{N-1}$, so $M_{\{1\}}/M_{\{2\}} = (\mu_1/\mu_2)e^{-T} = (\mu_1/\mu_2)N^{-1/4}$ exactly. For the start $\mu_2 = 1/(1 + 2N^{1/4})$ this ratio is 2 at every N, while Lemma 4.1 with the Perron tilt, whose constant is bounded uniformly in the start, gives $M_{\{1,2\}} \le CLN^{-1/2}M_{\{1\}}$. So $\{1\}$ wins at every large N, against the limiting prediction. The switch between $\{1\}$ and $\{2\}$ in the variable v_2 sits at $\log(\mu_1/\mu_2)/(\tau L) = 1 + 4\log 2/L$ instead of at $q_2 - q_1 = 0$. On $\mathcal{M}$ the same shift is at most $\log(1/\mu_*)/(\tau L)$, uniformly.

Remark 9.18 (the size of the correction). On the two-state class with $R = 3$ and $b = 1$, the chain $q_1 = q_2 = 1$ with the uniform start has $t_{12} = 6.16$ and $c_{12} = -16.5$, and at $N = 10^7$ the exact law gives $t_{12,N} = 4.65$, against 5.63 for $t_{12}(1 - \varepsilon_N)$ and 4.60 for the prediction of (b), so the term c_{ij}/L, which no correction fixed in advance removes, dominates $t_{ij}\varepsilon_N$ at accessible N.

10 Low temperature and the cycle hierarchy

When the rates are exponentially small in a temperature, the exit rates of the faces are exponentially small too, and the competition between dimension and exit rate takes place on the scale of the Freidlin and Wentzell cycles of the chain. Away from a finite set of critical heights the winners are cycles of least size among those deep enough to survive. At a critical height a finite spectral problem decides, with noncycle winners on open windows.

The cycles, their depths and the exit times of a domain are classical. They go back to Freidlin and Wentzell [40], and we use them in the form of Catoni [35], of Olivieri and Vares [43, Chapter 6] and of Cirillo, Nardi and Sohier [37]. The depth of a face (Theorem 10.1) is a short consequence of Catoni's theorem on the exit time of a domain. The asymptotics of the exit rates with their prefactors (Proposition 10.3) are classical in substance, and go back to Wentzell [52] and to Freidlin and Wentzell [40, Chapter 6]. What is new here is the score that weighs the dimension of a face against its exit rate, the finite competition of envelopes at a critical depth, the open window on which a noncycle wins, and the reading of the cycle tree from the winners.

10.1 Arrhenius families

Let $V_{ij} \ge 0$ for $i \ne j$ be fixed costs and let

$$q_{ij}(\eta) = e^{-V_{ij}/\eta}, \qquad \eta > 0,$$

so every off-diagonal rate is positive at every positive temperature. We write Q_η for the generator, and for a face F we write $\lambda_F(\eta)$ for the exit rate λ_F of F under Q_η. In this section the letter h is a height, not the step size T/N. For a proper face F define its depth

$$D(F) = \lim_{\eta \downarrow 0} \big(-\eta \log \lambda_F(\eta)\big),$$

whose existence is part of Theorem 10.1. Put $D(V) = \infty$. A fixed height $h > 0$ sets the time scale

$$\tau_\eta = e^{h/\eta}, \qquad T = \tau_\eta L.$$

Throughout the section we take $N \to \infty$ first at fixed η and then $\eta \downarrow 0$, and every threshold in N may depend on η. The start μ_η may depend on η but has full support and does not depend on N. Since all rates are positive, every face is irreducible, and with a full-support start every face

satisfies (H) by Lemma 2.2. So Theorem 4.2(iii) applies at every fixed η to every face. At fixed η the face maxima of Theorem 5.7 give

$$\log M_F = -\Phi_{\eta,h}(F)\, L + \frac{|F|-1}{2}\log L + O_\eta(1), \qquad \Phi_{\eta,h}(F) = |F| - 1 + e^{h/\eta}\lambda_F(\eta). \tag{31}$$

So $\Phi_{\eta,h}$ is the exponent Φ_τ of Theorem 4.2 for the generator Q_η at $\tau = e^{h/\eta}$. The constants in (31) may depend badly on η, and nothing below exchanges the two limits.

We use the cycles of Freidlin and Wentzell in Catoni's convention [35, Definition 4.2]. A nonempty proper set C is a cycle if it is a singleton or if, for all distinct $x, y \in C$, the probability of leaving C before hitting y, starting from x, is exponentially small in $1/\eta$. Its depth $\Gamma(C)$ is the exponential scale of its mean exit time, which does not depend on the starting point in C [35, Proposition 4.6], and $\Gamma(\{i\}) = \min_{j\neq i} V_{ij}$. For a reversible family write $V_{ij} = H_{ij} - U_i$ with H symmetric, so that $\pi_i(\eta) \propto e^{-U_i/\eta}$. Then a proper set C with at least 2 states is a cycle exactly when every two of its points communicate strictly below its boundary height $b(C) = \min_{i\in C, j\notin C} H_{ij}$, that is, are joined by a path whose saddles H are all less than $b(C)$, and $\Gamma(C) = b(C) - \min_{i\in C} U_i$ [37, Definition 3.5, Propositions 3.10 and 3.15]. We call this the energy criterion. The inequality is strict, since a weak one would admit sets at a common saddle level that are not cycles.

Theorem 10.1 (the depth of a face)**.** *For every nonempty proper face F the depth $D(F)$ exists and*

$$D(F) = \max\{\Gamma(C) : C \subseteq F \text{ a cycle}\}.$$

In particular $D(C) = \Gamma(C)$ for every cycle C, and F contains a cycle C with $D(C) = D(F)$. Moreover, with $\mathcal{G}_F = (-Q_F)^{-1}$ and $M_F^{\mathrm{exit}} = \max_{i\in F} \mathbb{E}_i \tau_{F^c}$,

$$\frac{1}{M_F^{\mathrm{exit}}} \le \lambda_F(\eta) \le \frac{|F|}{M_F^{\mathrm{exit}}},$$

so that $D(F) = \lim_{\eta\downarrow 0} \eta \log M_F^{\mathrm{exit}}$.

Proof. Fix η and a proper face F, and write λ_F for $\lambda_F(\eta)$. By Lemma 3.2 with $v = 0$, $\lambda_F > \lambda_V = 0$. So $\mathcal{G}_F = \int_0^\infty e^{tQ_F}\, dt$ converges, and since $(e^{tQ_F})_{ij} = \mathbb{P}_i(X_t = j,\ t < \tau_{F^c})$ for the chain X with generator Q_η,

$$(\mathcal{G}_F)_{ij} = \mathbb{E}_i \int_0^{\tau_{F^c}} \mathbf{1}\{X_t = j\}\, dt = \mathbb{P}_i(\tau_j < \tau_{F^c})\, (\mathcal{G}_F)_{jj} \le (\mathcal{G}_F)_{jj}$$

by the strong Markov property at τ_j.

The exit-time bounds. The row sums of $\mathcal{G}_F$ are the mean exit times $\mathbb{E}_i\tau_{F^c}$. The spectral radius ρ of a nonnegative matrix is at most its largest row sum and at least each of its diagonal entries. Hence

$$\rho(\mathcal{G}_F) \le M_F^{\mathrm{exit}} \le \operatorname{tr} \mathcal{G}_F \le |F| \max_j (\mathcal{G}_F)_{jj} \le |F|\rho(\mathcal{G}_F).$$

The Perron vector of $-Q_F$ is positive and is an eigenvector of $\mathcal{G}_F$ for $1/\lambda_F$. A positive eigenvector of a nonnegative matrix belongs to its spectral radius [1], so $\rho(\mathcal{G}_F) = 1/\lambda_F$, which gives the bounds.

The depth. Fix $c > d$ and put $P_\eta = I + Q_\eta/c$. Every exit rate is less than d, so P_η is stochastic with holding probabilities at least $1 - d/c > 0$. Then $-\eta \log P_\eta(i,j) \to V_{ij}$ for $i \neq j$ and $-\eta \log P_\eta(i,i) \to 0$. So (P_η) has rare transitions in the sense of [35, Definition 2.1], with $\beta = 1/\eta$ and the rate function equal to V_{ij} off the diagonal and to 0 on it, and this rate function is

irreducible since all rates are positive. The chain P_η visits the same states in the same order as X, so the two chains have the same cycles. The continuous chain is the discrete one run at the jump times of a Poisson clock of rate c, so its mean exit times are the discrete ones divided by c. For a domain D and $x \in D$, Catoni's Proposition 4.2 [35] gives the limit $H_D(x)$ of β^{-1} log of the mean exit time from D started at x, and his Proposition 4.11 states that for every domain D other than the whole space, $\max_{x\in D} H_D(x) = \max\{\Gamma(C) : C \in \mathcal{M}(D)\}$, where $\mathcal{M}(D)$ is the set of maximal cycles contained in D (his Definition 4.3) and $\Gamma(C)$ is the depth of his Proposition 4.6. With $D = F$ this reads

$$\lim_{\eta\downarrow 0} \eta \log M_F^{\text{exit}} = \max_{C\in\mathcal{M}(F)} \Gamma(C).$$

These are cycles of the whole chain, not of a chain restricted to F. By the exit-time bounds $-\eta \log \lambda_F$ differs from $\eta \log M_F^{\text{exit}}$ by at most $\eta \log |F|$, so $D(F)$ exists and equals this limit. Every cycle $C \subseteq F$ lies in some $C' \in \mathcal{M}(F)$, and $\Gamma(C) \le \Gamma(C')$ since $\tau_{C^c} \le \tau_{C'^c}$ pathwise. So the maximum over $\mathcal{M}(F)$ is the maximum over all cycles in F, attained at some $C \in \mathcal{M}(F)$. If F is a cycle, then $\mathcal{M}(F) = \{F\}$ and $D(F) = \Gamma(F)$. □

Theorem 10.2 (noncritical heights)**.** *Let $\mathcal{D}$ be the finite set of depths of proper cycles and let $h \notin \mathcal{D}$. Let $k(h)$ be the least size of a cycle of depth greater than h, where V counts as a cycle of infinite depth and size d. Then for all sufficiently small η, every minimizer of $\Phi_{\eta,h}$ is a cycle of size $k(h)$ and depth greater than h, and every eventual support of the global mode, in the limit $N \to \infty$ at that η, is such a cycle.*

Proof. By Theorem 10.1 the depth of every proper face lies in $\mathcal{D}$ and differs from h. If $D(F) < h$, then $e^{h/\eta}\lambda_F(\eta) = e^{(h-D(F)+o(1))/\eta} \to \infty$. If $D(F) > h$, then $e^{h/\eta}\lambda_F(\eta) \to 0$. The face V has the score $d - 1$. Call a face deep if it is V or a proper face of depth greater than h.

A deep face of least size is a cycle. Indeed, a deep proper face F contains a cycle C with $D(C) = D(F) > h$ by Theorem 10.1, and C is deep, so $C = F$ if F has least size. Every cycle of depth greater than h is deep. So the deep faces of least size are the cycles of size $k(h)$ and depth greater than h.

There are finitely many faces, so for all sufficiently small η every face that is not deep has $\Phi_{\eta,h} > d$, and every deep face of size $k(h)$ has $\Phi_{\eta,h} < k(h)$. A deep face of larger size has $\Phi_{\eta,h} \ge k(h)$, since $\lambda_F \ge 0$. Hence every minimizer is a cycle of size $k(h)$ and depth greater than h, and Theorem 4.2(iii) at $\tau = e^{h/\eta}$ puts the support of every global mode among the minimizers for large N. □

Theorem 10.2 is a theorem about least cardinality. It does not say that every cycle of size $k(h)$ and depth greater than h wins, and the cycles that are minimal for inclusion among the deep ones can have other sizes and are then never selected. Among faces of equal size, minimizing $\Phi_{\eta,h}$ is minimizing $\lambda_F(\eta)$ at fixed η, whatever h is. So the deeper cycle wins first, and at equal depth the smaller prefactor of Proposition 10.3. An exponentially small difference of rates still decides, because $N \to \infty$ first. If two scores agree exactly at some η, the terms of (31) below L decide. The full-support start is needed, since otherwise some cycles are not admissible faces.

Proposition 10.3 (prefactors)**.** *For every proper face F there is $a_F \in (0, \infty)$ with $\lambda_F(\eta) = a_F e^{-D(F)/\eta}(1 + o(1))$, namely $a_F = 1/\rho(B_F)$ with $B_F = \lim_{\eta\downarrow 0} e^{-D(F)/\eta}\mathcal{G}_F(\eta)$, and*

$$D(F) = \nu\big(\det(-Q_F)\big) - \min_{i\in F} \nu\big(\det(-Q_{F\setminus\{i\}})\big),$$

where $\nu(f)$ is the exponent b of the first nonzero term $c\,e^{-b/\eta}$ of a finite sum of exponentials f, and the determinant of the empty matrix is 1.

Proof. Every minor of $-Q_F$ is a finite real combination of exponentials $e^{-b/\eta}$. After equal exponents are collected, a combination that is not identically zero has a first nonzero term $c\,e^{-\nu/\eta}$ and equals $c\,e^{-\nu/\eta}(1+o(1))$. The determinant of $-Q_F$ is positive, since its real eigenvalues are at least $\lambda_F > 0$ and the others come in conjugate pairs. The entries of $\mathcal{G}_F$ are positive by the identity in the proof of Theorem 10.1. By Cramer's rule each is a cofactor divided by $\det(-Q_F)$, so

$$(\mathcal{G}_F)_{ij} = \beta_{ij} e^{e_{ij}/\eta}(1+o(1)), \qquad \beta_{ij} > 0.$$

The diagonal cofactor of j is $\det(-Q_{F\setminus\{j\}})$, so $e_{jj} = \nu(\det(-Q_F)) - \nu(\det(-Q_{F\setminus\{j\}}))$. The exit-time bounds give $\max_j(\mathcal{G}_F)_{jj} \le 1/\lambda_F \le |F| \max_j(\mathcal{G}_F)_{jj}$, so $D(F) = \max_j e_{jj}$, which is the determinant formula. Since $(\mathcal{G}_F)_{ij} \le (\mathcal{G}_F)_{jj}$, we have $e_{ij} \le D(F)$ for all i, j. Hence $e^{-D(F)/\eta}\mathcal{G}_F$ converges entrywise to the matrix B_F with entries β_{ij} where $e_{ij} = D(F)$ and 0 elsewhere. It has a positive diagonal entry, so $\rho(B_F) > 0$. The spectral radius is continuous in the entries, and $\rho(e^{-D(F)/\eta}\mathcal{G}_F) = e^{-D(F)/\eta}/\lambda_F$. Hence $e^{D(F)/\eta}\lambda_F \to 1/\rho(B_F)$. □

The asymptotics of principal eigenvalues of matrices with entries of order $e^{-V_{ij}/\eta}$, and their expressions through graphs and determinants, are due to Wentzell [52] and Freidlin and Wentzell [40, Chapter 6], and the proof above is a short version for killed generators. The proposition uses the exact form of the rates. A family that satisfies only the logarithmic limits $-\eta \log q_{ij} \to V_{ij}$ need not have finite prefactors.

Theorem 10.4 (critical windows)**.** *Fix a depth $\gamma \in \mathcal{D}$ and $x > 0$, and let $T = xe^{\gamma/\eta}L$. Define the limiting score*

$$J_{\gamma,x}(F) = \begin{cases} +\infty, & D(F) < \gamma, \\ |F| - 1 + x a_F, & D(F) = \gamma, \\ |F| - 1, & D(F) > \gamma, \end{cases}$$

with V in the last case. If $J_{\gamma,x}$ has a unique minimizer F_x, then for all sufficiently small η, F_x is the unique minimizer of the score $|F| - 1 + xe^{\gamma/\eta}\lambda_F(\eta)$ and the eventual support of the global mode as $N \to \infty$ at that η. The same holds, with one threshold for η, uniformly for x in a compact interval on which one face has a positive gap. The lower envelope $x \mapsto \min_F J_{\gamma,x}(F)$ is concave and piecewise linear, and its minimizers need not be cycles.

Proof. Let $S_\eta(F) = |F| - 1 + xe^{\gamma/\eta}\lambda_F(\eta)$, which is $\Phi_\tau(F)$ for Q_η at $\tau = xe^{\gamma/\eta}$. By Proposition 10.3, $e^{\gamma/\eta}\lambda_F(\eta) = a_F e^{(\gamma - D(F))/\eta}(1+o(1))$, which tends to ∞, a_F or 0 as $D(F) < \gamma$, $D(F) = \gamma$ or $D(F) > \gamma$, and $S_\eta(V) = d-1$. On a compact interval $I \subset (0,\infty)$ the convergence $S_\eta(F) \to J_{\gamma,x}(F)$ is uniform in x, divergence included, since x is bounded below on I.

Suppose that one face F satisfies $J_{\gamma,x}(G) \ge J_{\gamma,x}(F) + g$ for all $G \ne F$ and $x \in I$, with $g > 0$. There are finitely many faces, so there is η_0 with $S_\eta(G) > S_\eta(F)$ for all $G \ne F$, $x \in I$ and $\eta < \eta_0$. A unique minimizer F_x at a single x has such a gap on a small interval around x, because the scores are continuous in x. Theorem 4.2(iii) at $\tau = xe^{\gamma/\eta}$ then gives the support of the global mode, with a threshold in N that may depend on η and x.

The function $x \mapsto J_{\gamma,x}(F)$ is affine for every face with $D(F) \ge \gamma$, and $J_{\gamma,x}(V) = d - 1$ is finite. So the envelope is the minimum of finitely many affine functions, which is concave and piecewise linear. Example 10.5 shows a minimizer that is not a cycle. □

At a tie of the limiting lines, Theorem 10.4 identifies the candidates only, and the exact scores at each η decide. Along the envelope the slope a_F of the active line decreases with x and its intercept $|F| - 1$ increases, as for the hull of Theorem 4.4. Putting $h = \gamma$ and treating the faces of

depth γ as faces of greater depth would discard the term xa_F, which is of order 1 in the coefficient of L.

Example 10.5 (a noncycle winner on an open window)**.** Take $U = (0, 1, 1, 1)$ and the symmetric saddles $H_{12} = 2$, $H_{13} = H_{23} = H_{34} = 3$, $H_{14} = H_{24} = 5$, so that with $t = e^{-1/\eta}$

$$(q_{ij})_{i\neq j} = \begin{pmatrix} * & t^2 & t^3 & t^5 \\ t & * & t^2 & t^4 \\ t^2 & t^2 & * & t^2 \\ t^4 & t^4 & t^2 & * \end{pmatrix}.$$

The only proper cycle with more than one state is $\{1, 2\}$, of depth 3. The faces of depth 3 are $\{1, 2\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$, with prefactors $a_{12} = 2$, $a_{123} = \frac{2}{3}$ and $a_{124} = 2$, so at $T = xt^{-3}L$ the limiting scores are $1 + 2x$, $2 + \frac{2}{3}x$, $2 + 2x$ and 3 for V. Hence the eventual winner is $\{1, 2\}$ for $0 < x < \frac{3}{4}$, the noncycle $\{1, 2, 3\}$ for $\frac{3}{4} < x < \frac{3}{2}$, and V for $x > \frac{3}{2}$. At $h = 3$ and $x = 1$ the noncycle beats every cycle. Its window has width $\log 2$ in the rescaled height $(h - 3)/\eta$ and collapses onto the single height $h = 3$, which is why it does not contradict Theorem 10.2.

Proof. Here $q_{ij} = t^{H_{ij}-U_i}$ and $\pi \propto (1, t, t, t)$. The exit rates are

$$q_1 = t^2 + t^3 + t^5, \qquad q_2 = t + t^2 + t^4, \qquad q_3 = 3t^2, \qquad q_4 = t^2 + 2t^4.$$

Cycles and depths. The singletons have depths $\min_j V_{ij} = 2, 1, 2, 2$. We test the sets with 2 or 3 states by the energy criterion. The set $\{1, 2\}$ has boundary height 3 and internal saddle $H_{12} = 2 < 3$, so it is a cycle of depth $3 - U_1 = 3$. A set that contains exactly one of the states 1 and 2 has the edge $H_{12} = 2$ to its complement, and no two of its points communicate below 2. The remaining sets $\{3, 4\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$ have boundary height 3. In the first two the state 3 is joined to the others only by saddles of height 3, and in the third the state 4 is joined to the others only by saddles of height 5. So none is a cycle. By Theorem 10.1, a face that contains $\{1, 2\}$ has depth 3 and every other proper face has depth at most 2. So $\mathcal{D} = \{1, 2, 3\}$, and the faces of depth 3 are the three named.

Prefactors. We apply Proposition 10.3. For $\{1, 2\}$, $\det(-Q_{12}) = q_1 q_2 - t^3 = 2t^4 + O(t^5)$, and the first column of the adjugate is $(q_2, t) = (t, t) + O(t^2)$, while the second column is $(t^2, q_1) = O(t^2)$. Hence $t^3\mathcal{G}_{12} \to B_{12}$ with first column $(\frac{1}{2}, \frac{1}{2})$ and second column 0, so $\rho(B_{12}) = \frac{1}{2}$ and $a_{12} = 2$. For $\{1, 2, 3\}$,

$$-Q_{123} = \begin{pmatrix} t^2 + t^3 + t^5 & -t^2 & -t^3 \\ -t & t + t^2 + t^4 & -t^2 \\ -t^2 & -t^2 & 3t^2 \end{pmatrix}.$$

Its determinant is $2t^6 + O(t^7)$. The first column of its adjugate has the leading terms $3t^3, 3t^3, 2t^3$, and every other entry of the adjugate is $O(t^4)$. So B_{123} has first column $(\frac{3}{2}, \frac{3}{2}, 1)$ and zeros elsewhere. Its eigenvalues are $\frac{3}{2}$ and 0, so $a_{123} = \frac{2}{3}$. For $\{1, 2, 4\}$ the determinant is $2t^6 + O(t^7)$, the first column of the adjugate is $(t^3, t^3, 0) + O(t^4)$ and the other entries are $O(t^4)$. So B_{124} has first column $(\frac{1}{2}, \frac{1}{2}, 0)$ and $a_{124} = 2$. These leading terms are the output of an exact computation in rational arithmetic.

The winners. At $\gamma = 3$ the faces of depth less than 3 have $J_{3,x} = \infty$. The lines are $1 + 2x$, $2 + \frac{2}{3}x$, $2 + 2x$ and 3. The face $\{1, 2, 4\}$ is always 1 above $\{1, 2\}$. Then $1 + 2x < 2 + \frac{2}{3}x$ exactly when $x < \frac{3}{4}$, $2 + \frac{2}{3}x < 3$ exactly when $x < \frac{3}{2}$, and $1 + 2x < 3$ exactly when $x < 1$. So the minimizer is unique and equals $\{1, 2\}$ on $(0, \frac{3}{4})$, $\{1, 2, 3\}$ on $(\frac{3}{4}, \frac{3}{2})$ and V on $(\frac{3}{2}, \infty)$, and Theorem 10.4 turns

each into the eventual support of the global mode, uniformly on compact subintervals. We claim nothing at the ties $x = \frac{3}{4}$ and $x = \frac{3}{2}$.

The noncycle. At $x = 1$, that is at $h = 3$, the score of $\{1, 2, 3\}$ is $\frac{8}{3}$. The cycles of depth less than 3 have infinite limiting score, and the only cycles of depth at least 3 are $\{1, 2\}$ and V, both of score 3. So the noncycle beats every cycle. In the rescaled height $s = (h - 3)/\eta = \log x$ the window is $\log \frac{3}{4} < s < \log \frac{3}{2}$, of width $\log 2$. Its heights $h = 3 + \eta s$ tend to 3, so at every fixed $h \neq 3$ the window is eventually left. □

The mechanism is that state 3 costs one dimension but returns the chain to $\{1, 2\}$ before it is killed at state 4, which lowers the leading exit rate from $2t^3$ to $\frac{2}{3}t^3$. On the middle interval this gain is worth more than the dimension. The selected face is the support of the counts, not of the limiting occupation fractions. For $\{1, 2, 3\}$ the right Perron vector tends to $(1, 1, \frac{2}{3})$ up to scale, and with $\pi \propto (1, t, t, t)$ the quasi-ergodic law α^*_{123} tends to the point mass at state 1.

Example 10.6 (a branch switch)**.** Take $U = (2, 0, 0)$, $H_{12} = H_{13} = 5$ and $H_{23} = 1$. Then $\{2, 3\}$ is a cycle of depth 5, the singleton depths are $3, 1, 1$, and the winners are $\{1\}$ for $0 < h < 3$, $\{2, 3\}$ for $3 < h < 5$ and V for $h > 5$. The two winning cycles are disjoint nodes of the same cycle tree, so the selector jumps between branches and never selects the nodes $\{2\}$ and $\{3\}$.

Proof. With $t = e^{-1/\eta}$ the rates are $q_{12} = q_{13} = t^3$, $q_{21} = q_{31} = t^5$ and $q_{23} = q_{32} = t$. So $\lambda_{\{1\}} = 2t^3$, $\lambda_{\{2\}} = \lambda_{\{3\}} = t + t^5$ and $\lambda_{\{2,3\}} = t^5$. By the energy criterion $\{2, 3\}$ is a cycle, and $\{1, 2\}$, $\{1, 3\}$ are not, since each has an edge of height 1 to its complement. Hence $\mathcal{D} = \{1, 3, 5\}$. Theorem 10.2 gives the winner $\{1\}$ on $(0, 3) \setminus \{1\}$, since $2t^3 < t + t^5$, the winner $\{2, 3\}$ on $(3, 5)$, where $k(h) = 2$, and V above 5. At $h = 1$, $\Phi_{\eta,1}(\{1\}) = 2t^2 \to 0$ while every other face has a score that tends to at least 1, so Theorem 4.2(iii) applies directly. We claim nothing at $h = 3$ and $h = 5$. □

10.2 Reading the cycle tree

We now ask the converse question of Section 9 at low temperature. Suppose that we observe only the winning face, as a function of the height and of a field. How much of the cycle structure do we learn? The unperturbed path shows the depth of every face that it selects and nothing else. Suppressing the complement of a face does not show its depth either. A calibrated reward on the face does. The depths then determine the cycles and the cycle tree, and the critical windows determine the prefactors of the faces on their envelopes.

A field is a vector $v = v(\eta) \in \mathbb{R}^d$, which may depend on η but not on N. As in Section 9.1 we write $\lambda_F(v) = -\varrho(Q_{\eta,F} + \operatorname{diag} v_F)$, with $Q_{\eta,F}$ the killed generator of F under Q_η, and

$$\Phi_{\eta,h}(F; v) = |F| - 1 + e^{h/\eta}\lambda_F(v),$$

so that $\Phi_{\eta,h}(F; 0) = \Phi_{\eta,h}(F)$. In this subsection a height h may be any real number. We say that a face G *wins at* (h, v) if there is $\eta_0 > 0$ such that for every $\eta < \eta_0$ the face G is the unique minimizer of $\Phi_{\eta,h}(\cdot\,; v)$ over all nonempty faces. Where two faces of equal size have identical rates, for instance two disjoint cycles built from the same costs, no face wins and nothing is observed.

Proposition 9.1 turns a win into an observation. At a fixed η it applies to Q_η, to the start μ_η, to the fixed vector $v(\eta)$ and to the single time scale $\tau = e^{h/\eta}$. Its proof uses only results stated for an irreducible admissible face of a general generator, and it allows a field of either sign. So if G wins at (h, v), then for every $\eta < \eta_0$ the support of every maximizer of the weighted law is G for all large N. This is the only probabilistic input below.

Two elementary facts are used throughout. The Perron root of a Metzler matrix is nondecreasing in its entries [1, Ch. 2], so $v \le w$ entrywise gives $\lambda_F(w) \le \lambda_F(v)$, and $\lambda_F(v + \kappa\mathbf{1}) = \lambda_F(v) - \kappa$ for every real κ. So adding a constant to the field changes no winner. And by Lemma 3.2, $\lambda_G(v) < \lambda_F(v)$ whenever $F \subsetneq G$. In particular $D(F) \le D(G)$ for $F \subseteq G$.

Proposition 10.7 (the unperturbed path)**.** *Let $v = 0$, and let a proper face W win at some height $h_0 \notin \mathcal{D}$. Then W is a cycle with $D(W) > h_0$, W wins at every $h \in [h_0, D(W)) \setminus \mathcal{D}$, and W wins at no $h > D(W)$. Hence the supremum of the heights at which W wins is $D(W)$.*

Proof. By Theorem 10.2, W is a cycle of size $k(h_0)$ and depth greater than h_0. Let $h \in [h_0, D(W)) \setminus \mathcal{D}$. A face of depth greater than h has depth greater than h_0, so $k(h) \ge k(h_0) = |W|$, and $k(h) = |W|$ since $D(W) > h$. By Theorem 10.2 again, for small η every minimizer of $\Phi_{\eta,h}$ is a cycle G with $|G| = |W|$ and $D(G) > h_0$. For such $G \ne W$, $\Phi_{\eta,h}(G) - \Phi_{\eta,h}(W) = e^{h/\eta}(\lambda_G - \lambda_W)$ has the sign of $\Phi_{\eta,h_0}(G) - \Phi_{\eta,h_0}(W)$, which is positive for small η. Hence W is the unique minimizer. For $h > D(W)$ we have $e^{h/\eta}\lambda_W \to \infty$, while $\Phi_{\eta,h}(V) = d - 1$. □

So the path shows the depth of every face it selects, as the supremum of the heights at which it is selected. The noncritical heights at which a face wins form an interval with finitely many critical heights removed, and these may be depths of faces that never win, so only the supremum is $D(W)$. The path shows nothing about the other faces, and a noncycle winner of a critical window occupies a single height. In Example 10.5 the singleton $\{4\}$ wins on $(0,1) \cup (1,2)$, since $q_4 < q_1 < q_3$ and $q_4 < q_2$ for small η, then $\{1,2\}$ wins on $(2,3)$ and V above 3. The cut at 1 is the depth of the cycle $\{2\}$, which never wins. The path shows $D(\{4\}) = 2$ and $D(\{1,2\}) = 3$ and nothing about the cycles $\{1\}$, $\{2\}$ and $\{3\}$. It is read against an absolute reference, since $\lambda_V(0) = 0$ makes the score of V exactly $d - 1$.

Killing the complement of a face at a large rate M does not read its depth. The killing gives V an exit rate below λ_F, and the winner leaves F only at the height where the gap between λ_F and that rate, the flux that returns to F through the killed states, becomes visible. This gap can be of smaller order than λ_F. For two states, $F = \{1\}$ and $M = e^{g/\eta}$, the exit rate λ of V is the smaller root of $(q_{12} - \lambda)(q_{21} + M - \lambda) = q_{12}q_{21}$, so the gap is $q_{12}q_{21}/(q_{21} + M - \lambda) = e^{-(V_{12}+V_{21}+g)/\eta}(1 + o(1))$, and the winner leaves $\{1\}$ at the height $V_{12} + V_{21} + g$ and not at $D(\{1\}) = V_{12}$.

The cure is to keep a face whose rate is known. For the unperturbed path this is V, with $\lambda_V(0) = 0$. Every field $v \ge 0$ keeps a version of it, since then $\lambda_V(v) \le 0$ and $\Phi_{\eta,h}(V; v) \le d - 1$. We therefore reward the target face instead of killing its complement. For a proper face G, a fixed constant $c > 0$ and a real γ put

$$v^{G,\gamma} = c\, e^{-\gamma/\eta}\mathbf{1}_G.$$

The field is constant on G, so $\lambda_G(v^{G,\gamma}) = \lambda_G - c\, e^{-\gamma/\eta}$ exactly. Define the threshold of G as

$$\vartheta(G) = \sup\bigl\{\gamma \in \mathbb{R} :\ G \text{ wins at } (h, v^{G,\gamma}) \text{ for some } h > \gamma\bigr\},$$

with $\sup \emptyset = -\infty$. It is read from the map $(h, v) \mapsto$ winner alone, at heights where a winner exists. The observer needs no list of critical heights, which are not known before the depths are.

The next two lemmas are standard facts of cycle theory. Laminarity is Catoni's Proposition 4.4 [35] and the tree of cycles of Trouvé [51] and of Olivieri and Vares [43, Chapter 6], and the strict growth of depths along nested cycles can be read off Catoni's Proposition 4.18. We include the short proofs for completeness, in the convention of cycles through hitting probabilities used here.

Lemma 10.8 (laminarity)**.** *Two cycles are disjoint or nested.*

Proof. Let C_1, C_2 be cycles with a common state z, and suppose that neither contains the other. Take $x \in C_1 \setminus C_2$ and $y \in C_2 \setminus C_1$. Since $y \notin C_1$, $\mathbb{P}_z(\tau_x < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_{C_1^c})$, which tends to 1 because C_1 is a cycle. Since $x \notin C_2$, $\mathbb{P}_z(\tau_{C_2^c} < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_y)$, which tends to 1 and contradicts that C_2 is a cycle. □

Lemma 10.9 (depth with a calibrated killing)**.** *Let $K \subseteq [d]$, $\gamma \in \mathbb{R}$ and $s = c\,e^{-\gamma/\eta}$. For every proper face G the limit $D^K(G) = \lim_{\eta \downarrow 0}\big(-\eta \log \lambda_G(-s\mathbf{1}_K)\big)$ exists, and*

$$D^K(G) = \max\{D^K(C) : C \subseteq G \text{ a cycle}\}.$$

Moreover $D^K(C) = D(C)$ if $C \cap K = \emptyset$, and $D^K(C) \leq \gamma$ if $C \subseteq K$.

Proof. Fix Λ larger than every element of $\mathcal{D}$. Add a cemetery state ∂. Every state of K jumps to ∂ at rate s, every other state of $[d]$ jumps to ∂ at rate $\varepsilon = e^{-\Lambda/\eta}$, and ∂ jumps to every state of $[d]$ at rate ε. All rates of this enlarged chain are positive. If $\gamma < 0$ we multiply all its rates by $e^{\gamma/\eta}$, which shifts every depth by $-\gamma$ and changes no hitting probability, hence no cycle. The proof of Theorem 10.1 uses the rates only through the limits of $-\eta \log$ of the rates and through the exit-time bounds, once the uniformization constant exceeds the largest total rate, here at most $\max(d-1, c) + d\varepsilon$. So it applies to the enlarged chain.

For $G \subseteq [d]$ the killed generator of G in the enlarged chain is $Q_G - \operatorname{diag}(s\mathbf{1}_{G\cap K} + \varepsilon \mathbf{1}_{G\setminus K})$. By the two elementary facts its exit rate λ'_G satisfies $\lambda_G(-s\mathbf{1}_K) \leq \lambda'_G \leq \lambda_G(-s\mathbf{1}_K) + \varepsilon$. Since $\lambda_G(-s\mathbf{1}_K) \geq \lambda_G$ and $D(G) < \Lambda$, we get $\lambda'_G = \lambda_G(-s\mathbf{1}_K)(1 + o(1))$. So Theorem 10.1 for the enlarged chain gives the existence of $D^K(G)$ and the identity $D^K(G) = \max D^K(C)$ over the cycles $C \subseteq G$ of the enlarged chain.

Each such C is a cycle of Q_η. Indeed, the enlarged chain is the chain X of Q_η run until an independent killing time, and then ∂. If X leaves C before it hits y, so does the enlarged chain. So for distinct $x, y \in C$ the probability $\mathbb{P}_x(\tau_{C^c} < \tau_y)$ for X is at most the same probability for the enlarged chain, which is exponentially small. Hence $D^K(G)$ is at most the maximum of $D^K(C)$ over the cycles $C \subseteq G$ of Q_η, and the reverse inequality holds by Lemma 3.2. If $C \cap K = \emptyset$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C$. If $C \subseteq K$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C + s \geq s$, so $D^K(C) \leq \gamma$. □

Lemma 10.10 (strict growth along nested cycles)**.** *If $C' \subsetneq C$ are cycles, then $D(C') < D(C)$. Hence $D(S) < D(C)$ for every nonempty proper subset S of a cycle C.*

Proof. Fix $z \in C \setminus C'$. Since C is a cycle there is $\alpha > 0$ with $\mathbb{P}_w(\tau_{C^c} < \tau_y) \leq e^{-\alpha/\eta}$ for all distinct $w, y \in C$ and all small η. For each η choose $y \in C'$ with $\mathbb{E}_y \tau_{C'^c} = M^{\text{exit}}_{C'}$. Start at y and cut the path into rounds. A round runs until the chain leaves C'. If it then sits at a state $w \in C \setminus C'$, the round continues until the chain returns to y or leaves C. A round ends outside C only if the chain leaves C' straight into C^c, which forces $\tau_{C^c} < \tau_z$, or if it leaves C from w before it reaches y. By the strong Markov property each round ends outside C with probability at most $2e^{-\alpha/\eta}$, and the rounds are independent and identically distributed. So the number R of rounds up to the exit from C has $\mathbb{E}R \geq e^{\alpha/\eta}/2$. Each round lasts at least its first part, of mean $M^{\text{exit}}_{C'}$, and R is a stopping time for the rounds. Wald's identity gives

$$M^{\text{exit}}_C \geq \mathbb{E}_y \tau_{C^c} \geq \mathbb{E}R \cdot M^{\text{exit}}_{C'} \geq \tfrac{1}{2} e^{\alpha/\eta} M^{\text{exit}}_{C'},$$

and $D(C) \geq D(C') + \alpha$ by the last statement of Theorem 10.1. For the second claim, Theorem 10.1 gives a cycle $C'' \subseteq S$ with $D(C'') = D(S)$, and $C'' \subsetneq C$. □

Theorem 10.11 (depths from a calibrated attraction)**.** *Fix $c > 0$.*

(a) *Let G be a proper face. If G wins at $(h, v^{G,\gamma})$ for some $h > \gamma$, then $\gamma \le D(G)$. Hence $\vartheta(G) \le D(G)$.*

(b) *Let C be a cycle, and put $D^-(C) = \max_{i \in C} D(C \setminus \{i\})$, with $D^-(C) = -\infty$ for a singleton. Then $D^-(C) < D(C)$, and for every $\gamma < D(C)$ the cycle C wins at $(h, v^{C,\gamma})$ for every $h \in (\max(\gamma, D^-(C)), D(C))$. Hence $\vartheta(C) = D(C)$.*

(c) *For every proper face F,*

$$D(F) = \max_{G \subseteq F} \vartheta(G).$$

Proof. Write $s = c\, e^{-\gamma/\eta}$ and $\sigma = e^{h/\eta}$, and write $\Phi(G)$ for the score of G under the field at hand.

(a) Let $\gamma > D(G)$ and $h > \gamma$. Then s is of smaller order than λ_G, so $\sigma(\lambda_G - s) \ge \sigma\lambda_G/2 \to \infty$. Since $v^{G,\gamma} \ge 0$, the score of V is at most $d-1$, so G is not a minimizer for small η.

(b) The inequality $D^-(C) < D(C)$ is Lemma 10.10. Fix $h \in (\max(\gamma, D^-(C)), D(C))$ and write $v = v^{C,\gamma}$. The score of C is $|C| - 1 + \sigma\lambda_C - \sigma s$, where $\sigma\lambda_C \to 0$ because $h < D(C)$ and $\sigma s = c\, e^{(h-\gamma)/\eta} \to \infty$ because $h > \gamma$. On another face G the field is $s\mathbf{1}_{G\cap C} = s\mathbf{1}_G - s\mathbf{1}_{G\setminus C}$, so

$$\lambda_G(v) = \lambda_G(-s\mathbf{1}_{G\setminus C}) - s.$$

If $G \cap C = \emptyset$, then $\lambda_G(v) = \lambda_G > 0$, so $\Phi(G) \ge 0$ while $\Phi(C) \to -\infty$. *If $G \subsetneq C$*, then $\lambda_G(v) - \lambda_C(v) = \lambda_G - \lambda_C$, and $D(G) \le D^-(C) < h < D(C)$, so $\sigma(\lambda_G - \lambda_C) \ge \sigma\lambda_G/2 \to \infty$. *If $G \supsetneq C$*, then $\lambda_C(v) - \lambda_G(v) = \lambda_C - \lambda_G(-s\mathbf{1}_{G\setminus C}) \le \lambda_C$, so $\Phi(G) - \Phi(C) \ge |G| - |C| - \sigma\lambda_C \to |G| - |C| \ge 1$. The face V is of this kind. *Otherwise* G meets C without containing it or lying in it. Put $K = G \setminus C$. Then $\lambda_G(v) - \lambda_C(v) = \lambda_G(-s\mathbf{1}_K) - \lambda_C$. A cycle $C' \subseteq G$ that met both C and K would be nested with C by Lemma 10.8, which is impossible since C' meets K and $C \not\subseteq G$. So C' lies in $G \cap C$ or in K. In the first case Lemma 10.9 gives $D^K(C') = D(C') \le D(G \cap C) \le D^-(C)$, and in the second $D^K(C') \le \gamma$. By Lemma 10.9, $D^K(G) \le \max(D^-(C), \gamma) < h$, so $\sigma\lambda_G(-s\mathbf{1}_K) \to \infty$, while $\sigma\lambda_C \to 0$.

In every case $\Phi(G) - \Phi(C)$ tends to $+\infty$ or has limit inferior at least 1. There are finitely many faces, so C wins at (h, v). The interval of h is nonempty for every $\gamma < D(C)$, so $\vartheta(C) \ge D(C)$, and (a) gives equality.

(c) For $G \subseteq F$, (a) and the monotonicity of D give $\vartheta(G) \le D(G) \le D(F)$. By Theorem 10.1 there is a cycle $C \subseteq F$ with $D(C) = D(F)$, and $\vartheta(C) = D(F)$ by (b). □

The theorem needs no assumption of uniqueness. Under its own attraction the score of C tends to $-\infty$ and every other score stays above it, so C wins even when other faces tie, as for two disjoint cycles of equal rates. This is the main advantage of the design over the unperturbed path. The theorem uses only the threshold in γ and not the right end of the heights at which C wins, since above $D(C)$ the faces that contain C compete through $\sigma(\lambda_C(v) - \lambda_G(v))$, which can diverge. A noncycle may win under its own attraction, for instance in a window, and its threshold may reach $D(G)$. This is why (c) takes the maximum over all subfaces and needs no prior knowledge of which faces are cycles. In the deep-singleton family with $V_{12} = V_{21} = 1$, $V_{13} = V_{23} = 3$ and $V_{31} = V_{32} = 5$, the singleton $\{3\}$ of depth 5 wins on all of $(0, 5)$ along the unperturbed path, so the path never shows the cycle $\{1, 2\}$ of depth 3, while Theorem 10.11(b) reads $\vartheta(\{1, 2\}) = 3$.

Theorem 10.12 (cycles from depths)**.** *For a nonempty proper face C the following are equivalent.*

(i) C is a cycle.

(ii) $D(C') < D(C)$ for every nonempty proper subset C' of C.

(iii) $|C| = 1$, *or* $D(C \setminus \{i\}) < D(C)$ *for every* $i \in C$.

Moreover $\Gamma(C) = D(C)$ *for every cycle. Hence the depth function determines the set of cycles, their depths and the cycle tree, that is, the laminar family of cycles ordered by inclusion with root* V*, and depths increase strictly along every branch.*

Proof. (i) implies (ii) by Lemma 10.10, and (ii) and (iii) are equivalent since D is monotone. Suppose (ii). Theorem 10.1 gives a cycle $C^* \subseteq C$ with $D(C^*) = D(C)$, and (ii) forces $C^* = C$. The identity $\Gamma(C) = D(C)$ is in Theorem 10.1, laminarity is Lemma 10.8, and the strict growth is Lemma 10.10. □

No nondegeneracy is assumed. Ties between the depths of different faces are allowed, because the cycles are strict and Lemma 10.10 rules out a tie along nested cycles. The weak form of the criterion fails. Take three states with all $V_{ij} = 1$. Then $\lambda_{\{1\}} = 2e^{-1/\eta}$ and $\lambda_{\{1,2\}} = e^{-1/\eta}$, so $D(\{1\}) = D(\{2\}) = D(\{1,2\}) = 1$. The set $\{1,2\}$ is not a cycle, since from state 1 the chain leaves it before it hits 2 with probability $\frac{1}{2}$. Criterion (ii) rejects it, and the weak criterion $D(C') \le D(C)$ accepts it. In Example 10.5 the faces $\{1,2,3\}$ and $\{1,2,4\}$ tie in depth with the cycle $\{1,2\}$ and are rejected for the same reason. Together, Theorems 10.11(c) and 10.12 say that the map $(h, v) \mapsto$ winner, restricted to the fields $v^{G,\gamma}$, determines the depth of every face, the cycles, their depths and the cycle tree.

The prefactors are read at a critical depth from the unperturbed path. Fix $\gamma \in \mathcal{D}$, take $v = 0$ and $T = xe^{\gamma/\eta}L$, and let $J_{\gamma,x}$ be the limiting score of Theorem 10.4. This is the height $h = \gamma + \eta \log x$, which moves with η, so here we observe the winner as a function of the window coordinate x. Write $e(x) = \min_F J_{\gamma,x}(F)$ for the lower envelope, and put $\bar{a}_G = a_G$ if $D(G) = \gamma$ and $\bar{a}_G = 0$ if $D(G) > \gamma$, so that $J_{\gamma,x}(G) = |G| - 1 + x\bar{a}_G$ when $D(G) \ge \gamma$.

Theorem 10.13 (prefactors from window endpoints)**.** *Suppose there are* $0 = x_0 < x_1 < \cdots < x_m < x_{m+1} = \infty$ *and faces* $G_0, \dots, G_m$ *with* $G_{k-1} \neq G_k$ *such that* G_k *is the unique minimizer of* $J_{\gamma,x}$ *for every* $x \in (x_k, x_{k+1})$.

(a) $D(G_m) > \gamma$, *and* $D(G_k) = \gamma$ *for* $k < m$.

(b) $\bar{a}_{G_m} = 0$, *and for* $k = m, \dots, 1$,

$$\bar{a}_{G_{k-1}} = \bar{a}_{G_k} + \frac{|G_k| - |G_{k-1}|}{x_k}.$$

Hence the points x_k *and the sizes determine* a_{G_k} *for every* $k < m$.

(c) Let G *be a face of depth* γ *other than* $G_0, \dots, G_m$, *and put* $s_G = \sup_{x>0}(e(x) - |G| + 1)/x$. *Then* $a_G \ge s_G$, *and every value* $a > s_G$ *in place of* a_G *gives the same unique minimizers. If* $G' \subseteq G$ *has depth* γ, *then* $a_G \le a_{G'}$.

Proof. By Theorem 10.4, G_k is the eventual winner, uniformly on compact subintervals of (x_k, x_{k+1}), so the faces G_k and the points x_k are observed. The envelope is continuous, so the lines of G_{k-1} and G_k meet at x_k, that is, $|G_{k-1}| - 1 + x_k\bar{a}_{G_{k-1}} = |G_k| - 1 + x_k\bar{a}_{G_k}$. This is the recursion in (b). Since G_{k-1} lies below G_k to the left of x_k and above it to the right, $\bar{a}_{G_{k-1}} > \bar{a}_{G_k} \ge 0$, and a face with $\bar{a}_G > 0$ has depth γ. As $x \to \infty$ every line of positive slope diverges, while $J_{\gamma,x}(V) = d - 1$. So the last piece has slope 0, that is, $D(G_m) > \gamma$ and $\bar{a}_{G_m} = 0$. This proves (a) and (b). For (c), the envelope is the minimum of the lines, so $|G| - 1 + xa_G \ge e(x)$ for all $x > 0$, which is $a_G \ge s_G$. If $a > s_G$, then $|G| - 1 + xa > e(x)$ for all $x > 0$, and the unique minimizers do not change. Finally $\lambda_G \le \lambda_{G'}$ for $G' \subseteq G$, and at equal depths Proposition 10.3 turns this into $a_G \le a_{G'}$. □

The reference is absolute because the last piece has slope exactly 0, and V, with $\lambda_V(0) = 0$, guarantees a face with a flat line. In Example 10.5 at $\gamma = 3$ the pieces are $G_0 = \{1, 2\}$, $G_1 = \{1, 2, 3\}$ and $G_2 = V$, with $x_1 = \frac{3}{4}$, $x_2 = \frac{3}{2}$ and sizes $2, 3, 4$. The recursion gives $a_{123} = (4 - 3)/\frac{3}{2} = \frac{2}{3}$ and $a_{12} = \frac{2}{3} + (3 - 2)/\frac{3}{4} = 2$, the values found in the example. The face $\{1, 2, 4\}$ is never on the envelope. Here $s_{124} = \frac{2}{3}$, attained on the whole interval $(\frac{3}{4}, \frac{3}{2})$, and $\{1, 2\} \subseteq \{1, 2, 4\}$ gives $a_{124} \le a_{12} = 2$. So $\frac{2}{3} \le a_{124} \le 2$. At $a_{124} = \frac{2}{3}$ the line of $\{1, 2, 4\}$ would coincide with that of $\{1, 2, 3\}$ on the middle interval, so the hypothesis of Theorem 10.13 excludes the lower end, but the observed winners alone do not. The window thus recovers $a_{12} = 2$ and $a_{123} = \frac{2}{3}$ and only bounds a_{124}, whose value is 2. A calibrated field does not obviously help, because a field changes the rate of V to a value that depends on Q, and the slope of V is then unknown. In this example the limit of π and the limiting quasi-stationary laws of the three faces of depth 3 all sit at state 1. So to first order a field $c\,e^{-3/\eta}$ at state 1 moves every line of the window, that of V included, by $-cx$ and leaves the endpoints where they are, and a field of this order at another state moves no line. Whether some other calibrated field separates a_{124} from a_{12} is open.

Remark 10.14 (what the design needs). The attraction design needs three calibrated quantities. The first is the time unit, so that the height h, equivalently $T = e^{h/\eta}L$, and the window coordinate x are known. The second is the amplitude $c\,e^{-\gamma/\eta}$ of the field in the same unit. The third is the labelling of the states, which places the field on G and gives the sizes $|G|$. It needs no rate of Q, and no step of this subsection uses reversibility. Theorem 10.1 rests on Catoni's theorem and on the exit-time bounds, and the other lemmas use only the Green matrix, the definition of a cycle and the strong Markov property. Lemma 10.9 applies Theorem 10.1 to a chain with a cemetery, which is never reversible, so even for a reversible family the proof of Theorem 10.11 goes through the general theorem. The selection step is Proposition 9.1, which holds for a general generator.

Section 11 records a check of Theorem 10.11 with the finite-η scores on the 138 proper faces of 11 cost matrices, reversible, non-reversible and tied, with no violation.

11 Verification and open questions

Every theorem, proposition, lemma and corollary of this paper is proved in the text or in the appendices. The only statement we do not prove is Hypothesis L for facets with three or more states, and Theorem 6.7 uses it only for faces with four or more states. We quote results of [28] where we say so. No proof rests on a computation. The formal proofs and the computations below check parts of the paper independently.

Formal proofs

Many finite and algebraic steps are proved in Lean 4 (v4.33.1, with Mathlib) in the folders `lean/PaperC` and `lean/PaperB` of the repository named under Code availability, and the file `lean/PaperC/MAP.md` matches each statement of the paper to its Lean theorems. No proof there uses `sorry` or `admit`. The files cover the run expansion (Lemma 2.3), the lower bound and the reversible form of the rate (Proposition 3.3), the restriction identity (Proposition 3.1), the finite form of Theorem 4.4, the tree formula of Proposition 5.2(b) for two and three states, the exact parts of Example 5.14, the exact statements on the complete graph and near a vertex in Sections 7 and 8, the multiaffine expansion of R_F, the normalization (N) and the fact that a diagonal similarity keeping a Metzler matrix Metzler has entries of one sign, part (e) of Theorem 9.13 for positive rates, and the determinants, adjugates and prefactors of Example 10.5. Mathlib has no Perron

and Frobenius theorem for irreducible Metzler matrices, so the positive null vector in (N) is a hypothesis of the Lean statements, and the asymptotic results are not in Lean.

Computations

The scripts are in the folder named under Code availability. Each check is an assertion, and the checks in floating point use a tolerance. None of them is interval-certified. The script `verify_all.py` runs all eleven with `py -I` and writes each output, headed by the sha256 of its script, to the folder `data`, in about fifteen minutes on a laptop. The older scripts `verify_first_order.py`, `verify_second_order.py`, `verify_forest.py`, `verify_rate_flows.py`, `verify_weak_field.py` and `verify_examples.py` compute the law in two independent ways and check Sections 2 to 5 and 7. The script `verify_layer.py` checks the constants and kernels of Section 6, including the inequality $I_0/I_1 \le 1 + 2/\zeta$. The script `verify_reconstruction.py` checks Section 9. It recovers Q from the strict winners on 40 random reversible chains, checks the constant $\frac{20}{3} + \frac{28}{9} < 10$ in the proof of Theorem 9.8 and the bounds ϕ_* and g_* on 2000 chains, checks on 64 random generators with a rank-one cut that the generator point of the cut-transpose has the principal minors of $-Q$ (in 49 of them it is neither Q nor Q^*), and checks on 200 generators with three states that every generator with the same minors is Q or Q^*. The script `verify_finite_horizon.py` computes the entry times of Theorem 9.16 on the path $1 - 2 - 3$ for $N = 10^3$ to 10^6 from the run expansion. The script `verify_cycles.py` redoes Examples 10.5 and 10.6 in exact arithmetic, and `verify_attraction.py` tests Theorem 10.11 on 11 cost matrices and 138 proper faces at $t = 10^{-32}$ and 10^{-48}. Every cycle has the grid threshold $D(G) - \frac{1}{4}$, no noncycle exceeds it, and $\max_{G \subseteq F} \vartheta(G)$ gives $D(F)$ on every face. The figures illustrate the theorems, and no statement is deduced from them.

Open questions

A local law up to the lower-dimensional corners of a face is open, and so is the facet layer for faces with four or more states without Hypothesis L. For a reducible face we expect the power of T in M_F to be $(k^* - 1)/2 + k_F - k^*$, between the bounds of Theorem 4.2(ii). The crossings of Theorem 5.8 are known to be unique only in special cases. The crossing equation for the weighted law, which would read the amplitudes $K_F(v)$ of Theorem 9.13(f) off the boundaries, is not proved. At low temperature it is open whether some calibrated field separates a_{124} from a_{12} in Example 10.5.

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Data availability. The outputs of the computations described in Section 11 are in the folder `kagey-problems/problem131/paperC/data` of the repository named under Code availability, each headed by the sha256 of the script that wrote it.

Materials availability. Not applicable.

Code availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems. The scripts of this paper are in the subfolder `paperC`. To rerun the checks, run `py -I verify_all.py` in that subfolder, which takes about fifteen minutes and writes the outputs to `paperC/data`. The option `--full` adds the largest cases of three of the older scripts and takes much longer. Each script also runs on its own, and its docstring says what it checks and how long it takes. The scripts need Python 3 with `numpy`, `scipy`, `sympy` and `mpmath`. The Lean files are in the subfolder `lean` and build with Lean 4 v4.33.1 and Mathlib, and the file `lean/PaperC/MAP.md` matches the statements of this paper to their Lean theorems.

Author contributions. I am the sole author and am responsible for the whole paper.

A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 5.2(a), Lemma 4.1, Theorem 5.4, Lemma 5.5 and Theorem 4.2(ii). Its last section proves Lemma A.3, which is used for Proposition 5.11. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [1]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$, and $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M (Lemma 3.2).

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 5.1, with (G, π) in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (9). Put $D = \operatorname{diag} \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root)**.** *Let G, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \operatorname{diag} \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \tag{32}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ, which is Γ without the row and column of state 1, is positive definite.

Proof. We differentiate the eigenvalue equation twice at $\eta = 0$. Since $G + \operatorname{diag}\eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. We write subscripts for derivatives at $\eta = 0$. Differentiating $(G + \operatorname{diag}\eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, since $\pi^\top G = 0$. Then $v_j = Ze_j$, because by (9) both vectors solve $Gx = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i, gives $Gv_{ij} + \operatorname{diag}(e_i)v_j + \operatorname{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$. We multiply by $\pi^\top$ again. The terms with G, κ_i and κ_j vanish, and we get $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (32) let $\bar\eta = \eta - (\pi \cdot \eta)\mathbf{1}$. By (9), $-Gu = \bar\eta$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2\sum_i \pi_i \eta_i u_i = 2\sum_i \pi_i \bar\eta_i u_i = -2\sum_i \pi_i u_i (Gu)_i$. Now expand the squares in (32). By $G\mathbf{1} = 0$ and $\pi^\top G = 0$ the terms with u_i^2 and with u_j^2 each sum to $-\sum_i \pi_i G_{ii} u_i^2$, and the cross terms sum to $-2\sum_i \pi_i u_i (Gu)_i + 2\sum_i \pi_i G_{ii} u_i^2$. So both sides of (32) agree. Also $\Gamma\mathbf{1} = 0$ by (9). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of G, so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $\bar\eta = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ. □

Proof of Proposition 5.2(a). Step 1: the derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag}\vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$ (the shift property). It is a simple eigenvalue with positive right and left eigenvectors $\hat r, \hat l$, normalized by $\sum_i \hat r_i = 1$ and $\hat l \cdot \hat r = 1$, and by the implicit function theorem Λ, $\hat r$ and $\hat l$ are real-analytic in ϑ. Put $\pi = \hat l \circ \hat r$ and $G = \hat R^{-1}(Q_F + \operatorname{diag}\vartheta - \Lambda(\vartheta)I)\hat R$ with $\hat R = \operatorname{diag}\hat r$. Then $G\mathbf{1} = 0$ and $\pi^\top G = 0$ by the two eigenvector equations, so G is an irreducible generator with stationary law π. Since $\hat R$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \operatorname{diag}\eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from (G, π).

Step 2: the gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3: the tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 5.1 and (8). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4: the Hessian of J_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $J_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla J_F = \beta$ remains. So $\nabla^2 J_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 5.1. □

A.2 The torus representation

The generating function of the words in H is Good's resolvent formula [48, (2)], and the torus integral below is his constant-term formula, here for the killed matrix A_F and with the circle of one state integrated out. We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\ge 2}$, $b_1 = (q_{i1})_{i\ge 2}$ and $A' = (q_{ij})_{i,j\ge 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z') b_1$ and

$$b(z') = \big(\mu_1 + \mu'^\top X(z') b_1\big)\big(1 + a_1^\top X(z')\mathbf{1}\big).$$

Lemma A.2 (torus representation)**.** *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \ge s_i$ for every $i \ge 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \le g(s')$ and $|b(z')| \le b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Big(t_1 g(z') + \sum_{i\ge 2} t_i z_i\Big)\, b(z') \prod_{i\ge 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

In words, $H(t)$ is a Cauchy integral over $k-1$ circles, one for each state of F except state 1. The circle of state 1 has already been integrated out, and it leaves the factor $e^{t_1 g(z')}$.

Proof. Step 1: the series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. So the series $X(z') = \sum_{j\ge 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \ge s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2: a radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1} A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon \mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \ge 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3: k circles. Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) = \varrho(S^{-1} A_F) < 1$ (the two matrices are similar), expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}} \prod_{i\in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \ge 1$ and $c_j \ne c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi}\int_0^{2\pi} e^{t_i z_i} z_i^{1-m}\, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \ge 1$ and 0 if $m \le 0$. We multiply the series by $e^{\langle t, z\rangle} \prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z\rangle}\, \mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i\in F} z_i \, d\varphi. \tag{33}$$

Step 4: the first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \le g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z')\mathbf{1}.$$

We integrate over φ_1 in (33) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi}\int_0^{2\pi} e^{t_1 z_1} z_1\, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula. □

A.3 Proof of Lemma 4.1

We bound the sum of Lemma 2.3 term by term by a continuum sum H_σ with slightly larger rates, and H_σ by the largest modulus of the integrand of Lemma A.2.

Step 1: the radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \geq 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1, which reads $a_1^\top u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{34}$$

Step 2: a continuum sum. Fix n with support F and h with $hq_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{hq_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \leq n_i$ for every i are nonzero, and in them $h^{m_i - 1}\binom{n_i - 1}{m_i - 1} \leq t_i^{m_i - 1}/(m_i - 1)!$ and $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)} \leq e^{-q_i t_i} \sigma^{m_i}$. Note that $h^{M-1} = h^{k-1} \prod_i h^{m_i - 1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F (a word with M letters has $M - 1$ switches). So

$$P_{|n|}(n) \leq \sigma\, h^{k-1} e^{-\langle q, t\rangle} H_\sigma(t), \tag{35}$$

where H_σ is H for σA_F.

Step 3: the bound for H_σ. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus with radii $\sigma\theta'$ we have $|g_\sigma| \leq \sigma g(\theta') = \sigma\theta_1$ by (34), $|b_\sigma| \leq b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t, \theta\rangle} b(\theta') \sigma^{k-1} \prod_{i \geq 2} \theta_i$, and for every $\sigma > 0$

$$H_\sigma(t) \leq \sigma^{k-1} b(\theta') \prod_{i \geq 2} \theta_i\, e^{\sigma\langle t, \theta\rangle}. \tag{36}$$

With (35) this gives $P_{|n|}(n) \leq \sigma^k b(\theta') \prod_{i \geq 2} \theta_i\, h^{k-1} \exp(-h\langle n, q - \theta\rangle + (\sigma - 1)h\langle n, \theta\rangle)$. Finally $\sigma \leq e^{1/2}$ and $\sigma - 1 \leq 2hq_{\max}$, so the lemma holds with

$$C_\theta = \max\Big(e^{k/2} b(\theta') \prod_{i \geq 2} \theta_i,\ 2q_{\max} \max_i \theta_i \Big).$$

By Lemma A.2, b is continuous on the open set of s' with $\varrho(A' - \operatorname{diag} s') < 0$, which contains θ' for every $\theta \in V_F$ by (34). So C_θ is bounded on compact subsets of V_F. □

A.4 Proof of Theorem 5.4

The field cancels from the two sides of Theorem 5.4, so we take $v = 0$. For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \operatorname{diag}(\theta')$. By (34) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q, \alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha, \varphi)} \chi(\alpha, \varphi)\, d\varphi, \tag{37}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha, \varphi) = \alpha_1 g(z') + \sum_{i \geq 2} \alpha_i z_i$ and $\chi(\alpha, \varphi) = b(z') \prod_{i \geq 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k - 1$.

Hypothesis (iii). By (34), $\Psi(\alpha,0)=\langle\alpha,\theta\rangle$ is real. By Lemma A.2, $|g(z')|\le g(\theta')=\theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi)\le\Psi(\alpha,0)-\sum_{i\ge2}\alpha_i\theta_i(1-\cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi=0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'),z')=0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z\in\mathbb{R}^F$ near θ let $\lambda(z)=\varrho(A_F-\operatorname{diag}z)$. With $R=\operatorname{diag}r$ we have $A_F-\operatorname{diag}(\theta-\eta)=R(\widehat{Q}_\alpha+\operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta-\eta)=\varrho(\widehat{Q}_\alpha+\operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta)=-\alpha$ and $\nabla^2\lambda(\theta)=\Gamma_\alpha$. For real z' near θ' we have $\varrho(A'-\operatorname{diag}z')<0$, so $(1,X(z')b_1)\ge0$ by the first fact. This vector lies in the kernel of $A_F-\operatorname{diag}(g(z'),z')$ (rows $i\ge2$ by the definition of X, and row 1 by the definition of g). By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'),z')=0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'),z')=0$ in z_j gives $\lambda_1g_j+\lambda_j=0$, and differentiating again in z_i gives $\lambda_1g_{ij}+\lambda_{11}g_ig_j+\lambda_{1j}g_i+\lambda_{i1}g_j+\lambda_{ij}=0$. Put $w_j=e_j-(\alpha_j/\alpha_1)e_1\in\mathbb{R}^F$ for $j\ge2$. Then the first equation is $g_j=-\alpha_j/\alpha_1$, so $w_j=e_j+g_je_1$, and the last four terms of the second equation are $w_i^\top\Gamma_\alpha w_j$. So

$$g_j=-\frac{\alpha_j}{\alpha_1},\qquad \alpha_1g_{ij}=w_i^\top\Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j=\alpha_1g_j+\alpha_j=0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top\Gamma_\alpha V$, where V has columns $w_2,\dots,w_k$. Since $\partial/\partial\varphi_j=\mathrm{i}z_j\,\partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi\Psi(\alpha,0)=0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla_\varphi^2\Psi(\alpha,0)=\Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B=I+\mathbf{1}\alpha'^\top/\alpha_1$. For $y\in\mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy+(\alpha'^\top y/\alpha_1)\mathbf{1}=(0,By)$. Since $\Gamma_\alpha\mathbf{1}=0$, we may replace Vy by $(0,By)$ in $y^\top V^\top\Gamma_\alpha Vy$, which gives $\mathcal{H}=B^\top\Sigma_\alpha B$. The matrix determinant lemma gives $\det B=1+\alpha'^\top\mathbf{1}/\alpha_1=1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det\mathcal{H}=\det\Sigma_\alpha/\alpha_1^2$.

Regularity. By Proposition 5.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0\in\mathcal{K}$. By (34) and continuity there is $\delta\in(0,\min_i\theta_i(\alpha_0))$ with $\varrho(A'-\operatorname{diag}(\theta'(\alpha_0)-\delta\mathbf{1}))<0$. By Lemma A.2, X, g and b are holomorphic where $|z_i|>\theta_i(\alpha_0)-\delta$ for every $i\ge2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\}\times\mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ-derivatives are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$.

The constant. The matrix $\mathcal{J}=\Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det\mathcal{J}=\prod_{i\ge2}\theta_i^2\,\det\Sigma_\alpha/\alpha_1^2$. So the lemma gives, uniformly for $\alpha\in\mathcal{K}$ and $T\ge1$,

$$\int_{\mathbb{T}^{k-1}}e^{T\Psi}\chi\,d\varphi=e^{T\langle\alpha,\theta\rangle}\Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}}\frac{\alpha_1}{\prod_{i\ge2}\theta_i\sqrt{\det\Sigma_\alpha}}\Big(\chi(\alpha,0)+O\Big(\frac1T\Big)\Big).$$

Here $\chi(\alpha,0)=b(\theta')\prod_{i\ge2}\theta_i$. As in the derivation of (34), $r'=r_1X(\theta')b_1$. In the same way the columns $j\ge2$ of $l^\top(A_F-\operatorname{diag}\theta)=0$, which is (8), give $l'=l_1X(\theta')^\top a_1$. Hence $\mu\cdot r=r_1(\mu_1+\mu'^\top X(\theta')b_1)$ and $l\cdot\mathbf{1}=l_1(1+a_1^\top X(\theta')\mathbf{1})$. Since $l_1r_1=\alpha_1$,

$$b(\theta')=\frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{\alpha_1}.$$

As $r_\alpha,l_\alpha>0$ and $\mu(F)>0$ by (H), $\chi(\alpha,0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha,0)(1+O(1/T))$. Putting this into (37) and using $J_F(\alpha)=\langle q-\theta,\alpha\rangle$ gives $f_T(\alpha)=T^{(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}(1+O(1/T))$, uniformly on $\mathcal{K}$. □

A.5 Proof of Lemma 5.5

Idea of the proof. The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them term by term. The two terms for m differ by a factor κ_m that is close to 1 when the number of runs M is small compared with $\sqrt{N}$, and under the continuum weights a path has of order T runs.

Step 1: the ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i\in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \qquad m \in \mathbb{Z}^F_{\geq 1}.$$

Since $h^{m_i-1}\binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!}\prod_{1\leq j<m_i}(1-j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i\in F}\Big[\prod_{1\leq j<m_i}\Big(1-\frac{j}{n_i}\Big)\Big](1-hq_i)^{n_i-m_i}e^{hq_in_i}.$$

Let $w_m = v_m/\sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m / \sum_m v_m = \mathbb{E}_w \kappa_m$.

Step 2: bounds for κ_m. Let $n_{\min} = \min_i n_i$. We claim that for every m

$$\Big(1-\frac{M^2}{2n_{\min}}\Big)e^{-q_{\max}^2T^2/N} \leq \kappa_m \leq e^{hq_{\max}M}. \tag{38}$$

First let $m_i \leq n_i$ for every i. The factors $1-j/n_i$ are at most 1, and $(1-hq_i)^{n_i-m_i} \leq e^{-hq_i(n_i-m_i)}$, so $\kappa_m \leq e^{hq_{\max}M}$. For the lower bound, $\log(1-x) \geq -x-x^2$ for $0 \leq x \leq \frac12$ gives $(1-hq_i)^{n_i-m_i}e^{hq_in_i} \geq e^{-n_ih^2q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2T^2/N}$. The factors $1-j/n_i$ lie in $[0,1]$, so their product is at least 1 minus the sum of the j/n_i, and hence at least $1-\sum_i m_i^2/(2n_i) \geq 1-M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then $\kappa_m = 0$, while $M \geq n_{\min}+1$ gives $M^2/(2n_{\min}) \geq 2$, so the left side of (38) is negative.

By (38) it remains to bound $\mathbb{E}_w e^{hq_{\max}M}$ and $\mathbb{E}_w M^2$ from above.

Step 3: the number of runs under w. Since W for σA_F is $\sigma^{M-1}W(m)$, we have $\mathbb{E}_w\sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0) = 0$, as the logarithm of a moment generating function, and $\Upsilon(1) \geq 0$ since $M \geq 1$. So $\Upsilon(c) \leq c\,\Upsilon(1)$ for $0 \leq c \leq 1$. We show that $\Upsilon(1) \leq C_{\mathcal{K}}T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (36) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1}b(\theta')\prod_{i\geq 2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$. For the lower bound note that $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TJ_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $J_F(\alpha) = \langle q-\theta,\alpha\rangle$. By Theorem 5.4 this is at least $\frac12 T^{-(k-1)/2}C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1,T_1]\times\mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \leq C_{\mathcal{K}}T$.

Step 4: the two averages. Let $x = hq_{\max} \leq \frac12$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \leq e^{x(1+C_{\mathcal{K}}T)} = 1+O(T^2/N)$, because $x \leq q_{\max}T/N$ and $T^2/N \leq 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\Upsilon(1)-y}$ for $y \geq \Upsilon(1)$ and the bound 1 for $y < \Upsilon(1)$. This gives

$$\begin{aligned}\mathbb{E}_w(M-1)^2 &= \int_0^\infty 2y\,\mathbb{P}_w(M-1 \geq y)\,dy \\ &\leq \Upsilon(1)^2 + \int_{\Upsilon(1)}^\infty 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2 + 2\Upsilon(1) + 2,\end{aligned}$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_{\mathcal{K}} N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}} \min_i \alpha_i > 0$. So by (38), $\mathbb{E}_w \kappa_m$ lies between $e^{-q_{\max}^2 T^2/N}(1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}} N))$ and $\mathbb{E}_w e^{hq_{\max} M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$. □

A.6 Proof of Theorem 4.2(ii)

Let F be admissible, with components $C_1, \ldots, C_r$. If F is irreducible, then $r = 1$, and the argument below uses only Lemma 4.1 and an explicit family of paths, not the local law.

Idea of the proof. A path with support F visits the components one after the other, in r blocks, and each passage to the next block is one switch. For the upper bound we apply Lemma 4.1 to every block. For the lower bound we choose the block lengths ourselves. A component of least exit rate gets almost all the time, and every other component C gets only about N/T steps, the time scale of a single switch. Such a short block has probability bounded below, and it has only about $(N/T)^{|C|-1}$ compositions instead of $N^{|C|-1}$.

If $Q = 0$, only singletons with positive start mass are admissible, and the bounds follow from $M_{\{i\}} = \mu_i$. Otherwise let a be the least positive rate q_{ij}. Since $T \leq N^{1/3}$, we have $h \leq N^{-2/3}$ and $T^2/N \leq 1$, and we take N_0 so large that h is as small as we need. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j-th block lies in the component D_j and starts at y_j, and whose switch out of block j is $x_j \to y_{j+1}$ for $j < r$. So block j ends at x_j for $j < r$, and the pattern does not fix the end of the last block. Let $\operatorname{supp} n = F$. Block j has length $N_j = \sum_{i\in D_j} n_i$. By the Markov property the paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j<r} hq_{x_j y_{j+1}} \prod_{j\leq r} \varpi_j$. Here ϖ_j is the probability that the chain started at y_j stays in D_j for N_j visits, with the composition that n prescribes on D_j, and for $j < r$ also ends at x_j. We drop the condition on the end. Then ϖ_j is at most the probability that $X_1, \ldots, X_{N_j}$ have that composition for the chain started at y_j. If $|D_j| \geq 2$, Lemma 4.1 with start y_j and $\theta = q - \lambda_{D_j}\mathbf{1}$ gives $\varpi_j \leq \kappa_1 h^{|D_j|-1} e^{-h\lambda_{D_j} N_j + \kappa_1 h^2 N_j}$, where κ_1 is the largest of these finitely many constants. If $D_j = \{i\}$, then $\lambda_{D_j} = q_i$ and $\varpi_j = (1 - hq_i)^{N_j - 1} \leq e^{1/2 - h\lambda_{D_j} N_j}$. Now we multiply over j. The powers of h add up to $\sum_j (|D_j| - 1) + r - 1 = k_F - 1$. In the exponent, $\lambda_{D_j} \geq \lambda_F$ and $\sum_j N_j = N$ give $h \sum_j \lambda_{D_j} N_j \geq T\lambda_F$ and $\kappa_1 h^2 \sum_j N_j = \kappa_1 T^2/N \leq \kappa_1$. Summing over the patterns gives $P_N(n) \leq \kappa_2 (T/N)^{k_F - 1} e^{-T\lambda_F}$.

Lower bound. We write down a family of paths with support F, bound its probability from below and count its compositions. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A. Take a walk $y = v_0, \ldots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \leq |C|^2$. Consider the paths $X_1, \ldots, X_m$ that make exactly these J switches, in this order, at any J of the $m - 1$ transition times. There are $\binom{m-1}{J}$ of them, and each has probability at least $(ha)^J (1 - hq_{\max})^m$. For large N we have $m > |C|^2 \geq J$. Then $\binom{m-1}{J} h^J \geq (hm - Jh)^J/J! \geq (1 - Jh)^J/J!$ and $(1 - hq_{\max})^m \geq e^{-2hq_{\max} m} \geq e^{-4q_{\max}}$, since $1 \leq hm \leq 2$. So their total probability is at least $(1 - Jh)^J a^J e^{-4q_{\max}}/J!$, and $(1 - Jh)^J \geq 2^{-J}$ for small h.

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the event of Claim A, with C_{j^*} in place of C and ℓ visits in place of m, has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let

$w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*}. Since $B^s w = (1 - h\lambda_F)^s w$, the probability of staying in C_{j^*} for s steps from z is $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y, then $s = \ell - 2m + 1$ steps inside C_{j^*} to any state z', then Claim A for the last m visits, from z' to x. By the Markov property this gives at least $c_A^2 (\min w / \max w)(1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2\lambda_F^2$.

The family of paths. Give block $j \neq j^*$ the length m and block j^* the length $\ell = N - (r-1)m$, which is at least $2m$ for $T \geq T_0 = r + 2$ and $N \geq N_0$. Consider the paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j + 1$ by the switch $x_j \to y_{j+1}$. They all have support F. By the Markov property, and since $h\ell \leq T$ and $h^2\ell \leq T^2/N \leq 1$, their total probability is at least $\mu_{y_1}(ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. The composition of such a path is fixed by its restrictions to the blocks. On the block in C_j the restriction is a vector of $|C_j|$ positive integers whose sum is m for $j \neq j^*$ and ℓ for $j = j^*$. A vector of positive integers with a given sum is fixed by all but one of its entries, and each entry is at most the sum. Also $\ell \leq N$. So these paths have at most $(N+1)^{k^*-1} \prod_{j \neq j^*} (m+1)^{|C_j|-1}$ compositions, and $m + 1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r-1)$ and $h = T/N$, dividing the total probability by the number of compositions gives $M_F \geq c\, N^{-(k_F-1)} T^{k_F - k^*} e^{-T\lambda_F}$. □

A.7 A forest formula for the covariance

For weights $f_{ij} \geq 0$ on the ordered pairs $i \neq j$ of F and a state $i \in F$ let

$$\mathrm{arb}_i(f) = \sum_{\mathcal{T}} \prod_{a \neq i} f_{a\, p_{\mathcal{T}}(a)},$$

the sum over the spanning trees $\mathcal{T}$ of the complete graph on F, where $p_{\mathcal{T}}(a)$ is the neighbor of a on the path from a to i in $\mathcal{T}$. So $\mathrm{arb}_i(f)$ is the weighted count of the spanning trees oriented toward the root i. If f is symmetric, then $\mathrm{arb}_i(f) = \mathrm{tree}(f)$ for every i.

Lemma A.3 (a forest formula for the covariance)**.** *Let G be an irreducible generator on F with stationary law π, where $k = |F| \geq 2$, and let D, Z and Γ be as at the start of this appendix, and Σ as in Lemma A.1 (namely Γ without the row and column of state 1). Put $f_{ij} = \pi_i G_{ij}$ for $i \neq j$, the stationary flow from i to j.*

(i) The number $\mathrm{arb}_i(f)$ does not depend on i. We write $\mathrm{arb}(f)$ for it. Also $\det(-G)_{(i)} = \pi_i \,\mathrm{arb}(f) / \prod_j \pi_j$ for every i, where (i) means that row and column i are deleted, and

$$\det(\mathbf{1}\pi^\top - G) = \sum_{i \in F} \det(-G)_{(i)} = \frac{\mathrm{arb}(f)}{\prod_j \pi_j}.$$

(ii) Let $S = -(DG + G^\top D)$. Then S is the Laplacian of the conductances $f_{ij} + f_{ji}$, and $\Gamma = Z^\top S Z$.

(iii) With $\mathrm{tree}(f + f^\top)$ the weighted count of spanning trees for the conductances $f_{ij} + f_{ji}$,

$$\det \Sigma = \frac{\mathrm{tree}(f + f^\top)}{\det(\mathbf{1}\pi^\top - G)^2} = \frac{(\prod_j \pi_j)^2 \,\mathrm{tree}(f + f^\top)}{\mathrm{arb}(f)^2}.$$

For two states with rates a and b this gives $\det \Sigma = 2ab/(a+b)^3$.

Proof. We use three facts about the adjugate $\operatorname{adj} M$ of a $k \times k$ matrix M. First, $\operatorname{adj}(AB) = \operatorname{adj}(B)\operatorname{adj}(A)$ for all square matrices A and B, by the Cauchy–Binet formula. Second, $\det(M + uv^\top) = \det M + v^\top \operatorname{adj}(M)\, u$ for all vectors u and v. Third, suppose that M has rank $k-1$ and that $Mu = 0$ and $v^\top M = 0$ with u and v nonzero. Then $\operatorname{adj} M = m\, uv^\top$ for some number m. Indeed $M \operatorname{adj} M = \operatorname{adj}(M) M = (\det M) I = 0$, so every column of $\operatorname{adj} M$ lies in the kernel $\mathbb{R}u$ of M, and every row lies in the left kernel $\mathbb{R}v^\top$. A matrix with these two properties is a multiple of $uv^\top$.

(i) Let $\mathcal{L} = -DG$. Its off-diagonal entries are $-f_{ij}$, its row sums vanish because $G\mathbf{1} = 0$, and its column sums vanish because $\pi^\top G = 0$. The kernel of G is $\mathbb{R}\mathbf{1}$ because G is irreducible, so $\mathcal{L}$ has rank $k-1$. By the third fact with $u = v = \mathbf{1}$, all entries of $\operatorname{adj} \mathcal{L}$ are equal. By the matrix-tree theorem for directed graphs [32, 9], the diagonal entry $\det \mathcal{L}_{(i)}$ of $\operatorname{adj} \mathcal{L}$ is $\operatorname{arb}_i(f)$. So $\operatorname{arb}_i(f)$ does not depend on i. Deleting row and column i from $-G = D^{-1}\mathcal{L}$ gives $\det(-G)_{(i)} = \operatorname{arb}(f)/\prod_{j \neq i} \pi_j = \pi_i \operatorname{arb}(f)/\prod_j \pi_j$, and the sum over i is $\operatorname{arb}(f)/\prod_j \pi_j$ because $\sum_i \pi_i = 1$. Next, $-G$ has rank $k-1$, right kernel $\mathbb{R}\mathbf{1}$ and left kernel $\mathbb{R}\pi^\top$. So $\operatorname{adj}(-G) = g\, \mathbf{1}\pi^\top$ for some number g by the third fact. The trace of $\mathbf{1}\pi^\top$ is 1, so g is the trace of $\operatorname{adj}(-G)$, which is $\sum_i \det(-G)_{(i)}$. Since $\det(-G) = 0$, the second fact gives

$$\det(\mathbf{1}\pi^\top - G) = \pi^\top \operatorname{adj}(-G)\, \mathbf{1} = g\, (\pi^\top \mathbf{1})(\pi^\top \mathbf{1}) = g.$$

(ii) The matrix S is symmetric. For $i \neq j$ its entry is $-(\pi_i G_{ij} + \pi_j G_{ji}) = -(f_{ij} + f_{ji})$, and $S\mathbf{1} = -DG\mathbf{1} - G^\top \pi = 0$. So S is the Laplacian of the conductances $f_{ij} + f_{ji}$. By (9), $-GZ = I - \mathbf{1}\pi^\top$ and $\pi^\top Z = 0$. Hence

$$Z^\top(-DG)Z = Z^\top D(I - \mathbf{1}\pi^\top) = Z^\top D - (\pi^\top Z)^\top \pi^\top = Z^\top D,$$

where we used $D\mathbf{1} = \pi$. The transpose of this identity is $Z^\top(-G^\top D)Z = DZ$. Adding the two gives $Z^\top S Z = Z^\top D + DZ = \Gamma$.

(iii) Put $w = \det(\mathbf{1}\pi^\top - G)$, which is not 0 because Z exists. By the matrix-tree theorem [5] every diagonal entry of $\operatorname{adj} S$ is $\operatorname{tree}(f + f^\top)$. This number is positive because the graph of G is strongly connected. So S has rank $k-1$, and the third fact with $u = v = \mathbf{1}$ gives $\operatorname{adj} S = \operatorname{tree}(f + f^\top)\, \mathbf{1}\mathbf{1}^\top$. Now we turn to Z. By (9), $Z\mathbf{1} = 0$ and $\pi^\top Z = 0$. The matrix $Z + \mathbf{1}\pi^\top = (\mathbf{1}\pi^\top - G)^{-1}$ is invertible, so Z has rank $k-1$, and $\operatorname{adj} Z = z\, \mathbf{1}\pi^\top$ for some number z by the third fact. Since $\det Z = 0$, the second fact gives

$$\frac{1}{w} = \det(Z + \mathbf{1}\pi^\top) = \pi^\top \operatorname{adj}(Z)\, \mathbf{1} = z.$$

Also $\operatorname{adj}(Z^\top) = (\operatorname{adj} Z)^\top = z\, \pi \mathbf{1}^\top$. By (ii) and the first fact,

$$\operatorname{adj} \Gamma = \operatorname{adj}(Z) \operatorname{adj}(S) \operatorname{adj}(Z^\top) = z^2 \operatorname{tree}(f + f^\top)\, \mathbf{1}(\pi^\top \mathbf{1})(\mathbf{1}^\top \pi)\mathbf{1}^\top = \frac{\operatorname{tree}(f + f^\top)}{w^2}\, \mathbf{1}\mathbf{1}^\top.$$

The entry of $\operatorname{adj} \Gamma$ at state 1 is $\det \Sigma$. This is the first formula in (iii), and the second follows from (i). □

B A Laplace lemma with a parameter

This appendix proves the Laplace lemma that Appendix A uses for Theorem 5.4. There the parameter is $\alpha \in \mathcal{K}$. A field needs no parameter, since it enters the density as the exact factor

$e^{Tv\cdot\alpha}$, and the compact space $\mathcal{K}$ can include the rates and the start when constants are needed uniformly over a class of chains. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \operatorname{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m.

Lemma B.1 (Laplace integrals with a parameter)**.** *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha, 0)$ is real and $\nabla_\varphi \Psi(\alpha, 0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re}\Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha, \varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

Idea of the proof. By (iii) and compactness the part of the torus away from $\varphi = 0$ is exponentially small. Near 0 we substitute $\varphi = x/\sqrt{T}$ and expand. The integrand becomes a Gaussian weight times $a(\alpha, 0)$, plus a term of order $T^{-1/2}$ that is odd in x and integrates to 0, plus a remainder of order $1/T$. All constants depend on α only through bounds that hold on the whole of $\mathcal{K}$.

Proof. By compactness $B_\Psi = \max_{|\beta|\leq 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta|\leq 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $\sigma_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \leq (mB_\Psi)^m$. Below, C depends only on m, $\sigma_{\min}$, B_Ψ, B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0, 1)$ with $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$. The set of (α, φ) with $\|\varphi\| \geq \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re}\Psi(\alpha, \varphi) - \Psi(\alpha, 0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)} \leq C T^{-1-m/2} e^{T\Psi(\alpha,0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \leq \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \leq \varepsilon$, and it preserves measure. Let $R(\alpha, \varphi) = \Psi(\alpha, \varphi) - \Psi(\alpha, 0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6}\Psi_{ijk}\varphi^i\varphi^j\varphi^k + \omega_0$ with $|\omega_0| \leq C\|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha, 0)$. It also gives $|R| \leq \frac{m^3}{6} B_\Psi \|\varphi\|^3 \leq \frac{1}{4}\sigma_{\min}|\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \leq \varepsilon|\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha,0)} \int_{\|x\|\leq\varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J} x}\, e^{w(x)} a(\alpha, x/\sqrt{T})\, dx,$$

and $|w(x)| \leq \frac{1}{4}\sigma_{\min}|x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$, $a_0 = a(\alpha, 0)$ and $a_i = \partial_{\varphi_i} a(\alpha, 0)$. Then $w = T^{-1/2}P + \omega_1$ with $|\omega_1| \leq CT^{-1}|x|^4$, and $|w| \leq CT^{-1/2}|x|^3$. Since $|e^w - 1 - w| \leq \frac{1}{2}|w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2}P| \leq CT^{-1}(|x|^4 + |x|^6)e^{\sigma_{\min}|x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \leq CT^{-1/2}|x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2}a_i x^i| \leq CT^{-1}|x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\big(a_0 P(x) + a_i x^i\big) + \omega(x), \qquad |\omega(x)| \leq CT^{-1}(1 + |x|)^6 e^{\sigma_{\min}|x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \geq \sigma_{\min}|x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1+|x|)^6 e^{-\sigma_{\min}|x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon\sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \leq CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2}x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha,0)} T^{-m/2} \big[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})\big]$. Since $\det \mathcal{J} \leq (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha,0)}(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. □

Remark B.2 (the uniform form). The proof uses compactness and joint continuity only to obtain the bounds B_Ψ and B_a on the derivatives, the lower bound $\sigma_{\min}$ on the least eigenvalue of $\mathcal{J}$, and a gap $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0) \leq -\eta$ for $\|\varphi\| \geq \varepsilon$, where $\varepsilon \in (0,1)$ satisfies $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$. So the following holds with the same proof. Let $\mathcal{P}$ be any set, and let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ satisfy (i) and (ii) for every $\alpha \in \mathcal{P}$, with φ-derivatives of Ψ up to order 4 and of a up to order 2 bounded by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$, with $\mathcal{J}(\alpha) \geq \sigma_{\min} I$ for all α, and with the gap above for all α. Then the expansion of Lemma B.1 holds uniformly for $\alpha \in \mathcal{P}$ and $T \geq 1$, with a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

C Proofs for the complete graph

This appendix proves Theorems 7.5 and 7.6 and Corollary 7.7, with the notation of Section 7.2. The first lemma writes the law with blocks between refreshes in place of runs.

Lemma C.1 (the refresh form)**.** *Let $N \geq 1$ and let n be a composition with support F. Sums over m run over the integer vectors with $1 \leq m_i \leq n_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) With $0^0 = 1$,

$$P_N(n) = \sum_m \Big(\sum_{i\in F} \mu_i m_i\Big) \frac{(M-1)!}{\prod_i m_i!}\, h^{M-1} \sigma^{N-M} \prod_{i\in F} \binom{n_i - 1}{m_i - 1}.$$

(b) Let $1/d < p < 1$, so that $0 < u < 1$ and $v > 0$. Then

$$S_n(u) = (1-u)^{N-1} \sum_m M!\, v^{M-1} \prod_{i\in F} \frac{1}{m_i!} \binom{n_i - 1}{m_i - 1}, \tag{39}$$

which is a sum of positive terms, and

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i\in F} B_{n_i}(vt)\, dt, \tag{40}$$

where B_n is defined in (23).

Proof. (a) The transition matrix is $I + hQ = \sigma I + hJ$, so a word $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (\sigma \mathbf{1}\{x_s = x_{s+1}\} + h)$. We expand each factor into its two terms and call the steps where the term h is chosen the refreshes. If there are $M-1$ refreshes, they cut the word into M blocks, and a term of the expansion is not zero only if the word is constant on each block. So $P_N(n)$ is the sum, over the sequences of M block labels and over the block lengths, of $\mu_{\ell_1} h^{M-1} \sigma^{N-M}$, where ℓ_1 is the label of the first block. If m_i blocks carry the label i, there are $m_i(M-1)!/\prod_j m_j!$ sequences of labels that start with i. Consecutive labels may be equal. Given the labels, the

lengths of the m_i blocks of the label i form an ordered sum of n_i with m_i positive parts, in $\binom{n_i-1}{m_i-1}$ ways. Multiplying and summing gives (a).

(b) For the uniform start $\sum_i \mu_i m_i = M/d$. By (20) we divide (a) by $V_N = p^{N-1}/d$, and $h^{M-1}\sigma^{N-M}/p^{N-1} = u^{M-1}(1-u)^{N-M} = (1-u)^{N-1}v^{M-1}$, since $\sigma/p = 1-u$. This gives (39). We write $M! = \int_0^\infty e^{-t}t^M\,dt$ and sum over each m_i separately. All terms are nonnegative, so the sum and the integral can be exchanged, and this gives (40). □

When $\sigma \geq 0$, part (a) says that the chain keeps its state with probability σ and otherwise draws a fresh state uniformly from all d states, possibly the same one. The proof does not need $\sigma \geq 0$.

C.1 Proof of Theorem 7.5

Let F be a face of k states and n a balanced composition of F, so that $P_N(n)/V_N = S_n(u)$ by (20), with $u = (1-p)/((d-1)p)$.

Part (i), one crossing. A word with j switches is a sequence of $j+1$ run colors in which successive colors differ, together with run lengths. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. So every coefficient of S_n is nonnegative, and the first nonzero one is $W_n(k-1) = k!$, from the $k!$ orders of k single runs. Hence S_n increases strictly from 0 to ∞ on $u > 0$ and equals 1 exactly once. As p increases over $(0,1)$, u decreases over $(0,\infty)$. This gives $p_{N,k}$ and the two inequalities for the balanced composition.

Part (i), monotonicity in N. Choose the balanced compositions for increasing N by adding 1 to one of the smallest coordinates. Each coefficient $W_n(j)$ is then nondecreasing. Indeed, inserting an extra copy of the incremented color just before its first occurrence keeps the number of switches, and deleting the first letter of the first run of that color inverts it. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2}\sum_i (n_i - 1) = \frac{k!(k-1)}{2}(N-k),$$

since exactly one run color appears twice, and for each choice of it there are $(k+1)!/2 - k! = k!(k-1)/2$ admissible sequences of run colors (all orders, minus those with the two copies adjacent) and $n_i - 1$ ways to split its n_i visits into two runs. This coefficient increases strictly with N. So at every $u > 0$ the value $S_n(u)$ increases strictly with N, the root of $S_n(u) = 1$ decreases strictly, and $p = 1/(1+(d-1)u)$ increases.

Part (i), the first crossing. If $N = k$, every coordinate is 1 and $S_n(u) = k!u^{k-1}$, which gives $p_{k,k}$. Since the root decreases with N, every root has $u \leq (k!)^{-1/(k-1)} < 1$, and $u < 1$ means $h < p$, that is, $p > 1/d$.

Part (i), the face maximum. For $p \geq 1/d$, Theorem 7.6 gives $M_F = P_N(n)$. For $p < 1/d$ we have $u > 1$, which exceeds the root of $S_n(u) = 1$, so $M_F \geq P_N(n) > V_N$. Hence $M_F = V_N$ exactly at $p_{N,k}$, with the same two inequalities.

Part (ii). By (10) with $K_F = D_k/d$ and $\lambda_F = d-k$ (Corollary 7.3), uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \tfrac{k-1}{2}\log T + \log(D_k/d) - (d-k)T + O(1/T + T^2/N).$$

By (2), $\log V_N = -\log d - (d-1)T + O(T^2/N)$. Subtracting, and using $\log D_k = -(k-1)a_k$, gives the display. With $T = L - \frac{1}{2}\log L + t$ and t in a compact set, $T - L + \frac{1}{2}\log T = t + \frac{1}{2}\log(T/L) = t + O(\log L/L)$, and the error tends to 0.

Part (iii). Let G be a vertex, so that $M_G = V_N$, $\Delta = \delta = k-1$ and $\log(K_F/K_G) = \log D_k = -(k-1)a_k$. By Theorem 5.8(i) the function $\log(M_F/M_G)$ has a zero in $[L/2, 2L]$, and by (i)

this zero is $T_{N,k}$. Dividing (12) by $k-1$ gives the implicit equation. For the explicit form put $z_* = L - \frac{1}{2}\log L + a_k + e_L/L$ with $e_L = \frac{1}{4}\log L - \frac{1}{2}a_k$. Since $\log z_* = \log L - \frac{\log L}{2L} + \frac{a_k}{L} + O(\frac{(\log L)^2}{L^2})$, we get $z_* + \frac{1}{2}\log z_* = L + a_k + O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2}\log z$ is at least 1, so $T_{N,k} - z_* = O(1/L)$, which is (22).

Part (iv). We have $a_k = \frac{1}{2}\log(4\pi) - f(k)$ with $f(x) = (x+\frac{1}{2})\log x/(x-1)$. For $x \geq 2$ the sign of $f'(x)$ is the sign of $\Delta(x) = 2x^2 - x - 1 - 3x\log x$. Here $\Delta(2) = 5 - 6\log 2 > 0$, and $\Delta'(x) = 4x - 4 - 3\log x > 0$ for $x \geq 2$, since $\Delta'(2) > 0$ and $\Delta''(x) = 4 - 3/x > 0$. So f increases strictly and a_k decreases strictly. This order is checked in Lean. □

C.2 Proof of Theorem 7.6

Let $\mu_a = \mu_0$ for $a \in F$. By (3), $P_N(n) = \mu_0 p^{N-1} S_n(u)$ for every composition with support F, so it is enough to show that the move strictly increases $S_n(u)$. The rest follows, since S_n is symmetric in the coordinates of n and repeated moves reach a balanced composition. At $p = 1/d$ we have $u = 1$, $S_n(1) = N!/\prod_i n_i!$, and the move multiplies it by $n_b/(n_a+1) > 1$. Let $1/d < p < 1$, so that $v > 0$.

Log-concavity. We show that for each $w > 0$ the sequence $B_n(w)$, $n \geq 1$, is strictly log-concave. Write

$$\frac{B_n(w)}{w} = b_{n-1}, \qquad b_j = \sum_{i=0}^{j} \binom{j}{i} \rho_i, \quad \rho_i = \frac{w^i}{(i+1)!}, \qquad c_j = \sum_{i=0}^{j} \binom{j}{i} \rho_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the mean of the strictly decreasing sequence $f_i = \rho_{i+1}/\rho_i = w/(i+2)$ under the weights $\omega_i = \binom{j}{i}\rho_i$, $0 \leq i \leq j$, normalized. Let $\omega'_i = \binom{j+1}{i}\rho_i$, $0 \leq i \leq j+1$, be the weights for $j+1$, and put $\omega_{j+1} = 0$. On $0 \leq i \leq j$ the ratio $\omega'_i/\omega_i = (j+1)/(j+1-i)$ increases in i. So for $i < l$ we have $\omega'_i\omega_l - \omega'_l\omega_i \leq 0$, with strict inequality when $l = j+1$. Hence

$$\sum_i \omega'_i f_i \sum_l \omega_l - \sum_i \omega'_i \sum_l \omega_l f_l = \sum_{i<l} (\omega'_i\omega_l - \omega'_l\omega_i)(f_i - f_l) < 0,$$

since $f_i > f_l$ for $i < l$. So $c_{j+1}/b_{j+1} < c_j/b_j$, and $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j. By the Laguerre form in (23), this is a Turán-type inequality for the Laguerre polynomials.

The transfer. Let $n_b \geq n_a + 2$. By log-concavity, $B_{n_a+1}(w)B_{n_b-1}(w) > B_{n_a}(w)B_{n_b}(w)$. We apply this for $w = vt > 0$ in (40) and integrate. □

C.3 Proof of Corollary 7.7

For large N, $T \leq CL$ forces $h \to 0$ and so $p > 1/d$. By Theorem 7.6 every global mode is a vertex or a balanced composition. So we compare the ratios M_F/V_N, which equal $S_n(u)$ at the balanced compositions n of F and depend on F only through $k = |F|$, with 1 and with each other. Fix $0 < c < 1$.

Case $T \leq cL$. The ratio $S_n(u)$ increases with u, and so with T. So it is at most its value at $T = cL$, which is $\exp(-(k-1)(1-c)L + O(\log L)) \to 0$ by Theorem 7.5(ii). So the vertices are the only modes, and $p > p_{N,d}$.

Case $cL \leq T \leq CL$. Theorem 7.5(ii) gives $\log(M_F/V_N) = (k-1)(y - a_k) + o(1)$ uniformly, with $y = T - L + \frac{1}{2}\log T$ the same for every k. Fix $0 < \delta < (a_{d-1} - a_d)/2$ when $d \geq 3$, and any $\delta > 0$ when $d = 2$. If $y \leq a_d - \delta$, every face ratio is at most $e^{-(k-1)\delta + o(1)} < 1$, so only the vertices

are modes. If $y \ge a_d + \delta$, the full center beats the vertices, and it beats a balanced composition with $2 \le k < d$ positive coordinates by

$$(d-k)(y-a_d) + (k-1)(a_k - a_d) + o(1) > 0.$$

If $|y - a_d| < \delta$, a proper face with $k \ge 2$ states has $(k-1)(y-a_k) + o(1) \le -(k-1)\delta + o(1) < 0$, since $a_k - a_d > 2\delta$. So it loses to the vertices, and the comparison of the vertices with the full centers is decided by the unique crossing $p_{N,d}$ of Theorem 7.5(i). In each subcase the conclusion agrees with (i) to (iii).

The last claim follows from (22) and $p_{N,k} = 1 - (d-1)T_{N,k}/N$, since $a_2 > \cdots > a_d$. □

D Proofs near a vertex

We prove Lemma 8.1 and Theorem 8.2, with the notation of Section 8.

Proof of Lemma 8.1. By definition, $B_a(w)/w = \sum_{j=0}^{a-1} \binom{a-1}{j} w^j/(j+1)!$. So the coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\binom{a-1}{j} j!/a^j = \prod_{l=1}^{j}(1 - l/a)_+$, also for $j \ge a$, where it is 0. By the clipped products of [28, Lemma 3.2(b)], its deficit from 1 lies between 0 and $j(j+1)/(2a)$. Also $\sum_{j\ge1} j(j+1)x^j/[j!(j+1)!] = \sum_{j\ge1} x^j/[(j-1)!\,j!] = x\,\mathcal{I}(x)$. So

$$0 \le \mathcal{I}(aw) - \frac{B_a(w)}{w} \le \frac{1}{2a} \sum_{j\ge0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2}\,\mathcal{I}(aw).$$

□

Lemma D.1 (elasticity and concavity of the rare-count polynomial)**.** *The polynomial $\mathcal{B}$ is convex, $\log \mathcal{B}$ is concave on $(0,\infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1} \le y_a \le 1, \qquad \kappa(y) \le \ell + \sqrt{\ell|a|y} \quad (y > 0). \tag{41}$$

For $0 < y \le 1$ we also have $\kappa(y) \ge \max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$.

Proof. Step 1. One factor as a law. The polynomial $\mathcal{B}$ has nonnegative coefficients, so it is convex. For one factor B_a and $y > 0$, let J take values in $\{0, \dots, a-1\}$ with $\mathbb{P}(J = j)$ proportional to $\binom{a-1}{j} y^j/(j+1)!$. Since $B_a(y) = \sum_j y\binom{a-1}{j} y^j/(j+1)!$, the elasticity of B_a is $\kappa_a = yB_a'/B_a = \mathbb{E}[J+1]$. The ratio

$$\mathcal{R}(j) = \frac{(j+1)\mathbb{P}(J = j+1)}{\mathbb{P}(J = j)} = \frac{y(a-1-j)}{j+2}, \qquad \mathcal{R}(a-1) = 0,$$

is nonincreasing in j. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}\mathcal{R}(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)\mathcal{R}(J)]$. So $\operatorname{Var} J = \mathbb{E}J + \operatorname{Cov}(J, \mathcal{R}(J)) \le \mathbb{E}J$, since for an independent copy J' twice the covariance is $\mathbb{E}[(J-J')(\mathcal{R}(J) - \mathcal{R}(J'))] \le 0$.

Step 2. Concavity and monotonicity. Since $y^2 B_a''/B_a = \mathbb{E}[J(J+1)]$, we get

$$(\log B_a)'' = \frac{\operatorname{Var} J - \mathbb{E}[J+1]}{y^2} < 0, \qquad y\kappa_a' = \operatorname{Var} J \ge 0.$$

Summing over the factors proves the concavity of $\log \mathcal{B}$ and the monotonicity of $\kappa = \sum_i \kappa_{a_i}$.

Step 3. The elasticity bounds. The differential equation $yB_a'' + yB_a' = aB_a$, which we check on the coefficients, gives

$$ay = \kappa_a^2 + \operatorname{Var} J + (y-1)\kappa_a.$$

Using $0 \le \operatorname{Var} J \le \kappa_a - 1$ we get

$$\kappa_a \le 1 + \sqrt{ay}, \qquad \kappa_a^2 + y\kappa_a \ge ay + 1.$$

For $0 < y \le 1$ the second inequality gives $(\kappa_a + \frac{1}{2})^2 > ay$, and $\kappa_a \ge 1$. So $\kappa_a \ge \max\{1, \sqrt{ay} - 1/2\} \ge \frac{1}{2}\sqrt{ay}$. Summing over the factors proves the elasticity bounds, using the Cauchy–Schwarz inequality $\sum_i \sqrt{a_i} \le \sqrt{\ell |a|}$ for the upper bound and $\sqrt{\sum_i x_i} \le \sum_i \sqrt{x_i}$ for the lower one.

Step 4. The root. We have $B_a(1) \ge 1$, from its term $m = 1$, and $B_a(y) \le y\mathcal{I}(ay) \le ye^{ay/2}$ by Lemma 8.1 and $(j+1)! \ge 2^j$. So $\mathcal{B}(1) \ge 1$, while, since $|a| + 1 \ge 2$, $\mathcal{B}(1/(|a|+1)) \le (|a|+1)^{-\ell} e^{|a|/(2(|a|+1))} \le \frac{1}{2} e^{1/2} < 1$. □

Lemma D.2 (an exact compound-binomial representation)**.** *For $0 < u < 1$, put $v = u/(1-u)$. Let Γ_2 have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u. Take all these variables independent and set*

$$\mathcal{X} = \Gamma_2 + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})], \qquad \mathbb{E}e^{s\mathcal{X}} = (1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \tag{42}$$

In particular, set $y = bu^2$ and $\kappa = \kappa(y)$. For every $0 < u < 1$,

$$(1-u)^{|a|}\mathcal{B}(y) \le S_{(b,a)}(u). \tag{43}$$

If also $s = \kappa/[bu(1-u)] < 1$, then

$$S_{(b,a)}(u) \le (1-u)^{|a|}\mathcal{B}(y)e^{-\kappa}(1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1}. \tag{44}$$

Proof. For $0 < u < 1$ the stay probability $p = 1/(1 + (d-1)u)$ lies in $(1/d, 1)$, so by (40) with $N = b + |a|$,

$$S_{(b,a)}(u) = \frac{(1-u)^{b+|a|-1}}{v}\int_0^\infty e^{-t}B_b(vt)\,\mathcal{B}(vt)\,dt.$$

The integral of $e^{-t}B_b(vt)$ is $\sum_m \binom{b-1}{m-1} v^m = v(1+v)^{b-1}$, and $(1-u)^{b-1}(1+v)^{b-1} = 1$. So $e^{-t}B_b(vt)$, normalized, is the density $\sum_{j=0}^{b-1}\binom{b-1}{j} u^j(1-u)^{b-1-j}\, t^{j+1}e^{-t}/(j+1)!$, a mixture of gamma densities with shape $J+2$, where J has the binomial law with parameters $b-1$ and u. This is exactly the law of $\mathcal{X}$, which proves the identity and its moment generating function. Since $\mathbb{E}\mathcal{X} = 2 + (b-1)u$, we have $v\mathbb{E}\mathcal{X} - bu^2 = u(2 - u + bu^2)/(1-u) > 0$, so $v\mathbb{E}\mathcal{X} \ge y$. Jensen's inequality for the convex function $\mathcal{B}$, and the monotonicity of $\mathcal{B}$, prove (43). Concavity of $\log \mathcal{B}$ gives, for $w > 0$,

$$\mathcal{B}(w) \le \mathcal{B}(y)\exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = v\mathcal{X}$, note that $\kappa v/y = s$, and use the moment generating function to prove (44). □

Proof of Theorem 8.2. Write $S = S_{(b,a)}$, $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since S increases on $u > 0$, it suffices to show $\log S(u_+) > 0$ and $\log S(u_-) < 0$ at $u_\pm = (1 \pm \delta)u_0$. Here $\delta \le 1/2$ forces $\varepsilon \le 1/(32c_\ell) \le 1/96$, and $u_0 \le \sqrt{1/b} \le \varepsilon$.

Step 1. The upper point. We have $u_+ \le \frac{3}{2}\varepsilon < 1/4$. Since $\log \mathcal{B}(y_a) = 0$ and $(\log \mathcal{B})' = \kappa/y$, the monotonicity of κ gives $\log \mathcal{B}(bu_+^2) \ge \kappa_0 \log(bu_+^2/y_a) = 2\kappa_0 \log(1+\delta)$. The lower bound in Lemma D.1, applied at $y_a \le 1$, gives $|a|u_0/\kappa_0 \le 2\sqrt{|a|/b} \le 2\varepsilon$. Hence, by (43) and $-\log(1-x) \le x/(1-x)$,

$$\log S(u_+) \ge 2\kappa_0 \log(1+\delta) - \frac{|a|u_+}{1-u_+} \ge \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

Here we used $2\log(1+\delta) \ge \frac{4}{3}\delta$ for $\delta \le \frac{1}{2}$, and $|a|u_+/(1-u_+) \le \frac{4}{3} \cdot \frac{3}{2}|a|u_0 \le 4\kappa_0\varepsilon$. The last inequality holds since $\frac{4}{3}\delta = \frac{64}{3}c_\ell\varepsilon > 4\varepsilon$.

Step 2. The lower point. Write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \ge 1/(|a|+1)$, $\delta \le 1/2$ and the upper bound in (41) imply

$$u_- \le \varepsilon, \qquad bu_- \ge \frac{1}{2\varepsilon}, \qquad s = \frac{\kappa}{bu_-(1-u_-)} \le 2(2\ell + \sqrt{\ell})\varepsilon \le 4c_\ell\varepsilon \le \frac{1}{8}.$$

For the bound on s, note that $\kappa/(bu_-) \le \ell/(bu_-) + \sqrt{\ell|a|y}/(bu_-) \le 2\ell\varepsilon + \sqrt{\ell|a|/b} \le (2\ell + \sqrt{\ell})\varepsilon$, and $1 - u_- \ge \frac{1}{2}$. We use $\log(1+x) \le x$, $-\log(1-s) \le s/(1-s)$ and $u_-, s \le 1/4$ in (44), and we drop the nonpositive term $|a|\log(1-u_-)$. This gives

$$\begin{aligned}
\log \frac{S(u_-)}{\mathcal{B}(y)} &\le -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\
&= \kappa\left(\frac{1}{(1-u_-)(1-s)} - 1\right) + \frac{2s}{1-s} \\
&\le 5\kappa(u_- + s) \le 25c_\ell\kappa\varepsilon.
\end{aligned}$$

Here the equality uses $bu_-s = \kappa/(1-u_-)$. The next inequality uses $(1-u_-)(1-s) \ge \frac{21}{32}$, so the first term is at most $\frac{32}{21}\kappa(u_- + s)$, and $\kappa \ge \ell \ge 1$, so the second is at most $\frac{16}{7}\kappa s$. The last one uses $u_- + s \le 5c_\ell\varepsilon$. Again by the monotonicity of κ,

$$\log \mathcal{B}(y) = -\int_y^{y_a} \frac{\kappa(x)}{x}\,dx \le 2\kappa \log(1-\delta) \le -2\kappa\delta = -32c_\ell\kappa\varepsilon.$$

Hence $\log S(u_-) \le \kappa(-32c_\ell + 25c_\ell)\varepsilon < 0$. □

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